

2020 K-Innovation ODA Program with Indonesia

Policy Consultation on the Arrangement of Science, Technology and Innovation (STI) Governance: Coordination Mechanism, Institutional Framework & Financial System, and Investment in Indonesia

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Title	Policy Consultation on the Arrangement of Science, Technology and Innovation (STI) Governance: Coordination Mechanism, Institutional Framework & Financial System and Investment in Indonesia
Institute	Science and Technology Policy Institute (STEP I)
Partner Country	Republic of Indonesia
In Cooperation with	Indonesian Institute of Sciences (LIPI)
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Survey Research 2020-05-05

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Published on 30 December 2020

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www.stepi.re.kr

For more information, visit

iicc.stepi.re.kr

Printed by Mirae Media

Tel.: +82-2-815-0407 Fax: +82-2-822-1269

ISBN 978-89-6112-738-7 93300



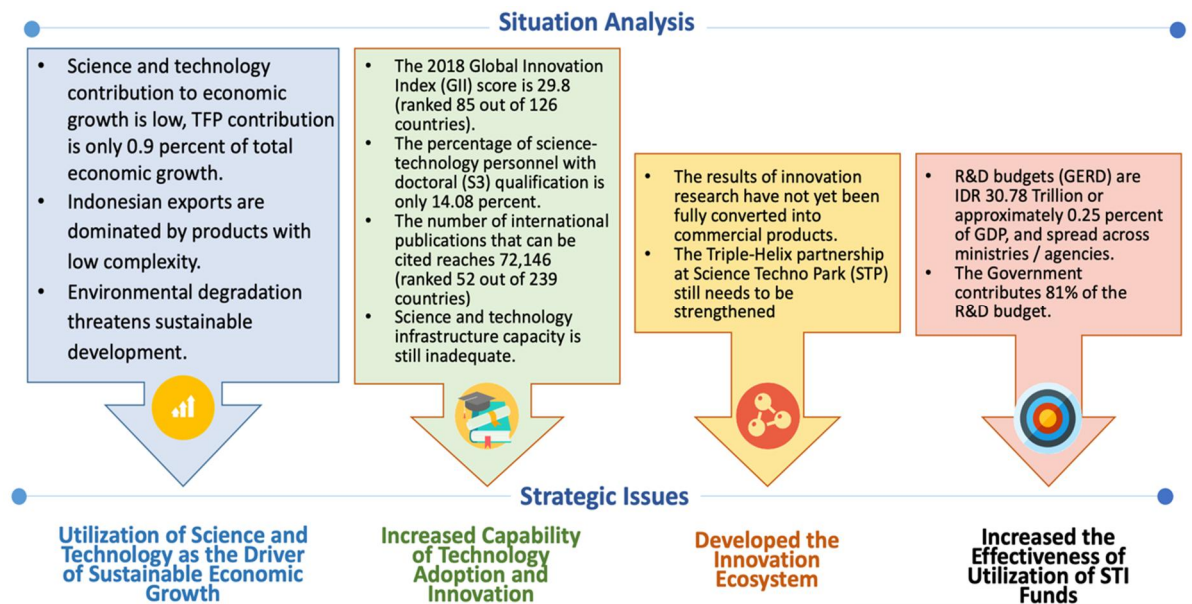
Executive Summary

1. Background

Indonesia has seriously endeavored to strengthen its National Innovation System (NIS). Nonetheless, more concerted efforts have to be made in order to have a better working Indonesian NIS so as to improve Indonesia's global competitiveness. The recent evaluation on the state of science and technology (S&T) development in Indonesia is officially incorporated into the National Mid-Term Planning 2020-2024. The situation analysis concluded that there are four major issues in the area of S&T development, which need to be addressed in Indonesia's NIS:

- How to increase the utilization of science and technology as the driver of sustainable economic growth
- How to increase the capability for technology adoption and innovation
- How to develop the innovation ecosystem
- How to increase the effectiveness of Science, Technology, Innovation (STI) Fund

[Figure 1] Situation Analysis of S&T Policy in Indonesia



Source: Bappenas (2019), p. 5.

Considering the mentioned problems and development objective of the Indonesian STI system, it is very timely to address three important leveraging questions:

- How a think tank can promote STI development, what type of STI think tank to establish, and how to operate it?
- As part of funding mechanism reform, how to manage a research endowment fund, including the institutional arrangement for research endowment fund in the Indonesian context?
- How to plan and manage national research and innovation programs (R&D programs) effectively, what are the structure and mechanisms to do it, including the actors and process of national R&D programs?

2. 2020 K-Innovation with Indonesia

The 2020 K-Innovation ODA Program with Indonesia has three objectives. The first objective is to support Indonesian policymakers and researchers in implementing policy recommendations on the Research and Development (R&D) strategy plan, Science and Technology Park, and evaluation of STI Policy and R&D governance from the 2018-2019 K-Innovation Program. The second objective is to increase capacity building and provide consulting on 'Arrangement of Science, Technology, and Innovation (STI) Governance: Coordination Mechanism; Institutional Framework & Financial System; and Investment in Indonesia'. The last objective is to strengthen STI cooperation between Korea and Indonesia as part of Korea's New Southern Policy.

The 2020 K-Innovation ODA Program consists of four activities: kick-off seminar, joint research to produce a policy consulting report, capacity building, and dissemination workshop (which changed to the form of seminar due to COVID-19). The first phase is to hold a kick-off seminar between the STEPI team and LIPI to discuss the details of activities. At the kick-off seminar, Korean experts give presentations about Korean practices on the themes with the participation of Indonesian experts from the Ministry of Research and Technology to the Ministry of National Development Planning (Bappenas).

Korean experts collaborate with LIPI researchers for joint research on three themes: building a new think tank for STI in Indonesia; R&D funding and Research Endowment Fund; and Management and Evaluation System of the National Research Program. Three teams supervised by Korean experts assign focal points for Indonesian researchers. The capacity-building program for Indonesian researchers in Korea is an important part of K-Innovation. Nonetheless, the COVID-19 pandemic hinders organizing a capacity-building program for Indonesian researchers in Korea. Thus, K-Innovation encourages Korean experts to utilize videoconferencing instead to share Korean practices and exchange ideas on the themes they are responsible for. The last activity of K-Innovation is to hold a dissemination seminar wherein the research teams can present and receive feedback on the outcomes of the policy consulting from the K-Innovation ODA Program from Indonesia's STI community. In addition to the dissemination seminar, Korean and Indonesian researchers work to publish academic

papers based on the outcomes of 2020 K-Innovation.

3. Outcomes of Policy Consulting

The 2020 K-Innovation with Indonesia focused on consulting to address three key issues for Indonesia's National Innovation System (NIS):

- How to establish a *think tank* for the Science-Technology-Innovation (STI) Policy at the national level
- Designing a funding agency responsible for managing a research endowment fund
- How national research programs are managed and evaluated (planning, implementing/organizing, output/outcome measurement, and evaluation)

The common theme of the issues above is institutional building of policymakers and researchers. The Indonesian government needs to design effective organizations and institutions to carry out particular functions: research planning and evaluation. To address the concerns above, the Korean team and Indonesian team conducted joint research with four major activities—such as conceptualization, focused group discussion (FGD), in-depth interview, and workshop—to disseminate the outcomes of research with the relevant stakeholders from Indonesia and Korea.

3.1 Policy Recommendations for Building a New Think Tank for Science and Technology Innovation in Indonesia

To realize the feasible governance of STI policy, the creation of a think tank in Indonesia should consider the existing regulations, norms, and systems. Still, creating a new think tank unit on the STI policy in a broader context will face big challenges or even clash with other regulations or agency interests. Indeed, designing an STI policy think tank unit is not separated from the intent of the top management, or even political will. This study provides four alternative STI policy think tank units that can be planned and designed in Indonesia.

[Table 1] Suggested STI Think Tank Governance in Indonesia

Option	Funding source	Governance	Status of the members	Function
1	Government budget (Ministry of Research and Technology/BRIN)	Government-affiliated	Government officers	Planning, Implementing, Evaluating
2	Budget of non-government organizations	Autonomous & independent body	Private sector	Planning, Implementing, Evaluating
3	Budget of government agencies and donors	Quasi-independent	Private sector	Planning, Implementing, Evaluating
4	Government budget under the Ministry of Finance (LPDP scheme)	Quasi-governmental	Contracted public officers	Planning, Implementing, Evaluating

Source: Developed by author

- The ideal condition cannot be reached due to the current obstacles in Indonesia. Before one of the options can be implemented, there are important issues that must be addressed. First, there is a need to reform the rigidity to enable more flexible bureaucracy. A less flexible system may affect finance, law, and administrative procedures, for instance. Managing funds from both government and other sources requires creating a new form of think tank such as quasi-independent and independent think tanks in Indonesia. Second, a favorable coordination system should be structured among public agencies. In addition, the STI policy think tank needs coordination with a wide range of stakeholders. Third, Indonesia needs to increase the number and capacity of human resources in STI policy. The limited number of researchers performing functions in STI policy studies and lack of data on STI mean lack of capacity of human resources in STI policy. Fourth, evidence-based policymaking on STI should be strengthened. In most cases, policymaking without evidence tends to lead to trial judgment and errors. Fifth, it is necessary to develop an enabling environment for promoting innovation. In Indonesia's case, government bodies should have the same frequency for synergy to create an innovation system. Ministry of Research and

Technology (MoRT)/ National Research and Innovation Agency (BRIN) can be the main actor for constructing NIS among the institutions.

3.2 Policy Recommendations for R&D Funding and Management of Endowment Fund

“The new” Law No. 11 Year 2019 has mandated the integration of all public R&D institutions in Indonesia under the auspices of the Ministry of Research and Technology/BRIN to ensure the sustainability of one-way R&D activities in order to increase R&D capacity in national development. Nonetheless, the meaning of "integration" in this mandate gives rise to some perceptions as to how it should be done. The STEPI-LIPI team produced three major options of governance with regard to the Research Endowment Fund.

[Table 2] Governance Options for the Research Endowment Fund

Governance Option	Option 1	Option 2	Option 3
Investment Manager	Ministry of Finance	Ministry of Finance	Ministry of Research and Technology/BRIN
R&D Program Manager	Ministry of Research and Technology/BRIN	Ministry of Research and Technology/BRIN	
Selection and Review Process		Independent Institution	
Opinion by Author*	-	Most Recommended	Least Recommended

Source: developed by Author.

*This does not reflect or represent the opinion of other parties.

There are two options for evaluating the Research Endowment Fund. As for Option 1, the evaluation process can be carried out by the Ministry of Research and Technology/BRIN. As an actor in charge of managing R&D affairs in the government, the Ministry of Research and Technology/BRIN has the responsibility of ensuring that the various programs and policies implemented can support R&D activities, leading to the

achievement of national development goals. The Ministry of Research and Technology/BRIN has adequate capacity and capability in conducting evaluations, supported by qualified resources. Option 2 means that the Ministry of Research and Technology/BRIN can assign an independent institution to conduct evaluations. This option is based on the view that evaluations carried out by an independent institution other than the policymakers themselves can guarantee a more objective evaluation process. This evaluation becomes an input for the Ministry of Research and Technology/BRIN in making plans for the next program. In this case, Academy of Indonesian Science (API) might be the one relied on to evaluate programs and projects funded by the research endowment fund. This should be a better choice than the previous one.

3.3 Policy Recommendations for the Management and Evaluation System of the National R&D Program

The Indonesian government has a long-and mid-term R&D policy such as National Research Master Plan (RIRN) (2017 ~ 2045), which is the Master Plan on the national R&D program and National Research Priority (PRN) to be revised every five years. This indicates that Indonesia has a structure for the formation, implementation, and evaluation of an R&D program and a strong will to implement an R&D policy. Still, it seems that Indonesia remains at the early stage of implementation of national R&D programs, and ex-ante and ex-post evaluation processes are not clear. In addition to this analysis, the investment target for R&D announced by the Indonesian government is not enough to operate R&D programs. Besides, funding sources to implement R&D programs are not determined. The cooperation mechanism between industry and national research institutes remains weak. Industries are not innovation-oriented due to their lack of R&D capacity. R&D activities in Indonesia are mainly driven by some state-owned companies.

Policy recommendations for Indonesia include: 1) R&D investment should be increased as long as there are clear indications as to the R&D funding sources, and programs should be developed to induce the private sector to make R&D investment; and 2) restructuring the governance for R&D policy and programs is required. By doing so, the institutional framework for ex-ante and ex-post evaluation and implementation of the R&D program and

projects can be effectively performed, and cooperation and coordination among research actors (national research institutes and universities) can be effectively facilitated.

4. Conclusion

Indonesia put its national priority on building STI capacity, aiming to achieve a knowledge-based economy in the mid- and long terms. The 2020 K-Innovation program contributes to Indonesia's STI development by providing policy consulting on the national innovation system of Indonesia, which is one of the key partner countries in Korea's southern policy. Policy directions are expected to be drafted from K-Innovation's consulting led by Korean experts through consultations with Indonesian researchers to improve the STI system in Indonesia. More specifically, Koreans shared insights and policy directions from the Korean and Indonesian practices regarding STI governance, management and evaluation system of the national R&D program, and management of R&D funding, which Indonesia considers in restructuring the framework on such focus areas. STEPI supported the capacity building of LIPI, Indonesia's largest research institute, over the last several years through the K-Innovation program. STI cooperation at the organizational level is believed to contribute to strengthening cooperation at the national level.

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CHAPTER 1

Project Overview

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Chapter 1. Project Overview

1. Introduction

1.1 STEPI's Support for Indonesia on STI Policy

Since 2014, STEPI has been partnering with Indonesia by providing policy consulting on STI policy and governance and supporting capacity building for Indonesian government officials and researchers from LIPI and BPTT who are responsible for policymaking on the STI policy of Indonesia. In the early days, STEPI's K-Innovation program was capacity-building program-oriented for Indonesian researchers and policymakers to enhance their research and STI policymaking capacity. From 2018, however, the policy consulting of the K-Innovation ODA Program includes evaluating the 2015-2019 Research Strategy Planning and establishing the National Research Priority 2020-2024.

[Table 1-1] STEPI's Program and Support for Indonesian Policymakers and Researchers

Year	Target	Activities
2014	STI policy-related government officials and researchers from Asian countries including Indonesia	- Operated the capacity-building program on STI policy
2015	Ministry of Research, Technology and Higher Education (RISTEK), Agency for the Assessment and Application of Technology (BPPT), UNESCO office in Jakarta	- Operated the capacity-building program for Asian researchers including Indonesia, Malaysia, and Sri Lanka, identifying policy consulting themes
	RISTEK and BPPT	- Operated the capacity-building program on Korea's policy for STI human resources, education system, and education based on

Year	Target	Activities
		demand analysis of education, development, and evaluation
2016	BPPT	- Operated the capacity-building program to increase STI capacity for Indonesian researchers and enhance the competency of officials as part of reforming bureaucracy
	BPPT	- Operated the capacity-building program to increase the capacity of developing and managing human resources on planning, design, and education, monitoring and evaluation of the education and training program
2018	LIPI	- Operated the capacity-building program and consulted on the R&D strategy plan, Science and Technology Park
2019	LIPI	- Operated the capacity-building program and consulted on the evaluation of STI Policy and R&D governance

STEPI developed the two-year program, “Policy Consulting on the Evaluation of LIPI R&D Strategic Planning 2015-2019, Development of LIPI R&D Strategic Planning 2020-2025 and Science and Technology Park” through the joint workshop with LIPI in Jakarta in December 2017.

- The 2018 K-Innovation ODA Program provided policy consulting on developing the 2020-2025 Research and Development Strategy, operation and maintenance of technopark and management of STI human resources, and strengthening of research capacity.
- The 2019 K-Innovation ODA Program focused on consulting on STI policy evaluation and R&D governance in Indonesia. The themes were chosen against the backdrop of STI governance emerging as a critical issue in the Indonesian research community with the launch of the new Ministry of Research and Technology and National Research and Innovation Agency (BRIN).

Amid such changes in STI governance, LIPI requested STEPI to share Korean experiences and to provide policy consulting on the management of research endowment fund and national research and development program. The Arrangement of STI Governance: Coordination Mechanism, Institutional Framework, Financial System, and Investment in Indonesia was conceptualized.

1.2 Status of Indonesia's National Innovation System

Science and Technology (S&T) development has become a serious concern for the government of Indonesia since the late 1970s. This fact can be shown by the rapid development of scientific infrastructure in Indonesia. The Science and Technology Park in Serpong, the satellite city of Jakarta, was built in 1978. The establishment of the Agency for Assessment and Application of Technology reflected the government's commitment to technology utilization for development purposes. The establishment of the National Aircraft Industry showed a serious attempt to master high technology. To foster S&T human resources development, the government began in 1984 a program to send abroad a significant number of students pursuing degrees in science and engineering. This was the first organized and sustained program to increase the availability of high-quality S&T human resources.

For almost two decades since the late 1970s, the S&T policy has been focusing mainly on S&T infrastructure development; the link of S&T to the economy was not explicitly stated as the policy objective. The policy orientation started to change when, in 1995, a World Bank Staff Report on the Indonesian science and technology system identified what it called a major disconnect between the research of the publicly funded research and development agencies and the needs of the Indonesian economy. To address such disconnect, the government launched the Industrial Technology Development Project, which aimed at the management strengthening of a number of government research institutes in order to be able to deliver services to industry. Since then, the government had developed and implemented various policies and programs to link government research institutes and universities with industry. This new policy orientation coincided with the widespread use of

the NIS framework in innovation policy.

Despite the widespread use of the NIS framework in innovation policy in many countries during the 1990s, however, the framework was still rarely explicitly discussed and used as a policy framework among Indonesian policy circles until 2008. This was reflected among others in the popular use of the term “National S&T System” and a clear distinction between S&T Policy and Industrial Policy. Similarly, rather than using the term National Innovation System, the S&T law (2002) to promote linkages between Government Research Institutes, Universities, and Industries was named “National System of Research, Development, and Application of Science and Technology.” The name is still centered on Science and Technology.

It was only in 2008, on the occasion of the National S&T Coordinating Meeting held by the Ministry of Research and Technology (MoRT), that the NIS framework was officially adopted in the science and technology community. Yet, the context is S&T policy. S&T policymakers still has to communicate the framework to other policymakers in other areas such as industrial policy and education policy. In 2008, the communication efforts by the S&T community started to be more intensified. Still, the nonexistence of a national innovation policy coordinating agency has made the communication and coordination efforts difficult. In the last few years, the NIS framework has slowly disappeared from the Indonesian STI Policy discourse. In 2019, the new law no. 11, which replaced the 2002 S&T law, barely mentioned the national innovation system, and it is called the law of Science and Technology System.

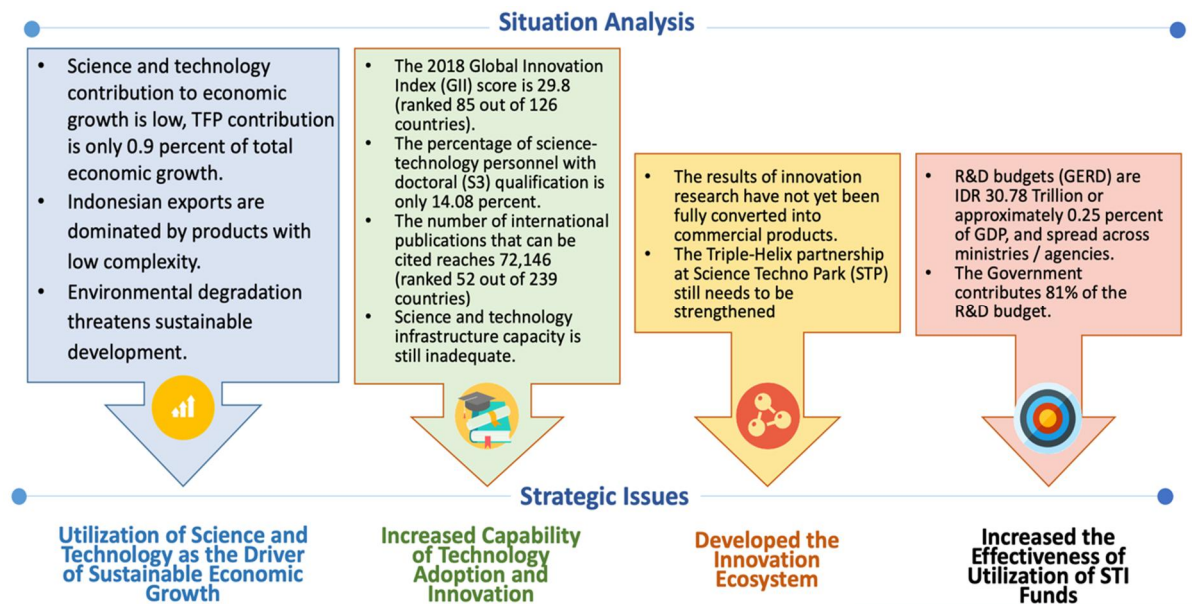
The elements of NIS are available and operational in Indonesia, but so far they are not interconnected properly so it has yet to enhance national economic competitiveness optimally. Incoherence between policies hindered the interconnection and interrelation between the elements of NIS. National policies on R&D, industry, education, and finance are mostly based on sectoral approaches. The problems of interconnection among the elements of NIS of Indonesia are not simple. There are disparities in understanding of and interest in technological innovation. Innovation performance depends not only on how specific actors perform, but also on how they interact with one another as elements of an

innovation system. This suggests that Indonesia needs concerted efforts to strengthen its innovation system. It needs a strong think tank that is able to formulate and promote a strategic approach for innovation system development. Meanwhile, reforms in the government R&D funding mechanism are urgently needed if the government is to improve linkages between government research institutes and industry and to promote innovation system development. The existing funding mechanism is very rigid, acting as a disincentive to researchers and research institutes as they cannot “make a profit” from their research contracts. More autonomy should be given to research institutes to manage their research funding. In particular, the line item budgeting system needs to be replaced by a performance-based budgeting system that gives more autonomy and flexibility to research institutes in managing research funding.

In short, Indonesia has seriously endeavored to strengthen its NIS. Nonetheless, more concerted efforts have to be made in order to have a better working Indonesian NIS so as to improve Indonesia’s global competitiveness. The recent evaluation on the state of S&T development in Indonesia is officially incorporated into the National Mid-Term Planning 2020-2014. The situation analysis concluded that there are four major issues in the area of S&T development, which need to be addressed:

- How to increase the utilization of science and technology as the driver of sustainable economic growth
- How to increase the capability for technology adoption and innovation
- How to develop the innovation ecosystem
- How to increase the effectiveness of STI Fund

[Figure 1-1] Situation Analysis of S&T Policy in Indonesia



Source: BAPPENAS (2019), p. 5.

Considering the mentioned problems and development objective of the Indonesian STI system, it is very timely to address three important leveraging issues:

- How a think tank can promote STI development, what kind of STI think tank should be established, and how to operate it?
- As part of funding mechanism reform, how a research trust fund should be managed, and what is the best institutional arrangement in the Indonesian context?
- How the national research and innovation program should be effectively planned and managed, what are the structure and mechanisms to do it, who are the actors, and what is the business process?

2. Objectives

The 2020 K-Innovation ODA Program with Indonesia has three objectives. The first objective is to support Indonesian policymakers and researchers in implementing policy recommendations on the R&D strategy plan, Science and Technology Park, and evaluation of STI Policy and R&D governance from the 2018-2019 K-Innovation Program. The second objective is to increase capacity building and provide consulting on 'Arrangement of Science, Technology, and Innovation (STI) Governance: Coordination Mechanism; Institutional Framework & Financial System; and Investment in Indonesia'. The last objective is to strengthen STI cooperation between Korea and Indonesia as part of Korea's New Southern Policy.

3. Project Framework

The 2020 K-Innovation ODA Program consists of four activities: kick-off seminar, joint research to produce a policy consulting report, capacity building, and dissemination workshop (which changed to the form of seminar due to COVID-19). The first phase is to hold a kick-off seminar between the STEPI team and LIPI to discuss the details of activities. At the kick-off seminar, Korean experts give presentations about Korean practices on the themes with the participation of Indonesian experts from the Ministry of Research and Technology to the Ministry of National Development Planning (Bappenas).

Korean experts collaborate with LIPI research for joint research on three themes: building a new think tank for STI in Indonesia, R&D funding and Research Endowment Fund, and Management and Evaluation System of the National Research Program. Three teams supervised by Korean experts assign focal points for Indonesian researchers. The capacity-building program for Indonesian researchers in Korea is an important part of K-Innovation. Nonetheless, the COVID-19 pandemic hinders organizing a capacity-building program for Indonesian researchers in Korea. Thus, K-Innovation encourages Korean experts to utilize videoconferencing instead to share Korean practices and exchange ideas on the themes they are responsible for. The last activity of K-Innovation is to hold a dissemination seminar wherein the research teams can present and receive feedback on the outcomes of the policy consulting from the K-Innovation Program from Indonesia's STI community. In addition to the dissemination seminar, Korean and Indonesian researchers work to publish academic papers based on the outcomes of the 2020 K-Innovation.

4. Project Team

More than 10 researchers from Korea and Indonesia participated in the implementation of the 2020 K-Innovation Project. Supervised by Dr. Deok Soon Yim, the Korean team invited three Korean experts for policy consulting on Coordination Mechanism, Institutional Framework & Financial System, and Investment in Indonesia. Among them, two experts, Prof. Seongsoo Kim and Dr. Youngjoo Ko, have participated in the 2018-2019 K-Innovation Project with Indonesia. Their participation as advisers for the 2020 K-Innovation project enabled facilitating the collaboration from LIPI as they have good understanding of the STI governance and policy in Indonesia. Prof. Kiseok Kwon, who possesses a great deal of expertise in STI policy, also participated in supervising the research on STI think tank. Elly Hyanghee Lee acted as project coordinator of the 2020 K-Innovation Project with Indonesia and conducted the research on the management and evaluation system of the national research program.

Korean advisers headed the research teams, each of which assigned a focal point for better communication among researchers. Dr. Dudi Hidayat, the Indonesian counterpart of the 2020 K-Innovation, also headed the Indonesian team on R&D funding and management of the research endowment fund. Dr. Galuh Syahbana Indraprahasta served as focal point for the research on the National Research Program. For the research on the STI think tank, Anugerah Yuka Asmara acted as focal point with the supervision of Korean team leaders. There are 3-5 core Indonesian researchers involved in this project from the Indonesian Institute of Sciences (LIPI) from various backgrounds (public administration, economy, engineering, and so on.) Still, the involvement of a number of supporting researchers and assistant researchers (estimated to be 8-10 Indonesian researchers/assistants) is a key factor for accelerating the project.

[Table 1-2] Project Team of the 2020 K-Innovation with Indonesia

Name	Position/Organization	Roles
Dr. Deok Soon Yim	Senior Research Fellow, STEPI	Project manager for the 2020 K-Innovation
Ms. Elly Hyanghee Lee	Researcher, STEPI	Project coordinator for the 2020 K-Innovation
Dr. Dudi Hidayat	Director, Research Center for STI Policy and Management (RCSTIPM), LIPI	Focal point for the 2020 K-Innovation
Team 1. Think Tank for Science-Technology-Innovation (STI) Policy		
Prof. Kiseok Kwon	Professor, Hanbat National University	Team leader
Anugerah Yuka Asmara	Researcher, RCSTIPM, LIPI	Focal point for Indonesian team 1
Budi Triyono	Researcher, RCSTIPM, LIPI	Participant
Maulana Akbar	Researcher, RCSTIPM, LIPI	Participant
Team 2. R&D Funding and Research Endowment Fund		
Prof. Seongsoo Kim	Professor, Hankook University of Foreign Studies	Professor, Hankook University of Foreign Studies
Dr. Dudi Hidayat	Director, Research Center for STI Policy and Management (RCSTIPM), LIPI	Focal point for Indonesian team 2
Aditya Wisnu Pradana	Researcher, RCSTIPM, LIPI	Participant
Anggini Dinaseviani	Researcher, RCSTIPM, LIPI	Participant
Indah Purwaningsih	Researcher, RCSTIPM, LIPI	Participant
Team 3. Management and Evaluation System of the National Research and Development Program		
Dr. Youngjoo Ko	President, Daejeon Institute of Science and Technology for Enterprise and People (DISTEP)	Team leader
Ms. Elly Hyanghee Lee	Researcher, STEPI	Participant
Dr. Youngsun Jang	Researcher, Korea Research Institute of Chemical Technology	Participant
Dr. Galuh Syahbana Indraprahasta	Senior Researcher	Focal point for Indonesian team 3
Karlina Sari	Researcher, LIPI	Participant
Ria Hardiyati	Researcher, LIPI	Participant
Luthfina Ariyani	Researcher, LIPI	Participant

5. Project Activities

5.1 Activity 1: Project Preparation Meetings

From early January 2020, the Korean team has been communicating with LIPI to elaborate project activities. In addition to detailing the 2020 K-Innovation, the composition of research teams for joint research and joint research were also determined for mutual understanding between STEPI and LIPI. LIPI played an active role in planning joint research for policy consulting on the Coordination Mechanism, Institutional Framework & Financial System, and Investment in Indonesia by providing the background of the joint research.

5.2 Activity 2: Kick-off Seminar, Field Study & Special Lecture for the Minister of Research and Technology

The Korean and Indonesian teams had a kick-off seminar and conducted a field study to diagnose the status of the three themes in Jakarta in February 2020. This was the first phase of the project activities of the 2020 K-Innovation. During the visit to Indonesia, STEPI had several meetings with the relevant stakeholders from the Korean Embassy in Indonesia to Mr. Bamnag Brodjonegoro, who is serving as the Minister of Research and Technology of Indonesia.

[Table 1-3] Kick-off Seminar Program of the 2020 K-Innovation with Indonesia

09.00 – 10.00	Registration	
10.00 – 10.05	Opening	MC
10.05 – 10.10	Welcome Remarks	Mrs. Wati Hermawati P2KMI LIPI
10.10 - 10.15	Opening Remarks	Dr. Mego Pinandito, Deputy Chairman, LIPI
10.15 – 10.20	Group Photos	
10.20 - 12.00	Panel Session Moderated by Mrs. Karlina	

2020 K-Innovation ODA Program with Indonesia

10.20 - 10.30	Work Plan of the 2020-2022 K-Innovation ODA Program with Indonesia & Major Outcomes of the 2018-2019 K-Innovation ODA Program	Dr. Deok Soon Yim, Senior Research Fellow, STEPI
10:30 - 11:00	Korea's S&T Development and Policy Trends	Dr. Deok Soon Yim, Senior Research Fellow, STEPI
11:00 - 11:20	Status and Challenges of Indonesia's STI Governance	Dr. Ikbal Maulana P2MB, LIPI
11:20 - 11:40	Status and Challenges of the Institutional Framework of Indonesia's STI Governance	Mr. Sjaeful Ministry of Research and Technology
11:40 - 12:00	Open Discussion: Q & A	Moderator
12:00 - 13:30	Lunch	
13:30 - 17:00	Group Discussion on the Activities of the 2020 Program - Contextualizing the concept of "Technological Catch-up" into the case of Indonesia, particularly with regard to Indonesia's effort to get out of the middle-income economy trap - Review on STI Governance (Coordination mechanism; institutional framework & financial system and investment) in Indonesia and design of effective governance - Arrangement and development of STI Governance (coordination mechanism; institutional framework & financial system and investment) in Indonesia - Series of Focus Group Discussion in the making of policy recommendation of STI Governance (coordination mechanism; institutional framework & financial system and investment) in Indonesia - Capacity building for researchers and policymakers	Dr. Deok Soon Yim, Senior Research Fellow, STEPI, Dr. Youngjoo Ko, Principal Researcher, Korea Research Institute of Chemical Technology, & Prof. Seongsoo Kim, Dean, Hankuk University of Foreign Studies & LIPI researchers



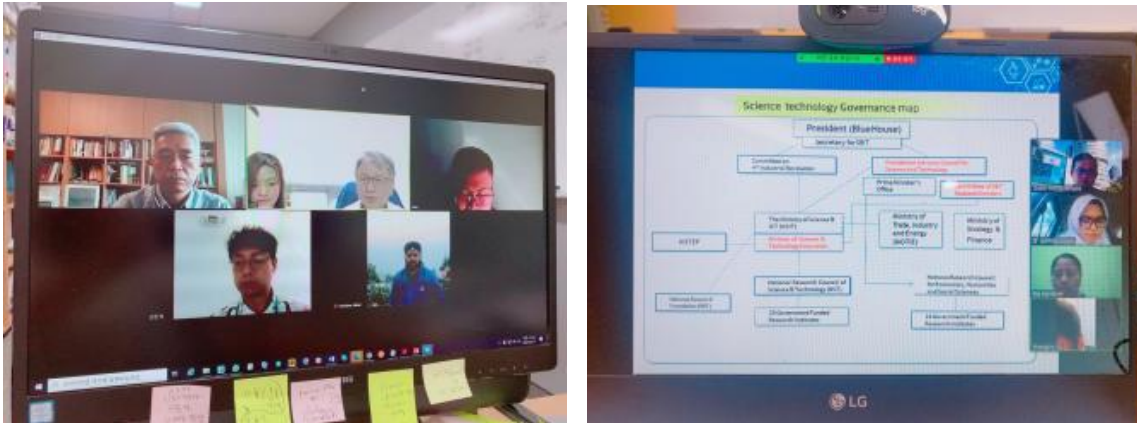
Invited by the Korean Embassy, Dr. Deok Soon Yim, project manager of K-Innovation with Indonesia, gave a special lecture to the Minister and heads of research institutes of Indonesia on the Trends of Korea's Science and Technology Development and Policy. Dr. Yim's presentation covered Korea's S&T development and S&T policy, National Research Council and so on. He also shared insights and addressed questions on how to promote industrial technological competitiveness, how to identify and consolidate the national research resources, and how to consolidate actors and networking in the national strategic research project.





5.3 Activity 3: Joint Research to Produce a Policy Consulting Report

Three teams were organized to produce a policy consulting report covering three themes: building a new think tank for STI in Indonesia; R&D funding and Research Endowment Fund; and Management and Evaluation System of the National Research Program. Korean experts led the teams, and LIPI assigned focal points for each team to communicate with the Indonesian researchers. Each team had online meetings to conduct joint research biweekly, and STEPI researchers organized a monthly meeting with all research teams.



5.4 Activity 4: Special Session on Indonesia's NIS to Disseminate the Project Outcomes

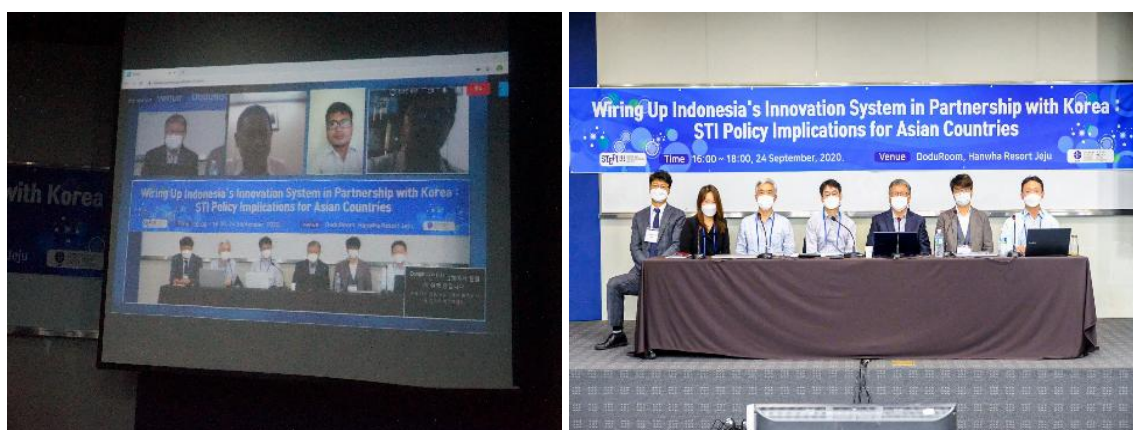
Sharing the outcomes of the policy consulting with the Asian STI community including Indonesia R&D policymakers and researchers is one of the key activities in the K-Innovation project. As part of advocacy activities, STEPI organized a special session on Indonesia's innovation system during the international conference of the Asian Journal of Innovation and Policy held in Jeju on 24 September 2020. The session provided an opportunity for Korean and Indonesian teams to get feedback on the outcomes of joint research from STI policy experts. Indonesian experts' recorded presentations were aired on YouTube during the session. Korean advisers and panels participated in the conference for the discussion and live Q&A time. The special session was successful, with more than 50 Indonesian researchers from LIPI joining the session via YouTube.

[Table 1-4] Program of STEPI Special Session

16:00 - 16:05	Congratulatory remarks	Dr. Dudi Hidayat, LIPI
16:05 - 16:10	Opening remarks and Introduction	Dr. Deok Soon Yim, STEPI
16:10 – 16:30	Presentation 1: Identifying think tank units for the science-technology-innovation policy in Indonesia: functions and potential to promote the national innovation system	Mr. Yuka Anugrah Asmara, LIPI Prof. Kiseok Kwon, Hanbat National University

2020 K-Innovation ODA Program with Indonesia

16:30 – 17:00	Presentation 2: Designing the management of Research endowment fund and the National Research Agency in Indonesia	Dr. Dudi Hidayat, LIPI, Prof. Seongsoo Kim, Hankuk University of Foreign Studies
17:00 -17:30	Presentation 3: Evaluation system of national R&D programs: learning from Korea and Indonesia	Dr. Galuh Indraprahasta, LIPI
17:30 – 18:00	Q & A and Panel discussion	Moderator: Dr. Deok Soon Yim, STEPI Panelists: Prof. Jiyoung Park, SUNY Buffalo/ Seoul National University, Prof. Jonghwan Jeon, Gyeongsang National University





[Table 1-5] Indonesian Participants of the STEPI Special Session

No.	Name	Title	Institution
1	Aditya Wisnu Pradana	Researchers	LIPI
2	Agus Fanar Syukri, PhD	Mr.	LIPI
3	ANGGINI DINASEVIANI	Junior researcher	P2KMI LIPI
4	Anugerah Yuka Asmara	Mr.	LIPI
5	Budi Triyono	Mr.	LIPI
6	Dadang Ramdhan	PhD	LIPI
7	Dudi Hidayat	Head of PPKMI LIPI	P2KMI LIPI
8	Galuh Syahbana Indraprahasta	PhD	LIPI
9	Husein Akil	Mr.	LIPI

No.	Name	Title	Institution
10	I Gede Mahatma Yuda Bakti	SE, MSM	LIPI
11	Indah Purwaningsih	Researcher	Indonesian Institute of Sciences (LIPI)
12	Indri Juwita Asmara	MTI	P2KMI LIPI
13	Ishelina Rosaira P.	SE	LIPI
14	Karlina Sari	Ms.	LIPI
15	Leni Purwaningsih	Subdivision Head of Data Processing	Ministry of Research and Technology
16	Luthfina Ariyani	Mrs.	Lembaga Ilmu Pengetahuan Indonesia
17	Manaek Simamora	Specialist Business Engineer	Research Center for Policy and Management of Science and Technology and Innovation-LIPI
18	Maulana Akbar	Mr.	P2KMI
19	Mesnan Silalahi	Dr.	LIPI
20	NURTRIA RENDI RAHMADI	Mr.	Centre for the Development of Vocational Training in Serang, Banten, Indonesia
21	Rachmini Saparita	Prof.	PPKMI LIPI
22	Ria Hardiyati	Assistant Researcher	P2KMI LIPI
23	Rislima Sitompul	Dr.	P2KMI - LIPI
24	Sigit Setiawan	Mr.	RC for STIPM - Indonesian Institute of Sciences
25	Syafrizal Maludin	Dr	Indonesian Institute of Sciences
26	Adityo Wicaksono	Mr.	LIPI
27	Andi Budiansyah	.	P2KMI LIPI
28	Arief Ameir Rahman Setiawan	Researcher	LIPI
29	Augy Syahailatua	Dr.	Research Centre for Oceanography - LIPI
30	Budi Nugroho	Dr. Eng.	P2I LIPI
31	Carolina	Ms.	Pusat Penelitian Kebijakan dan Manajemen IPTEK dan Inovasi LIPI
32	Darmawan Alisaputra	Mr.	UIN Sunan Kalijaga

No.	Name	Title	Institution
33	Edi Iswanto Wiloso	Dr.	PPKMI-LIPI
34	Eki Karsani Apriliyadi	Researcher	Pusat Penelitian Kebijakan dan Manajemen Iptek dan Inovasi -Lembaga Ilmu Pengetahuan Indonesia (PPKMI-LIPI)
35	Febtri Wijayanti	M.Ec.Dev	P2KMI LIPI
36	Nany Grace Simamora	M.Hum	LIPI
37	Nidya Judhi Astrini	Mrs.	LIPI
38	Ona Martha Nurron	Mrs.	Lembaga Ilmu Pengetahuan Indonesia
39	Rahmi Kartika Jati	Mrs.	P2KMI LIPI
40	Rendi Febrianda	Dr.	P2KMI LIPI
41	Ria Jayanthi	MSi	P2KMI LIPI
42	Savitri Dyah	Dr.	PPKMI-LIPI
43	Sofia Yuniar Sani	Ms.	BKHH LIPI
44	Sri Hartinah	Mrs.	P2KMI - LIPI
46	Subkhi Abdul Aziz	Researcher	P2KMI LIPI
47	Tri Hadi Jatmiko	MT	BPTBA
48	Tri Handayani	Mrs.	LIPI
49	Wati Hermawati	Researcher	P2KMI LIPI
50	Winda Anestya Ayunda	M.P.	P2KMI LIPI
51	Zainal Arifin	Prof. (research)	Research Centre for Oceanography - LIPI

6. Project Schedule

The 2020 K-Innovation with Indonesia was implemented according to the timeline below. The project schedule was adjusted over the project cycle according to internal factors such as delay of the contract for joint research in Indonesia and external factor, i.e., COVID-19 pandemic.

[Table 1-6] Project Schedule

Activities/ Month	1	2	3	4	5	6	7	8	9	10	11	12
Consultations on project activities and so on												
Literature review												
Kick-off seminar and field-study (in Indonesia)												
Joint capacity-building workshop with Indonesia in Korea	(canceled due to COVID-19)											
Joint research for policy consulting report												
Mid-term report												
Dissemination seminar (virtually)												
Academic paper												
Final report												

7. Project Outputs

Actionable policy recommendations of STI Governance in Indonesia are the key outcomes of the 2020 K-Innovation in the areas of R&D coordination mechanism, institutional framework & financial system, and investment in Indonesia. Throughout the project cycle, the project is expected to have the following deliverables:

- Deliverable 1: Presentation materials from the kick-off seminar
- Deliverable 2: Concept paper on joint research on the three themes
- Deliverable 3: Presentation materials for the dissemination seminar
- Deliverable 4: Policy consulting report of the 2020 K-Innovation (in English and Korean)
- Deliverable 5: Academic papers on the three themes





CHAPTER 2

Overview of Policy Consulting on the Arrangement of Science, Technology and Innovation (STI) Governance: Coordination Mechanism, Institutional Framework & Financial System and Investment in Indonesia

Dr. Deok Soon Yim
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Elly Hyanghee Lee
Researcher, STEPI

Dr. Dudi Hydiyat
Director, LIPI

Chapter 2. Overview of Policy Consulting on the Arrangement of STI Governance: Coordination Mechanism, Institutional Framework & Financial System and Investment in Indonesia

1. Overview of Policy Consulting

1.1 Conceptualization

The 2020 K-Innovation with Indonesia focused on consulting to address three key issues for Indonesia's National Innovation System:

- How to establish a *think tank* for the science-technology-innovation policy at the national level
- Designing a funding agency responsible for managing a research endowment fund
- How national research programs are managed and evaluated (planning, implementing /organizing, output/outcome measurement, and evaluation)

The common theme of the issues above is institutional building of policymakers and researchers. The Indonesian government needs to design effective organizations and institutions to carry out particular functions, i.e., research planning and evaluation. To address the concerns above, the Korean team and Indonesian team conducted joint research with four major activities such as conceptualization, focused group discussion (FGD), in-depth interview, and workshop to disseminate the outcomes of research to the relevant stakeholders from Indonesia and Korea.

The STEPI team did the conceptualization work based on Korean and/or other countries' experiences with regard to:

- How to establish an effective national STI think tank and to suggest an effective one in the Indonesian context;
- Design of an organization along with the necessary institutions and regulations to manage a research endowment fund.

The STEPI team prepared a concept paper for joint research to address the three topics, taking into account the Indonesian context. During the process, the Indonesian team also provided assistance in understanding the Indonesian context.

1.2 Methodology

1.2.1 Focused Group Discussion (FGD)

After the desk review, the LIPI research team plans to hold Focused Group Discussions (FGD) virtually due to the COVID-19 situation. The purpose of the FGD is to capture various perspectives, opinions, and ideas on the three themes from key stakeholders. Key participants for FGD can be categorized based on the type of participants and their institutions. The key participants listed in the table below were only candidates, not fixed actors, subject to change based on the conditions. The composition of FGD is not restricted to the actors listed in table below but can be extended to other actors after tracing previous informants.

[Table 2-1] Categorization of Participants for FGDs

Type of participants	Description	Affiliations/Institutions
National policy experts	Those who have capability and/or authority in influencing national policies	- Indonesian Presidential Office - National Research Council - National Economy Board
Key or Experienced researchers	Those who are interested in a specialized field of research and involved in STI activities	- Indonesian Institute of Sciences (LIPI) - Agency for Assessment and Application of Technology (BPPT) - National Nuclear Agency (BATAN) - National Space Agency (LAPAN) - Information Geospatial Board (BIG) - National Standardization Agency (BSN) - Indonesian Researchers Union (Himpenindo)
Decision makers/stakeholders	Those who have full authority and resources in decision making or policymaking at the national or regional level	- Particular directors or general directors at the Ministry of National Planning and Development - Particular directors or general directors at the Ministry of Research and Technology - Particular directors or general directors at the Ministry of Coordinating Economy Affairs - Particular directors or general directors at the Ministry of Finance - Particular directors or general directors at the Ministry of Industry - Particular directors at local governments (provincial/municipality/regency)
Politicians	Those who are fully involved in political practices	- Members of house of public representatives (DPR)
Public officials	Those who are in street-level bureaucrats or front-line staff who execute policies	- Staff or division chiefs at the Ministry of National Planning and Development

Type of participants	Description	Affiliations/Institutions
	from the top level at a ministry	- Staff or division chiefs at the Ministry of Research and Technology - Staff or division chiefs at the Ministry of Coordinating Economy Affairs - Staff or division chiefs at the Ministry of Finance - Staff or division chiefs at the Ministry of Industry - Staff or division chiefs at local governments (provincial/municipality/regency)
Shareholders/Business actors	Those who have interest in profit-seeking rather than achieving social goals	- Manufacturing industries – private and state –owned enterprises - Start-up enterprises - Small ~ medium enterprises - National business associations - National trade and industry association (KADIN)
Funding-related actors	Those who have the capability, authority, and resources to gain/collect funding sources and to spend/allocate funds	- State-owned banks - Private banks - State-owned financial services unit
Academe	Those who are capable and expert in the academe and who contribute to the practical field	- Lecturers or association on public policy, public administration, engineering, economy, law, etc.
Non-Government Organizations (NGOs)	Those who are frequently interested and involved in many discourses in science-technology-innovation issues, notably located inside the Indonesian geography	- Knowledge Sector Initiative (KSI) - Smeru Institute - Center for Innovation, Policy, and Governance (CIPG) - Other NGOs

1.2.2 In-depth Interview

In-depth interview is aimed at gaining deeper and holistic results based on the view and perspective of several key informants like national policy experts, key or experienced researchers in specialized fields, decision makers/stakeholders, politicians, public officials, shareholders/business actors, funding-related actors, academicians dealing with national policy, economy, engineering, and non-government organizations (NGOs). Each actor will be involved in several discussions during both formal and informal meetings or in one-on-one meetings or two-three-by-one meetings. Each of the key informants is expected to share his/her ideas/opinion to enrich the plan/design of the three themes.

1.2.3 Dissemination Seminar

The STEPI and LIPI research teams planned to hold National Workshops or conferences to communicate the results of the research in order to reach as many stakeholders as possible. Dissemination workshops/conferences were aimed at creating new and extended networks that are useful in developing new designs of STI policy and institutional structure with regard to the three themes.

Due to the COVID-19 situation, however, the dissemination workshop took on the form of a virtual seminar. The special seminar was held with the virtual participation of Indonesian stakeholders during the International Conference of Asian Society for Innovation and Policy. Indonesian researchers who are actively engaged in the joint research presented their paper addressing the three issues to get contributive feedback from the audience via YouTube.

1.3 Timeline

The timeline was arranged starting from conceptualization in April to November 2020 and was adjusted depending on the conditions and needs of both parties based on mutual agreement. From April to September 2020, researchers collected qualitative data through FGDs and in-depth interviews as key methodologies to understand the status of the three

topics because of the lack of reliable quantitative sources about the STI policy in Indonesia. Researchers prepared the policy consulting report covering the possible design of a think tank for STI policy, research endowment, and management scheme of the national research and development program.

[Table 2-2] Timeline for Joint Research

Month/ Activity	Apr.	May	Jun.	Jul.	Aug.	Sep.	Oct.	Nov.
Conceptualization								
In-depth Interview								
FGD								
Mid-term Consulting Report								
Dissemination Seminar								
Final Consulting Report								
Academic Paper								



CHAPTER 3

Building a New Think-tank for Science, Technology and Innovation in Indonesia

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Professor, Hanbat National University

Anugerah Yuka Asmara

Researcher, LIPI

Budi Triyono

Researcher, LIPI

Maulana Akbar

Researcher, LIPI

Chapter 3. Building a New Think-tank for Science, Technology and Innovation in Indonesia

1. Introduction

Science, Technology, and Innovation (STI) is a relatively new term used to accelerate industrial growth, especially in the 1970s (Pack, 2000). Proceeding in the early 1980s-1990s, STI has contributed much to leveraging economic development both in developed and developing countries (see Dodgson, 2000; Lee, 2000; Lall, 2000; Mani, 2002). STI is widely acknowledged to be one of the key factors in realizing the economic growth of a country—especially from a developing country to a developed one like South Korea, Hong Kong, Singapore, and Taiwan, which are called Newly Industrialized Economies (NIEs) (Pack, 2000, Dodgson, 2000)—as well as China (mainland), which has emerged as a new global player in the innovation-based economy (Yip & Mckern, 2016).

Implementing STI is not a special domain of academics or business. Basically, STI is one of public affairs that needs government intervention. Many studies like Dodgson (2000), Lee (2000), Mani (2002), and Kuhlmann, et al (2010) revealed that government policy functions to boost STI practices, so it is generally called STI Policy or innovation policy¹ as one of the

¹ In this study, Science-Technology-Innovation (STI) policy and innovation policy are interchangeable terms. Referring to Dodgson (2000) by which innovation policy consists of a broad definition of science policy and technology policy

key factors for those countries classified as NIEs and developed countries. In fact, STI policy has contributed significantly in mobilizing existing resources (manpower, fund, facilities, etc.) to accelerate the national economy based on the research and engineering results in a country.

STI policy is one of the key instruments in creating innovative environments at the national level. It is aimed at boosting the creation of a national innovation system (NIS) leading to national competitiveness (Senker, 1996; Lundvall, 2005). According to Watkins, et al (2015: 1,408), NIS is defined as “a system wherein governments and collective activities can and do play a central orchestrating role in the generation and dissemination of innovation in a national economy.” This statement clearly reveals that the role of the government in establishing NIS in a country through STI policy lies not in technical affairs but in coordinating actors and mobilizing all existing resources focusing on achieving the economic goals in a country.

In an NIS framework, although the STI policy is mainly implemented by the government, other actors such as academicians, businesses, non-government organizations (NGOs), politicians, and local communities are involved in its policy formulation process. In other words, there is no single actor in determining the content and direction of STI policy in a country (Dodson, 2000; Chaminade & Edquist, 2010; Sun & Chao, 2018). Technology creation and diffusion in an NIS are not separated from how the government interacts with political actors through the establishment of institutional capacity building and governance (Watkins, et al, 2015). It means that the final stage of executing the STI policy in a country depends on the political process, not on researchers or academicians (Chaminade & Edquist, 2010).

STI policy is a part of public policy characterized by a collective decision of many actors with special or private interests in realizing a policy. In public policy literature, making a public policy must involve various and related actors with their interests (Andersson, 2011; Birkland, 2015). Similarly, formulating an STI policy must involve actors like academicians, business/industries NGOs, politicians, local communities, and definitely government agencies. Each actor involved has special interest (vested interests), and it is only logical for

them to want to join in the STI policymaking process. Therefore, at each STI policymaking process, the government must consider and accommodate the interests of the actors involved.

Orchestrating actors in realizing NIS in a country is the main duty of the government (Watkins, et al, 2015), but it is not a modest work in practice. Orchestrating actors is an effort to accommodate the interests of each actor involved in formulating an innovation policy. Many issues of various actors are inputted to the innovation policymaking process, not only science-technology interests but also business and political interests are importantly considered in the agenda setting process. According to Andersson (2011), agenda setting is a critical stage in the public policymaking process. In this stage, all issues are considered in selecting the main issue that is in turn delivered to the policy formulation process.

In the agenda setting stage, every interest group can play a role in influencing the content of the policy. They must compete with other pressure groups to influence the government and other groups to follow their ideas (Andersson, 2011). In the case of STI policy, for instance, universities and research and development (R&D) institutes are the most profitable actors benefited by this policy. On the other hand, industry is a pressure group influencing the government through lobby and negotiation in order to shape regulation and institutions supporting industries' interests (Watkins, et al, 2015). Government agencies are actors that are fully concerned with advancing the national economy and increasing the prosperity of citizens.

Aligning the same interests of actors is not a linear, simple work in the innovation policymaking process. Though political interest is essential in this stage, support of a robust justification is needed. Such justification is based on empirical evidence to assure the government and other interested actors involved in the STI policymaking process. Therefore, each country needs a special unit functioning to provide policy input or recommendation based on empirical evidence. This is called the STI policy think tank. According to Dakowska (2016), think tanks are "laboratories of ideas." In this case, the STI policy think tank is related to laboratories of ideas on STI, which will be explored in detail in the next section.

One of best practices in the design of an STI policy think tank was established by the government of South Korea. The studies of Lee (2000), Kim (2000), Lim (2008), Choung, et al (2013), Lee & Kim (2016), and Yoon (2018) show that Korea's STI policy has contributed significantly to national development, especially in catching up with other countries and boosting its national economy today. Historically, the Republic of South Korea is one of NIEs that has successfully realized economic growth from a low-income, underdeveloped country to a mid-income, industrialized country for thirty years since 1960s by paying attention to STI matters. In 1997, South Korea joined the Organization for Economic Cooperation and Development (OECD) (Lee, 2000; Lim, 2008). The country set up the institutions and organizations for systematic STI policy design and implementation that contribute to STI policymaking, such as the Korean Institute of Science and Technology (KAIST) (Yoon, 2018), government-funded research institutes (GRIs) that contribute considerably to advanced industrial R&D in Korea (Kim, 2000), and Science and Technology Policy Institute (STEPI, 2020).

STEPI was established in January 1987 (initially called the Centre for Science and Technology Policy/CSTP), an STI policy think tank that successfully delivered evidence-based policy in STI areas. STEPI has contributed a lot to South Korea's NIS development. Based on the NIS approach, the South Korean case can serve as a general model for a catch-up NIS. In order to build this model into an Indonesian NIS effectively, some contextual points need to be considered. When building the agencies in the Indonesian NIS, all other factors in the system (e.g., governments, industry, other agencies, and all other institutional settings) will be related to the new agencies. Based on this idea, the establishment of a think tank can be further developed coherently. FGI and interviews need to be carried out considering this concept and the elements below.

The rest section of the research is organized as follows: Part 2 analyzes the status of STI think-tank in Indonesia; Part 3 discusses Korea's best practice case with regard to STI think tanks; Part 4 address the need for a new think tank on STI in Indonesia; Part 5 is about planning a new STI think tank in Indonesia; and Part 6 provides policy recommendations for new STI think tanks in Indonesia.

2. Status of STI Think-tank in Indonesia

Indonesia has started building an NIS environment at the end of the 1970s when Mr. Habibie was the Minister of Science and Technology (Menristek). During his term, STI-based industries were built, such as PT INKA Corp. (train manufacturer), PT Krakatau Steel Corp. (steel manufacturer), PT Barata Corp. (Engineering, Procurement & Construction, Manufacturing, & Foundry), PT Boma Bisma Indra Corp. (Engineering, Procurement & Construction, Manufacturing, & Foundry), PT Telkom Corp. (telecommunication), PT IPTN Corp. (aircraft manufacturer), PT PAL Corp. (shipbuilding and yard), PT Pindad Corp. (weapon and military devices manufacturer), PT LEN Corp. (electronics company), and PT Dahan Corp. (weapon manufacture); these were classified as strategic high-technology industries (BUMNIS)/anchor industries (Supardi, 2013). During the period of 1965 - early 1990s, Indonesia was called “Asian Tiger” in parallel with the Philippines, Malaysia, and Thailand among Southeast Asian Countries (ASEAN) (Kozlova & Noguera-Santaella, 2017).

Due to the economic crisis in Indonesia in 1997-1998, almost all economic sectors collapsed², including BUMNIS. This period was characterized by political upheaval and change of president, with the government regime dramatically changing from an authoritarian regime led by President Soeharto to a reformation regime led by President Habibie. Unfortunately, the leadership of Mr. Habibie lasted only from mid-1998 to mid-1999 (Habibie, 2006). During this short period, STI policy to support NIS was less neglected, and the main priority was to maintain stability in national politics and economy.

Attention to NIS creation in Indonesia was renewed after the 2000s when Mrs. Megawati was elected president of the Republic of Indonesia (2002-2004), followed by subsequent presidents such as Mr. Susilo Bambang Yudhoyono (2004-2014) and Mr. Joko Widodo (2014-up to recent). During this period, STI policies were formulated by the Ministry of

² Small-medium enterprises (SMEs) characterized by low capital, low-capability human resources, and modest management are sustaining their operations in the national economy crisis in 1997-1998, but those large-scale enterprises characterized by high foreign and domestic investment, skilled labor, and complex management collapsed, including BUMNIS

Research and Technology (MoRT), such as national research agenda (ARN), national research incentives (Insinas), and national main research planning (RIRN). Besides, the Ministry of National Planning (Bappenas) formulated the mid- to long-term national STI programs plans. All those policies should be adhered to by all institutions regarding STI policy implementation like universities, R&D institutions, or STI-related NGOs at the national and regional levels.

Almost all ministries in Indonesia have an R&D unit to support the main duty and to produce evidence-based policy for related stakeholders, especially to their ministers. Practically all R&D activities conducted in those R&D units are directed by authorized internal management (manager or director level) according to the ministry's interests and goals, not by STI programs formulated by MoRT. For instance, the Ministry of Agriculture has the Research Centre for Social and Economy on Agriculture. This research center strictly follows the instruction and guidance of the Ministry of Agriculture, not Kemenristek since there is no special guidance for agricultural social-economy research from Kemenristek. This case is common among other R&D units of the ministries in Indonesia. In fact, this research center rarely cooperates with similar R&D centers related to agriculture and social-economy such as in universities, LIPI, Agency for the Assessment and Application of Technology (BPPT), or other private R&D units.

Small public budgets supporting R&D³ (Ariana, et al, 2014) are classic problems faced by the government, especially Ministry of Research and Technology and Bappenas, in directing nationally integrated STI policy in Indonesia. As another issue, it is widely acknowledged that innovation is the domain of not only academicians, researchers, or public agencies dealing with STI but also other actors like industries, association or NGOs, and politicians. Practically, the Ministry of Research and Technology and is the single authority in formulating and executing the national STI policy in Indonesia. Although the Ministry of Industry has special units for research on industry and technology, including its policy, both ministries are not closely interlinked. Consequently, STI policy is on policy documents only, and it is difficult to

³ Indonesia's R&D budget is 0.09% of gross domestic product (GDP) in year 2014 (Pappiptek LIPI, 2015)

implement among private firms and state-owned firms.

In case of the mid- to long-term national STI programs plans formulated by Bappena, by which such policy must be adhered by all R&D actors including Ministry of Research and Technology, the study of Asmara, et al (2019) revealed that the actors involved in the mid-to long-term national STI programs are limited, such as Ministry of Research and Technology and public R&D institutions under the coordination of Ministry of Research and Technology like LIPI, BPPT, National Nuclear Energy Agency (BATAN), National Institute of Aeronautics and Space (LAPAN), and Nuclear Energy Regulatory Agency (BAPETEN). Actors of R&D suppliers like R&D units in ministries, private R&D institutes, and universities are not fully engaged in this policy formulation process. Even industries as R&D users are not actively involved in this STI policy formulation process facilitated by Bappenas.

According to Pattinson, et al (2015) and Edler, et al (2016), innovation policy is a combination of the supply and demand of science and technology. In this case, supply-side policy is more dominant than demand-side policy. Ministry of Research and Technology, LIPI, BPPT, BATAN, LAPAN, BAPETEN, and Bappenas are key actors in formulating STI policy in Indonesia. They are not closely connected to R&D units in ministries, private R&D institutes, and universities, even to industries as end users of R&D products. Consequently, the national STI goals and outputs are not achieved. For example, the study of Asmara & Maninggar (2019) revealed that the government of Indonesia's target to build 100 science and techno parks (STP) within five years (2015-2019) was not accomplished. In fact, only eight STPs were built, and all of them are previously existing science and technology laboratories in public R&D institutions developed to be STP.

The dominant role of R&D supplier side and S&T government agencies in formulating STI policy proves evidence that innovation policy is not importantly considered for industrial actors. Kemenristek is the focal point to in formulating STI policy technically, whereas Bappenas is the main partner of Ministry of Research and Technology in providing the long-to mid-term STI policy in Indonesia. In addition, the number of national R&D research units focusing on STI policy is limited in Indonesia. There are only three research agencies conducting STI policy research in Indonesia: Centre for Science-Technology-Innovation

Policy and Management (PPKMI) at LIPI; Centre for Specific Technology Area and Innovation System (PTKSSI) at BPPT; and Centre for Innovation Policy and Governance (CIPG). PPKMI-LIPI and PTKSSI-BPPT are public R&D agencies, whereas CIPG is a private agency.

STI policy is not an “alluring” policy in Indonesia compared to finance and defense security⁴. The limited public R&D fund allocation, limited R&D agencies and human resources such as STI policy researchers, and limited actors who are involved in STI deals making up the STI policymaking process as explained above are rational reasons why the NIS environment in Indonesia is not well-established. It is widely acknowledged that the STI policymaking process is not merely STI deals between STI-dealing government agencies and researchers, academicians, or R&D units but involves various actors, especially those with political and economic interests. Therefore, STI-funding actors, STI users, and politician are key actors that must be actively involved in the STI policymaking process.

To involve and convince many key actors such as the Ministry of Finance, Ministry of Industry, private and state-owned enterprises, politicians, local and specific communities/organizations, and other key actors dealing with STI areas, a robust evidence-based policy is needed. The dissemination, socialization, and advocacy of STI policy are a must and are important. To make evidence-based STI policies, STI policy think tanks are very critical.

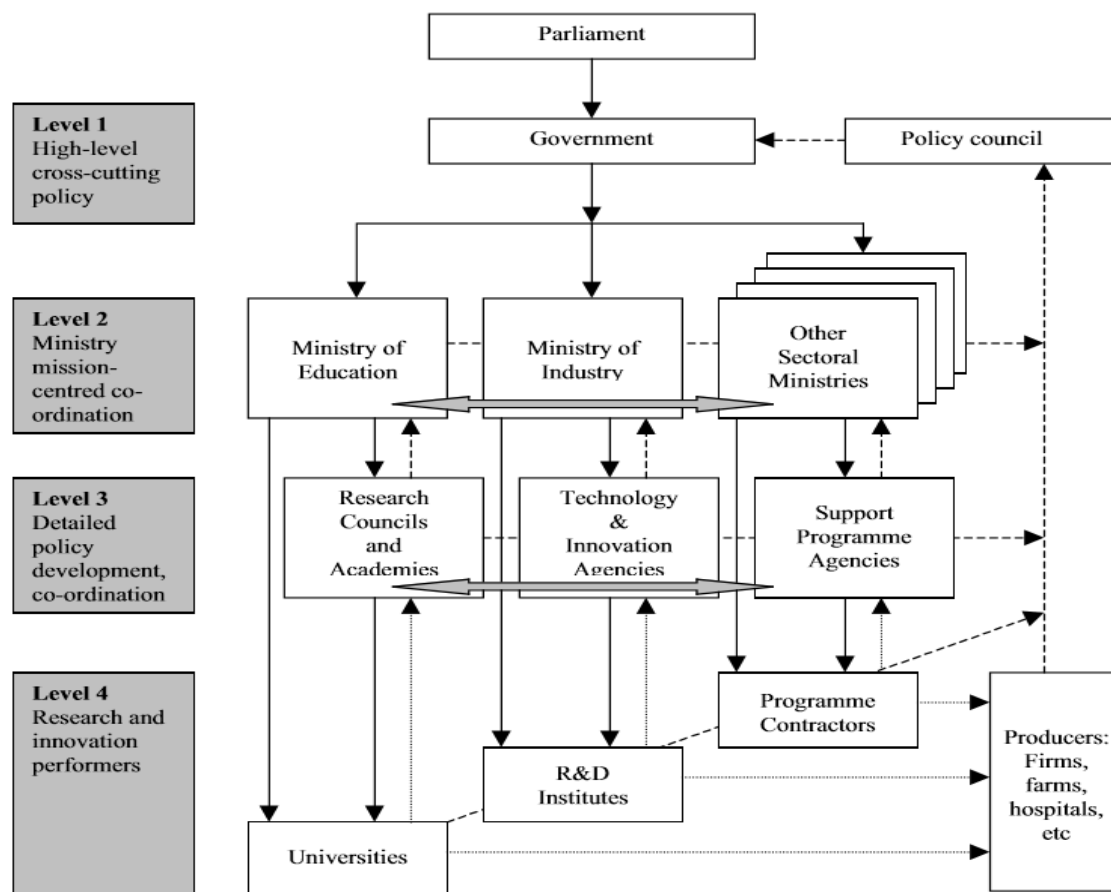
⁴ Interviews were conducted with two prominent law experts of the University of Indonesia in 2017. They said that the law of science and technology (S&T) and S&T-based industry does not have detailed provisions to regulate or reinforce actors to do something related to the S&T-based industry. In contrast, the finance law and defense laws have detailed provisions including how to implement the content of provisions technically to their users. This difference shows that S&T actors involve only limited government agencies, especially public R&D agencies, not business actors.

3. Korea's Experience on STI Think-tank

3.1 Overview and Operation of STI Policy System in South Korea

In order to understand the policy system in South Korea, we review the R&D governance framework proposed by Erik, et al (2003), which analyzes eight European countries' research and innovation system governance vertically and horizontally as shown in the figure below.

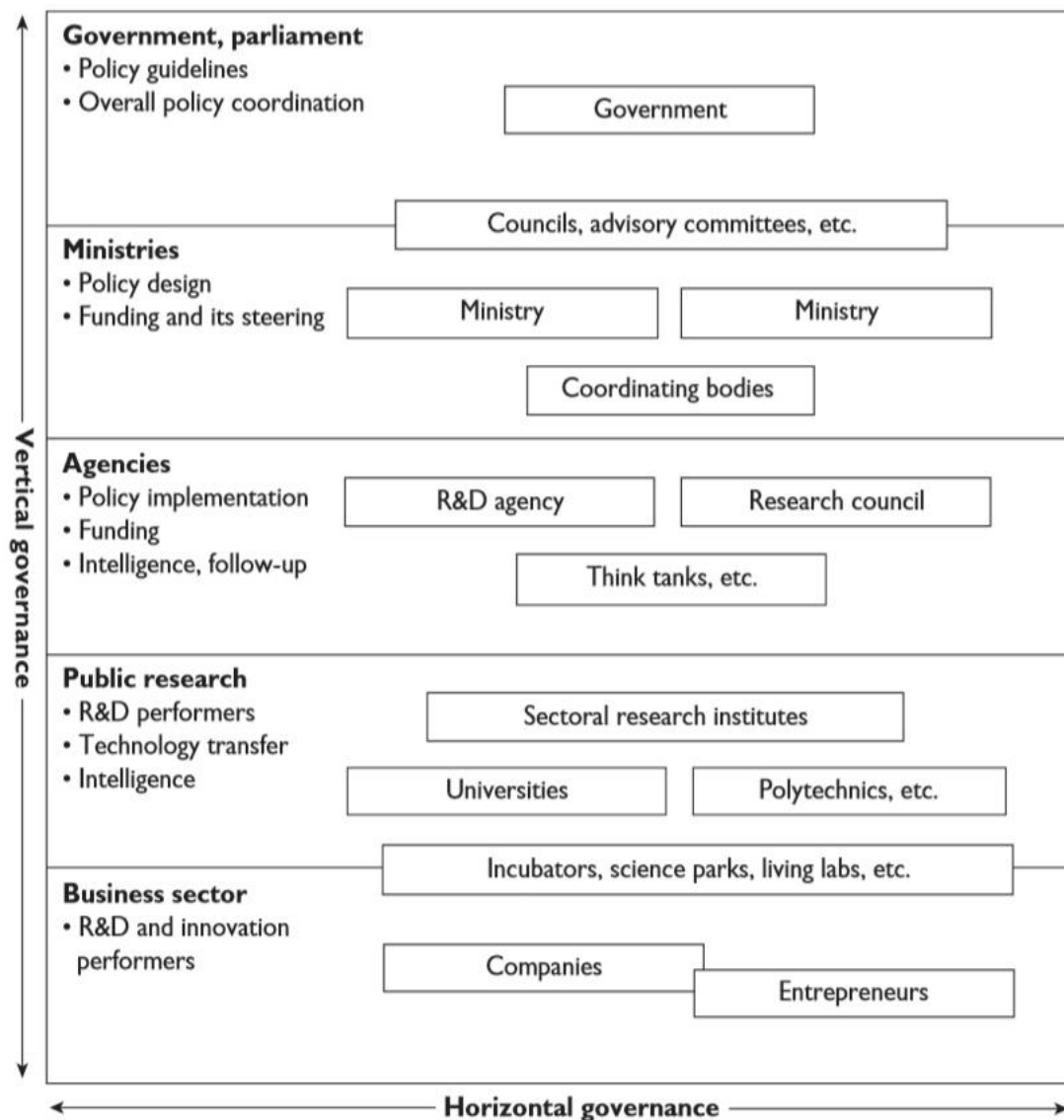
[Figure 3-1] Generic Organizational Structure for Research and Innovation Policy



Source: Arnold et al. (2003)

They establish vertical governance in four levels: high-level cross-cutting policy (i.e., Level 1); ministry mission-centered coordination (i.e., Level 2); detailed policy development, coordination (i.e., Level 3); and research and innovation performer (i.e., Level 4). In addition, they explain horizontal governance as the coordination and integration between different actors at the same level.

[Figure 3-2] General Framework for STI Governance



Source: Palmberg & Lemola (2012), p. 472.

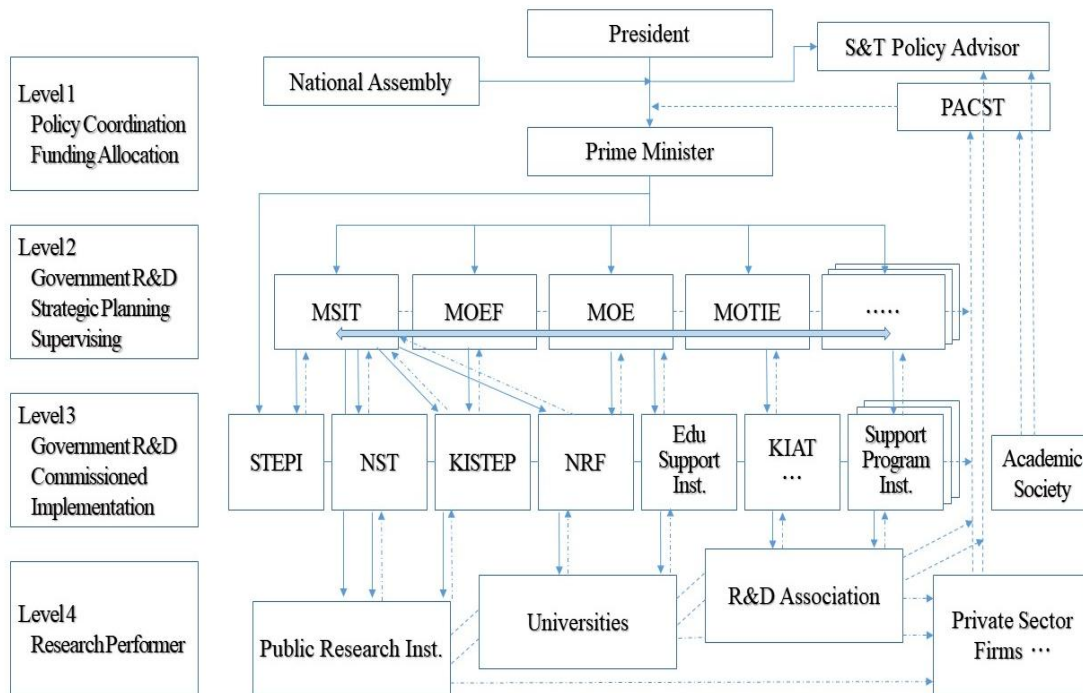
In a similar vein, Palmberg & Lemola (2012) break down the vertical governance into five levels such as government and parliament, ministries, agencies, public research, and business sector as shown in the figure below. While Erik, et al (2003) focused on the different scope of policy according to decision level such as cross-cutting policy and detailed policy development, Palmberg & Lemola (2012) designated different functions according to policy process such as policy guideline, policy design, and policy implementation.

Applying those frameworks to the Korean case, the Korean governance system can be suggested as shown in the figure below. At level 1 of policy coordination and funding allocation, Presidential Advisory Council on Science and Technology (PACST) controls funding allocation and priority setting in the national R&D agenda. At level 2 of government strategic planning, ministries such as Ministry of Science and ICT (MSIT), Ministry of Economy and Finance (MOEF), Ministry of Education (MOE), and Ministry of Trade, Industry, and Energy (MOTIE) plan and implement their R&D program in their own policy areas. The various agencies including STEPI, National Research Council of Science & Technology (NST), Korea Institute of S&T Evaluation and Planning (KISTEP), and National Research Foundation (NRF) at level 3 carry out the allocation of funding for R&D projects and evaluation under the supervision of their affiliated ministries. Finally, R&D performers such as universities, firms, and public institutes implement the R&D project at the street level.

Considering those frameworks and application to the Korean case, the Korean R&D governance system can be characterized as a centralized, fragmented system. Regarding level 1, the overall policy coordination and cross-cutting policy formulation have been strongly restricted by Ministry of Economy and Finance through strong budget control on other ministries (Hong, 2016). As a result, overall policy coordination by a higher-level actor has not been implemented successfully. With regard to level 2, the Korean ministries have developed an R&D system independently as silos. Moreover, overlapping between each ministry's R&D agenda has been recognized as a main barrier for the efficient implementation, whereas coordination between ministries' R&D policy and funding, mainly by the Innovation Division, is quite limited until now. The autonomy of agencies and performers at levels 3 and 4 and delegation issue from the ministries regarding R&D decision making have been recognized as bottleneck for the advancement of the Korean innovation

system. In addition, the optimal institutional structure encouraging university-industry-government relations regarding competition and cooperation has been one of the main issues in the enhancement of the Korean national innovation system up to now.

[Figure 3-3] R&D Governance System in South Korea



Source: Authors and adapted from Yang et al. (2016).

3.2 History of STEPI and Main Achievements

3.2.1 Birth of STEPI: Late 1980s

STEPI originated from the establishment of Center of S&T Policy (CSTP) as an affiliated institute of Korea Advanced Institute of Science and Technology (KAIST) in 1987. In 1982, the Korean government initiated a specific R&D program as National R&D Program (NRDP). Moreover, in the process of implementation of a long-term plan for the advancement of science and technology, consensus on the needs for expertise in systematic evaluation as well as the creation of a specialized organization for science and technology policy has spread widely in the S&T policy communities. Against this background, CSTP was established

as a think tank for S&T policy following the model of Science Policy Research Unit (SPRU) in Sussex University and Center for Policy Alternatives (CPA) in Massachusetts Institute of Technology (MIT).

The initial functions of CSTP can be explained as follows: first, as a think tank, CSTP supports the ministries' planning and implementation of science and technology policy; second, CSTP provides the government, public agencies, and private sector with information on R&D and science and technology; third, CSTP establishes and supports an academic hub for science and technology studies as newly emerging areas; fourth, CSTP develops the methodology and theoretical grounds for evaluation for R&D investment; fifth, CSTP encourages international cooperation in order to gain advance knowledge from overseas prestigious research institutes and international agencies; sixth, CSTP trains human resources as S&T administrators and research managers.

During this initial period, the major achievements of CSTP can be summarized as follows: first, CSTP contributed the long-term vision and basic plan at the national level; second, in order to increase the efficiency of national R&D resources, CSTP has implemented a study on a "long-term plan for S&T human resources" and an "investment plan for S&T"; third, CSTP carried out research on tax, finance, and procurement system for R&D activities in the private sector; fourth, public acceptance of science and technology is also part of their mission, so CSTP has explored the role of science and technology for balanced development between regions at the national level. During this project, researchers in CSTP have established strong cooperative relations with government officers. Additionally, collection of overseas S&T information and international conference with advanced countries have increased the research capacities of CSTP.

3.2.2 Strengthening the Planning and Evaluation Function: The 1990s

In 1989, KAIST was divided into Korea Institute of Science and Technology (KIST) as a research-oriented institute and KAIST as a university. Therefore, CSTP was reorganized under KIST. In 1993, CSTP was renamed STEPI and established as an independent organization, because STEPI's role is "an umpire," whereas KIST is "a player" regarding the

planning and evaluation of national R&D programs. Such planning and evaluation function of STEPI has been separated into KISTEP later in 1999.

In 1991, CSTP had only 25 research staff, and 3 staff had doctoral degrees. In 1996, STEPI had 77 researchers (50 doctoral degree holders) out of a total of 168 staff, a seven times' increase compared to 1987 when it was established. Regarding the disciplines of research staff, 45% of them are from science and engineering, and 35% are from economics and management studies. This means that STEPI has a balanced disciplinary structure for interdisciplinary research. In 1995, the center for international S&T cooperation of KIST moved into STEPI. Since this integration, STEPI has established a basic organizational structure consisting of three functions: research on S&T policy, R&D management and planning, and international S&T cooperation.

Among those three missions, S&T policy research has been regarded as a leading function of the institute. For example, STEPI implemented research on national R&D plans, R&D statistics in OECD countries, financial and legal system for R&D, and socio-economic effect of R&D investment. STEPI also carried out various short-term commissioned research projects from ministries and other clients. Regarding R&D management and planning, STEPI established Project-Based System (PBS) in the national R&D program and planned the "G7 Project," a landmark big national program for mission-oriented R&D. In other words, STEPI contributed remarkably to building a planning, implementation, and evaluation system for big national R&D programs. With regard to international cooperation, STEPI initiated a cooperative program with OECD and other international agencies. Moreover, STEPI has trained policymakers from developing countries based on a commissioned project from Korea International Cooperation Agency (KOICA).

In 1999, STEPI was established as an independent legal body pursuant to the "Act on the Creation and Management of Public Research Institutes." According to this law, the jurisdiction of STEPI has been moved to the Prime Minister from the Ministry of Science and Technology, which aims to enhance the autonomy and expertise of public research institutes. In the same year, the function of industrial policy research from Institute of Technology Enterprise Policy (ITEP) has been integrated into STEPI. Therefore, STEPI has

been recognized as a unique institute in the area of S&T policy domestically and internationally.

3.2.3 Expansion of Studies on S&T Policy: Mid-1990s and Mid-2000s

During this period, various organizations dealing with the S&T policy have been newly established in universities (e.g., technology policy program in Seoul National University and department of science studies in JNU), and academic societies such as Korean Society of Innovation Management and Economics (KOSIME) and Korean Technology Innovation Society (KOTIS) have been created. Moreover, other public agencies such as KISTEP and ITEP have also implemented S&T policy studies. Even private think tanks such as Samsung Economic Research Institute and LG Economic Research Institute have actively participated in S&T policy studies.

STEPI has played the role of a hub for those actors regarding exchange of experts as well as knowledge in the area of S&T policy studies. Against this backdrop, STEPI has become a national think tank beyond a supporting agency for the ministry of science and technology, having dominated the S&T policy projects commissioned from the ministries. Moreover, STEPI researchers have served various advisory roles for government agencies such as adviser for ministers as well as for the president.

3.2.4 Searching for the Future of S&T: After 2011

Post-catch up issues emerging after 2010 have stimulated the creation of the center for future studies in STEPI. This center produced various national S&T agenda items based on the foresight methodologies. In addition, R&D for social problem solving has been conceptualized, with various projects planned during this period. Convergence studies, S&T Official Development Assistance (ODA), startups, and regulation will become important policy issues in the upcoming era. In 2015, STEPI has moved into the government policy complex in the Sejong Special Self-governing City, which encourages opportunities for interdisciplinary approach.

3.3 STEPI's Current Status and Function and Role in Korea's NIS

STEPI plays essential roles in several areas.⁵ Regarding the research functions, STEPI leads STI studies in Korean academic communities (e.g., “Research on S&T and R&D activities and technological innovations,” “Research on Global S&T Policy,” and “Study on the Relationship between S&T, Economy, and Society”). Moreover, as a strategic planner, STEPI supports not only government officers but also work with industrial partners (e.g., “Development of S&T policy alternatives” and “Consultancy on technology management strategy”). Collaboration is also an important mission for STEPI, and domestic actors such as universities as well as international partners are all important in terms of disseminating the research outcome, sharing data, and inspiring the next generation in the fields (e.g., “International S&T cooperation,” “Collaboration with government, industry, academe, and foreign institute,” “Dissemination of research outcomes,” and “Education and Training on its expertise”).

- Research on S&T and R&D activities and technological innovations
- Development of S&T policy alternatives
- Consultancy on technology management strategy
- Study on the relationship among S&T, economy, and society
- International S&T cooperation
- Research on global S&T policy
- Collaboration with government, industry, academe, and foreign institutes
- Dissemination of research outcomes
- Education and training on expertise

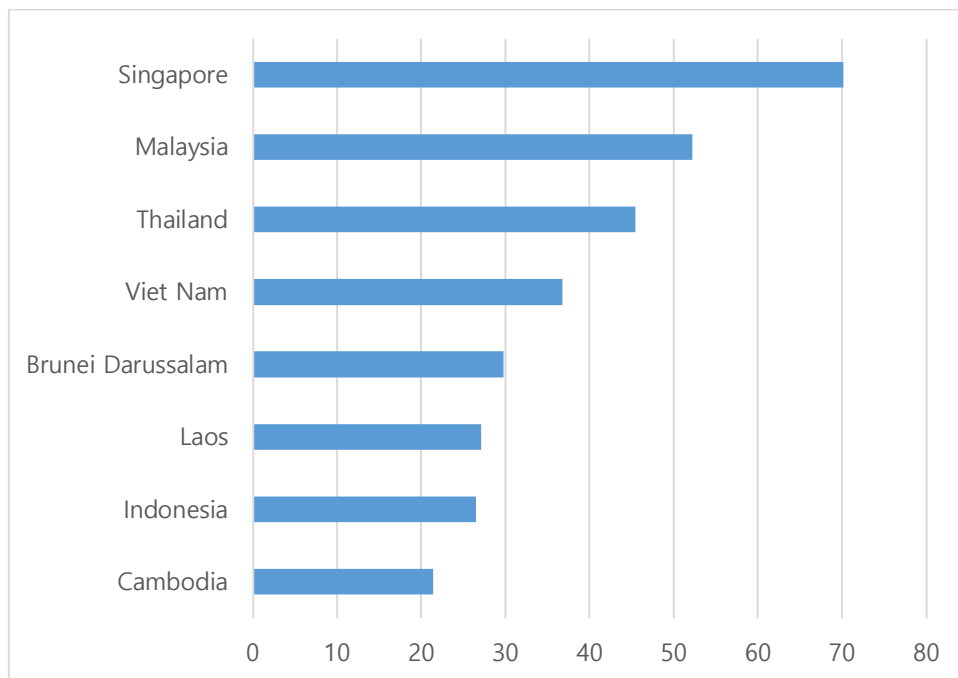
⁵ Homepage, STEPI, <http://www.stepi.re.kr/eng/app/ePart/main.jsp?cmsCd=CM0143>

4. Need for a New Think-tank on STI in Indonesia

4.1 Importance of STI Policy to Develop the National Innovation System in Indonesia

The innovation environment in Indonesia is still not in a satisfactory position. SC Johnson College of Business, Cornell, INSEAD, and World Intellectual Property Organization (WIPO) released Global Innovation Index (GII) in 2020, describing the innovation process in Southeast Asian countries (Soumitra Dutta, et al, 2020). The index is ranked by seven pillars: institutions, human capital & research, infrastructure, market sophistication, business sophistication, knowledge & technology output, and creative outputs.

[Figure 3-4] Innovation Ranking of Indonesia among ASEAN Countries



Source: Soumitra Dutta et al. (2020).

In 2020, Indonesia ranked 85th out of 131 countries. In the group of ASEAN countries (Figure 7), the innovation index of Indonesia is slightly above Cambodia yet below most Southeast Asian countries. The report views Indonesia as having moderate market sophistication (ranked 62nd) but poor institution and business sophistication (ranked 111th and 114th).

In detail, based on GCI and GII in the last nine years, comparing to the ASEAN countries [Table 3-1], Singapore is ranked first, followed by Malaysia, Vietnam, Thailand, Philippines, Brunei Darussalam, Indonesia, and then Cambodia. After 2012, the economic growth of Indonesia has stood at 5% per year since 2014 up to 2019 but is declining dramatically in recent years particularly due to the pandemic-covid COVID-19 situation.

[Table 3-1] Development of Global Competitiveness Index and Global Innovation Index

Year	Global Competitiveness Index (GCI)		Global Innovation Index (GII)	
	Ranking	Score	Ranking	Score
2010	44	4.43	72	2.95
2011	46	4.37	99	27.78
2012	50	4.39	100	28.10
2013	38	4.53	85	31.95
2014	34	4.57	87	31.81
2015	37	4.52	97	29.79
2016	41	4.51	88	29.07
2017	36	4.68	87	30.10
2018	45	64.9	85	29.80
2019	50	64.6	85	29.72

Source: WEF (2012); WEF (2010-2019).*

On the other hand, the fourth mid-term national development plan (RPJMN) (2020-2025) agenda of STI focuses on raising the quality and competitiveness of human resources by strengthening education and enhancing science and technology capabilities and creating innovation (PPN/Bappenas, 2019, p. 110). This agenda is an approach of the government to enrich the innovation level to realize a more established national innovation system.

Besides, the roles of various actors are needed to develop Indonesia's NIS, especially the

integration of stakeholders. One of them is Ministry of Research and Technology/BRIN, which sets research priorities based on sectoral and policy interventions through research funding. As the body with authority and coordination with universities and government research and development agencies, Ministry of Research and Technology (MoRT)/National Research and Innovation Agency (BRIN) can be a central actor in constructing NIS among universities, public agencies, ministries, and other institutions.

Nonetheless, the challenge lies in the implementation as the actors involved, including think tank that focuses on the STI Policy area, are still hard to find. There are few among public and private institutions possessing expertise in STI Policy. The main actors are Ministry of Research and Technology/BRIN and LIPI, which produce the knowledge of STI Policy. In wider communities, the concentration spread individually in universities and other institutions.

4.2 Overview of STI Governance and Policy System in Indonesia

The national-scale policymaking in Indonesia goes through two main processes: the development of short- and mid-term planning and development regulations and laws (Jackson, Purnawati and Shaxson, 2018, p. 67). The Ministry of National Development Planning/National Development Planning Agency (PPN/Bappenas) has a role of developing the long-term national development plan 2005-2025 (RPJMN 2005-2025) consisting of five-year mid-term national development plan (RPJMN). The fourth RPJMN (2020-2025) agenda of STI is focused on raising the quality and competitiveness of human resources by strengthening education and enhancing science and technology capabilities and creating innovation (PPN/ Bappenas, 2019, p. 110).

In the national-scale STI policymaking, the MoRT/BRIN is authorized to handle government affairs in the fields of research, technology, and higher education to assist the President in organizing the government of the country. The determination of policies and implementation of national research are based on the National Research Master Plan 2017-2045 (RIRN 2017-2045), which sets research priorities based on sectoral and policy interventions through research funding. On the other hand, research and studies on science

and technology policies are conducted LIPI, which consists of researchers and policy analysis. The main function of LIPI is to carry out technical research activities, including those in the field of science and technology, but it is not directly involved in policy advocacy based on the results of its research.

4.3 Policy Think Tanks and STI Policy Units in Indonesia

Think tanks in Indonesia have a dynamic role that relies on the changes in the political environment in Indonesia. Pane, *et al* (2018) argue that the role of think tank in Indonesia is a Non-Government Organization (NGO) that evolves in three waves: service delivery (1970s-1980s), Advocacy Work (1990s-2000s), and Knowledge Production (2010-present). Think tanks that previously conducted policy research at the time of pre-implementation are now in each cycle of policymaking. Pane, *et al* (2018, p. 40) mentioned that there are several main functions in Indonesia, including (1) Strategic Actors in Community Development, (2) Provision of Data to Support the CSO Coalition Advocacy Agenda, (3) Creation of Public Pressure by Advocating Through the Media and Social Media, and (4) Focus on Core Research Agenda to Build Credibility. They assume that think tanks in Indonesia have different functions when engaging in the policymaking process. Still, what is key here is that the ability of think tanks to establish collaborative relationships with international agencies, develop credible research, and influence public views is an important factor in influencing policymaking effectiveness.

There is no accurate number of think tanks in Indonesia. The current mapping and ranking conducted by the University of Pennsylvania and McGann titled “Global Go To Think Tank Index” since 2008 show the growth of the number of think tanks in Indonesia. In general, think tanks in Indonesia are classified as independent (semi-independent) or quasi-independent. Only a few agencies are affiliated with the government and universities. The number of think tanks in Indonesia based on Global Go To Think Tank Index has increased by more than half compared to ten years ago, from 20 institutions in 2008 to 31 institutions in 2019.

Among non-government-affiliated think tanks, two prominent institutions are the Centre

for Strategic and International Studies (CSIS) of Indonesia and the Economic Research Institute for ASEAN and East Asia (ERIA). CSIS is an independent think tank based in Washington, United States established in 1964 and first formed in Indonesia in 1971. It focuses on the study and advocacy of foreign policy, especially in politics, economics, and security. CSIS Indonesia was ranked as the no. 1 think tank in South Asia, ASEAN, and Asia Pacific in 2018. This is closely related to the performance of the United States CSIS, which is considered very good as the third-best think tank in the world based on the Global Go To Think Tank Index report.

ERIA is a quasi-independent think tank based in Jakarta, the result of the agreement among 16 ASEAN leaders in 2008. One of ERIA's tasks is to conduct research and dissemination of policy research with three pillars: deepening economic integration, narrowing development gaps, and pursuing sustainable development. ERIA stands out for its international economic policy studies and ranks 13th in the "Top International Economic Policy Think Tanks" category in Global Go To Think Tank Index in 2019. In anti-corruption advocacy and governance transparency, two institutions that consistently wield an impact are Indonesian Corruption Watch (ICW) and Centre for Regional Information and Studies (PATTIRO). The two institutions ranked 26th and 34th, respectively, in Global Go To Think Tank Index in 2019. An independent institution that often did advocacy on the practical problems of eradicating corruption in Indonesia often made it to the list.

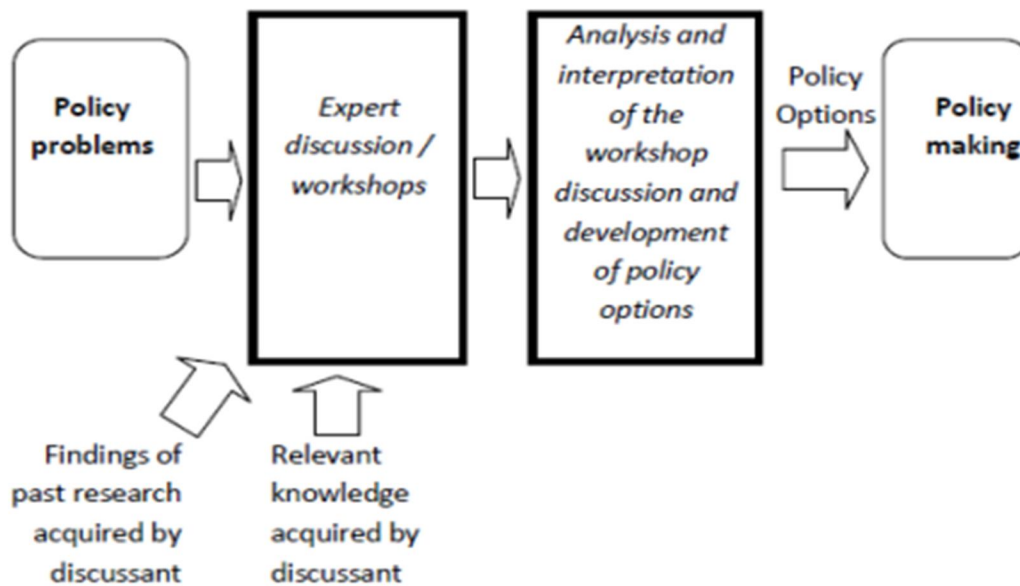
Among government-affiliated think tanks, some government agencies that have functioned as think tanks are still very limited. PPN/ Bappenas is a planning ministry for which we are strengthening its functions as a think tank with the establishment of the Policy Analysis Center (PAK) (Pellini, *et al*, 2018, p. 62). PPN/ Bappenas PAK- Bappenas has the task of carrying out analysis and harmonization of policies on the implementation of programs and activities of the Ministry of National Development Planning/ National Development Planning Agency, carrying out the preparation of competencies and materials as well as substantive assistance and coordination of follow-up directives or activities of the minister, and coordinating the planning, implementation, monitoring, and evaluation of performance analysis of the assignments.

Other institutions were created to promote coordination across ministries/agencies to improve the implementation of poverty reduction programs, improve the living standards of the poor and the vulnerable, and reduce inequality among income groups: the National Team for the Acceleration of Poverty Reduction (National Team for the Acceleration of Poverty Reduction or TNP2K); Fiscal Policy Agency (BKF), which also serves as a think tank under the ministry of finance; Presidential Advisory Council (WANTIMPRES), which has a function of advising the president on practicing state power in various fields; and other research and analysis policy institutions. Nonetheless, there is no specific think tank in the field of STI under the government. The function of providing evidence and ideas in the policymaking process in STI is conducted by government research institutions, such as LIPI and national research council (DRN).

According to Cislowski & Purwadi (2011), Bappenas, as the Planning Coordination Ministry, does not have an R&D unit; the function is embedded in all units. While the Policy Analysis unit appears to be very successful in addressing this function, it also sees the need to develop standard operating procedures for the Ministry, manuals, and ongoing training to support staff in all units in providing quality advice. This is in keeping with the idea of Bappenas itself being a think tank for the government.

The Policy Analysis Unit in Bappenas reported that it more frequently uses the experts' knowledge approach for the generation of knowledge instead of commissioning an external research unit to conduct policy-oriented research. The usual methodology involves convening an experts' meeting or workshop to analyze the problems, identify the relevant factors impacting on it, and then develop feasible solutions and policy to address the issue or prevent similar problems from occurring in the future.

[Figure 3-5] Policy Options Development: The Think Tank and Bappenas Policy Analysis Unit Approach



Source: Cislowski & Purwadi (2011).

National Team for the Acceleration of Poverty Reduction (TNP2K) is a new Think Tank established under the Vice President to accelerate poverty reduction. It draws on some of the principles underpinning SMERU, described in a separate Diagnostic. The Coordinator of this Think Tank describes its role as “an interface between knowledge and decisions.” It produces knowledge by outsourcing, advocating, liaising, and communicating with many sources: university, government, research institutes, multi-nationals, development partners, experts, and private sector. It then interprets, scans, and synthesizes research with the aim of providing a two-page policy brief with clear options to the Minister. The recognition that policy advice should be succinct is a breakthrough in the Indonesian context wherein many important documents appear to be valued by weight.

In relation to Balitbang in Ministries, the Coordinator commented that “Balitbang can’t behave like a Think Tank because they are bound by bureaucracy and procurement is a problem.” This makes them process-oriented, not output-oriented. With good leadership, however, it is possible to have a good Balitbang, and there will always be a place for

Balitbang and think tank models existing side by side.

There are a few think tank units for STI policy in Indonesia. Specifically, there are nine STI policy think tank units in Indonesia consisting of seven units of Indonesian government agencies, 1 unit of private agency, and the rest, one unit of semi-government agency funded by the Australian Government.

[Table 3-2] Think Tank Units for the Science-Technology-Innovation Policy in Indonesia

Institutions	Category	Main Function	Professionally Managed by
Centre for Science-Technology-Innovation Policy and Management (PPKMI) - Indonesian Institute of Sciences (LIPI)	Government-Affiliated	Researching/Studying/Analyzing the STI Policy and Providing STI Policy Recommendations (policy brief)	Civil servants (researchers and policy analyst)
Centre for Specific Technology Area and Innovation System (PTKSSI) at the Agency for Technology Assessment (BPPT)	Government-Affiliated	Analyzing and Mentoring on STI Policy in Practical/Technical ways	Civil servants (engineer and policy analyst)
National Research Council (DRN)	Government-Affiliated	Advising the Minister of Research and Technology on STI Policy	Senior experts from the academe, bureaucrats, and businesses (ABG actors)
Directorate of Science-Technology, Education, and Culture (Iptekdikbud) at the Ministry of National Planning and Development (Bappenas)	Government-Affiliated	Planning the STI Policy Through Mid-and Long-term Programs	Civil servants (technocrats and policy analyst)
Academy of Indonesian Science (AIPI)	Government-Affiliated	Advising related stakeholders (Minister of Research and Technology, Minister of	Senior lecturers and researchers who are in

Institutions	Category	Main Function	Professionally Managed by
		Bappenas, President, International scientific communities) on the STI Policy	Indonesia and overseas
Agency for Research and Development of Industry (BPPI) at the Ministry of Industry (Kemenperin)	Government-Affiliated	Researching/Studying/Analyzing/Advocating industrial policy to the Minister of Industry	Civil servants (researchers, policy analyst, and bureaucrats)
Centre for Innovation Policy and Governance (CIPG)	Autonomous and Independent	Researching/Studying/Analyzing/Advocating STI policy and providing STI policy recommendation to stakeholders, especially to the Minister of Research and Technology	Non-government staffs (researcher, policy analysts)
Knowledge Sector Initiative (KSI)	Quasi-Governmental (supported by the Australian Government)	Studying/Analyzing/Advocating Public Policy (including STI Policy)	Non-government staff (researcher, policy analysts)
National Committee of Economics and Industry (KEIN)	Government-Affiliated	Advising the President on Economic & Industrial Policy	Senior experts from academe, bureaucrats, and businesses (ABG actors)

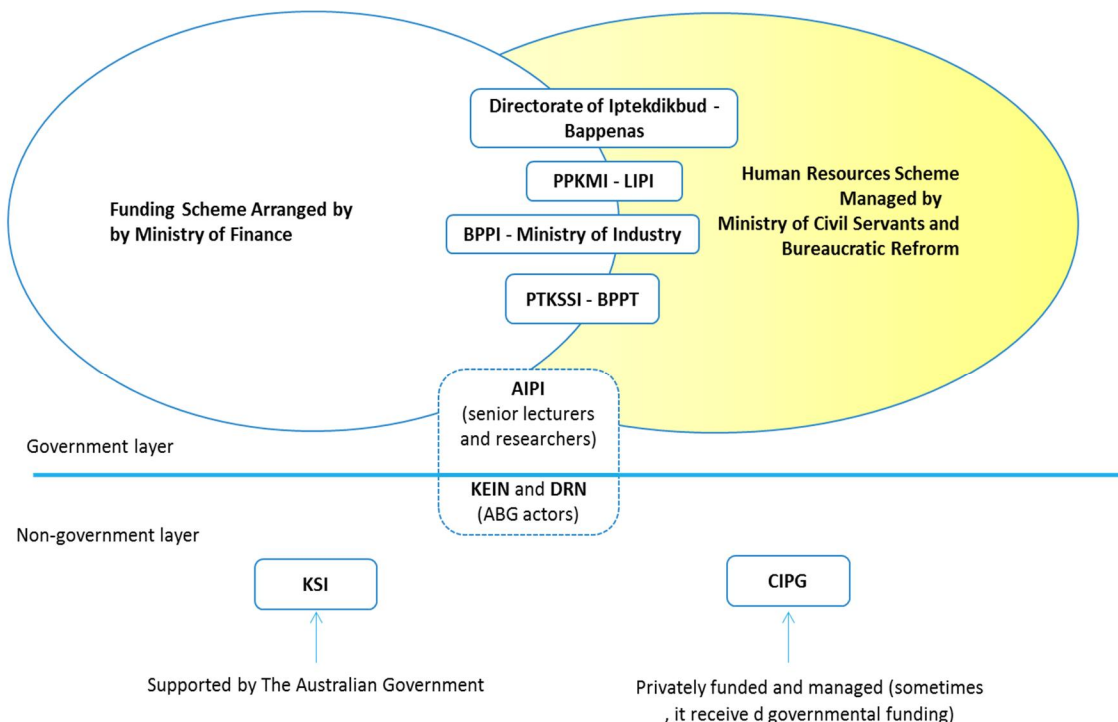
Source: developed by author

5. Planning a STI New Think Tank in Indonesia

5.1 How do STI Policy Think Tanks work? – The Current Condition

From nine STI policy think tank units in Indonesia, there are seven units with the same condition as government-affiliated ones, and four of them are fully funded by the government and filled by civil servants in terms of researchers, engineers, policy analysts, technocrats, and bureaucrats. They are the Directorate of Iptekdikbud – Bappenas, PPKMI – LIPI, BPPI - Ministry of Industry, and PTKSSI - BPPT. The funding law from the Ministry of Finance and human resources law from the Ministry of Civil Servants and Bureaucratic Reform are often hindrances of their activities. In other words, some government laws set a strict boundary for their actions.

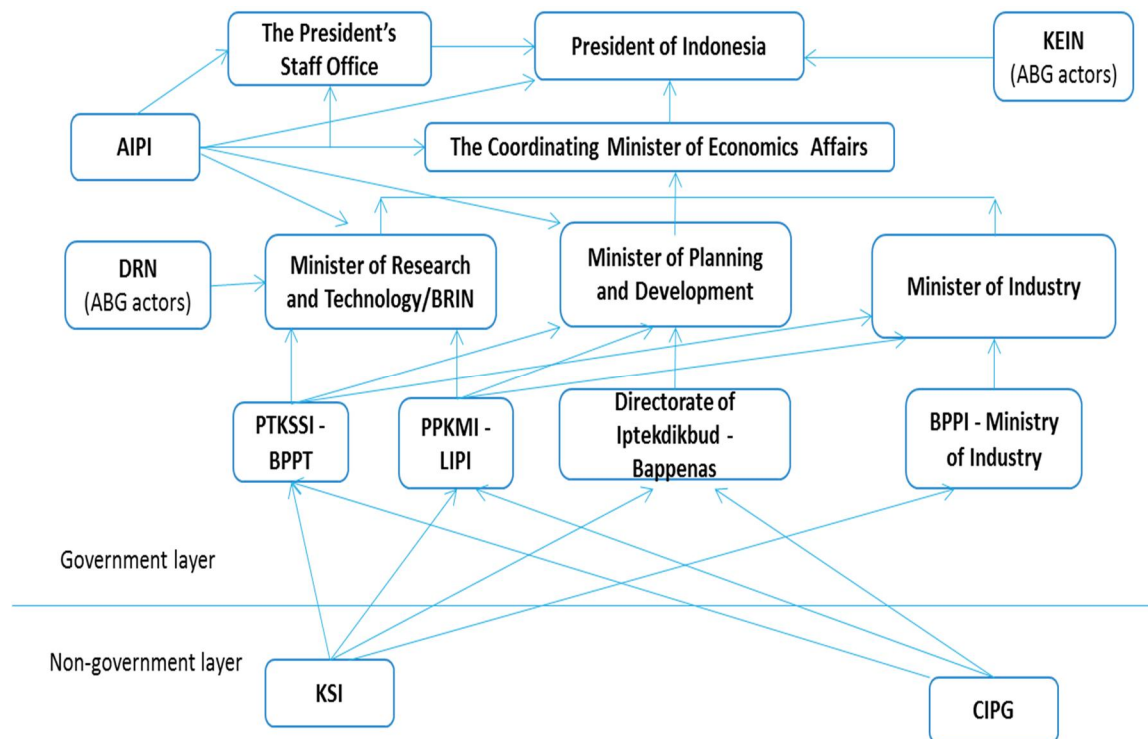
[Figure 3-6] Organization Mode & Governance of STI Think Tank Units of Indonesia



Source: developed by authors

For AIPI, KEIN, and DRN, these units are supported by government budgets, although they are filled by senior experts consisting of academicians, government, and business actors. They are from government and non-government representatives. In this regard, private funding is allowed, but it must follow the government funding management because they are public institutions. Unfortunately, they are filled by top officials or public leaders in government and non-government organizations, so the time devoted to discussing STI issues is very limited. In addition, the rest is constituted by KSI and CIPG, both of which are private STI policy think tank units that are privately funded and filled by private researchers and staff. Nonetheless, they often get funding from the Indonesian government like CIPG and the Australian government like KSI.

[Figure 3-7] Governance of STI Policy Think Tank Units



Source: developed by authors

Indonesia has four coordinating ministries, and one of them is the Coordinating Ministry of Economic Affairs. This ministry coordinates all ministries related to economics and innovation, such as the Ministry of Research and Technology/BRIN, Ministry of Planning and Development or Bappenas, and Ministry of Industry or Kemenperin. The coordinating ministry is directly responsible to the President of Indonesia with regard to tasks and duties. In this term, the President of Indonesia is assisted by the President's Staff Office to formulate policies in Indonesia, including economic and STI policies. The President of Indonesia has strong power to execute all policies, including STI policy.

The think tank units categorized as government-affiliated, such as the Directorate of Iptekdikbud – Bappenas, PPKMI – LIPI, BPPI - Ministry of Industry, and PTKSSI – BPPT, KEIN, DRN, and AIPI, have direct access to key stakeholders in those ministries and the President's Staff Office, whereas KSI and CIPG are categorized as non-government-affiliated, having no direct access to key stakeholders in the STI policy. To gain government access, they often do Indonesian government projects, implement an independent project related to current public issues, and involve government researchers/ academicians/ technocrats/ policy analysts in this project.

6. Policy Recommendations

6.1 STI Think Tank Governance in Indonesia

For the feasible governance of STI policy, the creation of a think tank in Indonesia should consider the existing regulations, norms, and systems. Still, creating a new think tank unit on STI policy in the broader context will face big challenges or even clash with other regulations or agency interests. Indeed, designing an STI policy think tank unit is not separated from the intent of the top management or even political will. This study provides four alternative STI policy think tank units that can be planned and designed in Indonesia.

The table below shows four alternative STI policy think tank units in Indonesia. All think tank units have the main function of planning, implementing, and evaluating policy activities and its results. The sequence of numbers shows the priority of the most feasible plan to be applied in Indonesia. Numbers 1 and 2 are suggested designs to create new STI policy think tank units in Indonesia. Number 3 refers to the previous suggestions to be conducted, and number 4 is the most difficult design to be implemented in Indonesia. Number 1 is the existing condition by which STI policy units are government-affiliated, managed and operated by government officials including government researchers. Increasing the quality of human resources and implementing good organizational management practice according to existing laws and constraints are the main duties. Number 1 is mostly suggested by this study because government officers have the most understanding of internal and external conditions to make the STI policy implementable.

[Table 3-3] Suggested STI Think Tank Governance in Indonesia

Option	Funding source	Governance	Status of the members	Function
1	Government budget (Ministry of Research and Technology/BRIN)	Government-affiliated	Government officers	Planning, Implementing, Evaluating
2	Budget of non-government organizations	Autonomous & Independent body	Private sector	Planning, Implementing, Evaluating
3	Budget of government agencies and donors	Quasi-independent	Private sector	Planning, Implementing, Evaluating
4	Governmental budget under the Ministry of Finance (LPDP scheme)	Quasi-governmental	Contracted public officers	Planning, Implementing, Evaluating

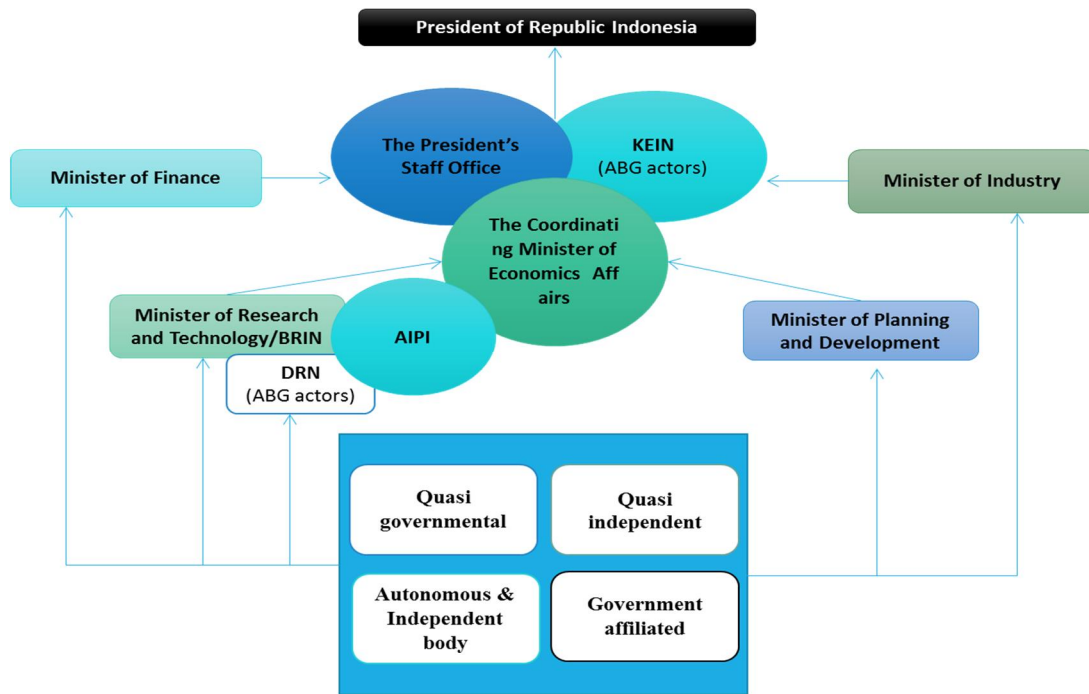
Source: developed by author

Number 2 is the autonomous and independent body funded by non-government organizations. In this pattern, all organization management, human resources, and funding management are purely the responsibility of private sectors, not government agencies. Though the funding is often derived from government sources, research funding management is autonomously managed by private organizations. Number 3 is a Quasi-independent pattern by which research budgets can be accessed from any government agencies and donors. In this case, private management is the main scheme for managing business organization, human resources (researchers and non-researchers), and other supporting activities. In this pattern, government management and private management are integrated under a private management system. It is a little bit difficult to implement in Indonesia due to the existing law and bureaucracy system.

Number 4 is the quasi-governmental pattern wherein private sectors are involved through the contracted public officers. In this case, government funding is the main source; the government is responsible only for budgeting reports, and the operators or research activities are private agencies. In this case, research activities including researchers' human

management scheme are not restricted by government laws. This pattern is difficult to apply in Indonesia due to the existing constraints and conditions there. Still, this pattern can be implemented provided a national innovation ecosystem is first created in Indonesia.

[Figure 3-8] Suggested Working Pattern of STI Policy Think Tank in Indonesia



Source: developed by author

There are four governance types of STI policy think tank units: quasi-governmental, autonomous & independent body, quasi-independent, and government-affiliated. According to Indonesian laws and institutions, each think tank unit has advantages and disadvantages. In the future, combining the advantages and minimizing the disadvantages are suggested. For instance, those categorized as government-affiliated have advantages to the stakeholders in accessing funding and network directly, but they are restricted by government funding management schemes and civil servants' laws. On the other hand, those categorized as autonomous & independent bodies and quasi-independent have advantages in managing funding and human resources but lack access to the networks for key stakeholders.

For that reason, there is a need for a new scheme to create a feasible framework related to how all STI policy think tank units can access government and non-governmental fund, manage agile and adaptive human resources to respond to public issues especially STI issues, and have wide and strong networks with knowledge suppliers like academicians or researchers and policymakers like ministers, committee, and even the president. Therefore, all STI policy think tank units can access and contribute to key STI policymakers like the Minister of Finance, Minister of Research and Technology or BRIN, Minister of Industry, and Minister of Planning and Development.

Furthermore, the recommendation of STI policy think tank units is widely accepted by the Coordinating Minister of Economic Affairs, President's Staff Office, and KEIN, and finally the President of Indonesia. In this context, the role of DRN and AIPI is a policy catalyst not only to the Minister of Research and Technology but also to the key stakeholders in other ministries. Still, this optional scheme is suggested by our team, and changes are possible after thorough discussion and field survey in the future.

6.2 Remaining Issues

The ideal condition as in the previous discussion is still difficult to reach due to the current obstacles. There is a pre-condition that those issues must be solved before the option is implemented. First, the Indonesian bureaucracy systems tend to be rigid. Such less flexible system may affect finance, law, and administrative procedures, for instance. Moreover, in the case of a quasi-independent or an independent think tank, flexibility to manage its funds from both government and other sources is required. Therefore, there is a need to reform the rigidity to have more flexible bureaucracy.

Second, there is no favorable coordination system among public agencies. They occasionally have overlapping coordination that may cause misinterpretation among agencies. In addition, the STI Policy think tank needs coordination with a wide range of stakeholders. Understanding among stakeholders and STI policy think tank should be established clearly.

Third, there is an insufficient number of human resources in STI policy. The current number

of research centers with a function of conducting STI policy studies is extremely limited. Moreover, there are few researchers with research concentration on STI policy even though there is no definite data, but the authors believe that the number is less than other social fields. There is also no university major that focuses on STI policy, which affects the supply of experts in STI policy.

Fourth, not all policymakers and stakeholders habitually and fully understand the use of evidence-based policymaking. In most of the agencies, the present capabilities of researchers and policy analysts in the phase of policymaking are still deemed insufficient. The policymaking process is sometimes based on the policymaker's judgment and experiences that tend to be trial and error. This reflects well the findings of studies, i.e., the result may not reach the correct recipients if evidence-based policymaking has no strategic position in the policymaking.

Fifth, there is a need to develop the environment for promoting innovation. It is established not only by the STI policy think tank but also by other public agencies. In this case, government bodies should have the same frequency with regard to synergy in creating an innovation system. Kemenristek/BRIN can be a main actor in constructing NIS among all institutions.



CHAPTER 4

R&D Funding and Governance of Research Endowment Fund in Indonesia

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Chapter 4. R&D Funding and Governance of Research Endowment Fund in Indonesia

1. Introduction: Issues Related to R&D Funds

In 2018, the Gross Expenditure on Research and Development (GERD) in Indonesia accounted for only approximately 0.28% of the total Gross Domestic Product (GDP) or Rp41.43 trillion. This value is still meager compared to other ASEAN countries such as Singapore (1.94%; data for year 2017), Malaysia (1.44%; 2016), Thailand (1.00%; 2017), and Vietnam (0.53%; 2017) and is way below the world average of around 2.27% (World Bank, 2020)⁶.

Even so, there has been a continual increase of research funds in recent years by looking at the GERD to GDP ratio. During the period 2013 – 2018, the percentage increased in a row; 0.09% to 0.14% in 2014 (up by 0.05 points), 0.20% in 2015 (up by 0.06 points), 0.25% in 2016 (up by 0.05 points), and 0.26% in 2017 (up by 0.01 points), becoming 0.28% in 2018 (up by 0.02 points) (*multiple sources*⁷).

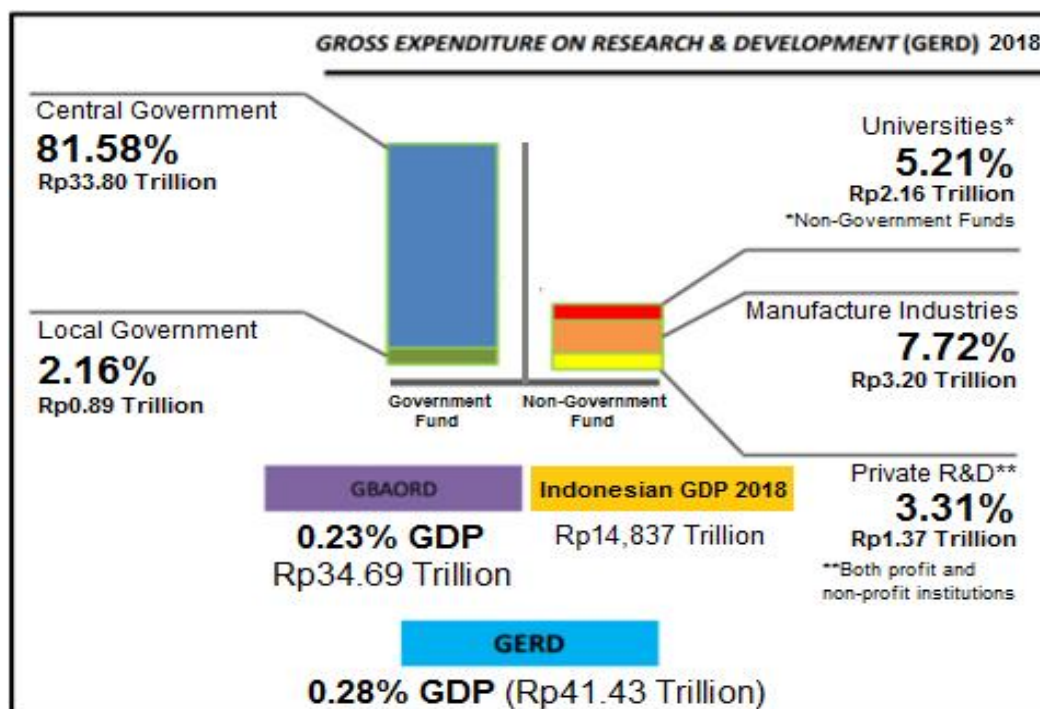
Regarding the sources of funding, about 83.73% of the GERD in 2018 or Rp34.69 trillion (0.23% of GDP) comes from government funds, both from the central and regional government budgets. The total of non-government R&D spending was only 16.27% of the GERD, with around 7.74% coming from the manufacturing industry, 5.21% from the university's non-government funds, and 3.31% from private R&D institutions (Ministry of Research and

⁶ data.worldbank.org (accessed July 2020).

⁷ data.oecd.org (accessed July 2020); data.worldbank.org (accessed July 2020); Ministry of Research and Technology/BRIN (2019); and Ministry of Research, Technology, and Higher Education (2017).

Technology/BRIN, 2019).

[Figure 4-1] Indonesia's Gross Expenditure on R&D for Year 2018



Source: Ministry of Research and Technology/BRIN (2019)⁸

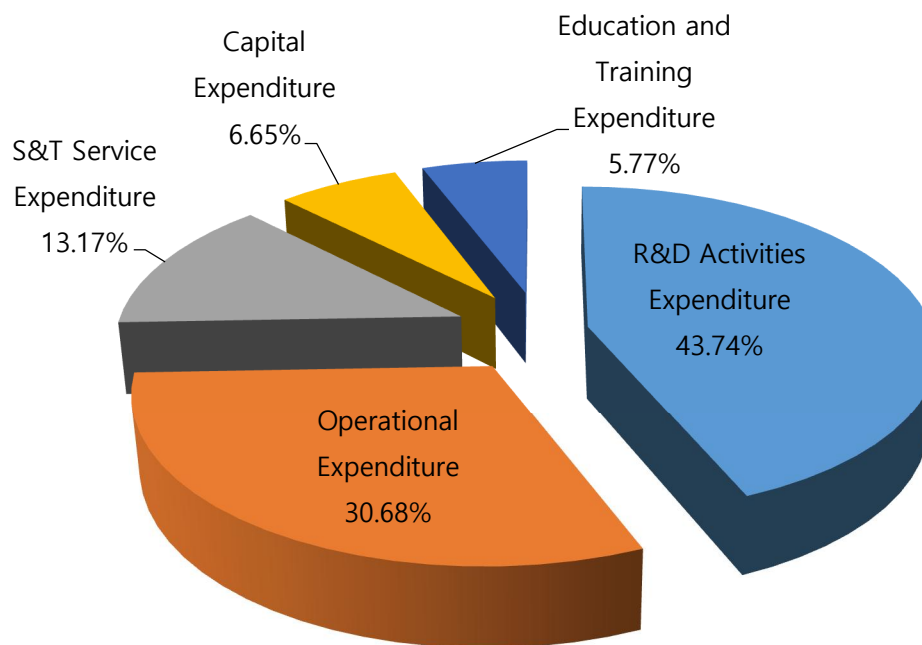
In addition, based on the data released by the Ministry of Research, Technology, and Higher Education⁹ (2017), from the total GERD of Rp30.78 trillion (0.25% of PDB) in 2016, only approximately 43.74% of the budget is allocated for research activities. The rest is used for non-research activities.

⁸ Presented by the Deputy for Strengthening Research and Development – MoRT/BRIN, November 21, 2019 in Jakarta

⁹ There was a change in nomenclature from the Ministry of Research, Technology, and Higher Education to the Ministry of Research and Technology/BRIN in line with the change of government in 2019. The function of higher education was integrated back into the Ministry of Education and Culture. BRIN, *Badan Riset dan Inovasi Nasional* (in Indonesia); National Research and Innovation Agency (in English).

- Operational expenditure: includes employees' salaries and operation of R&D (30.68%)
- Science and technology service expenditure: includes science and technology information services, data collection, conservation, testing and standardization, patent and license work, and dissemination of R&D results (13.17%).
- Capital expenditure: includes R&D laboratory building and research infrastructures (6.65%)
- Education and training expenditures, including workshops and other competency development activities (5.77%)

[Figure 4-2] Allocation of R&D Expenditure in Indonesia for Year 2016



Source: Ministry of Research, Technology, and Higher Education (2017)¹⁰

By looking at the numbers above, it can be concluded that government funds still dominate and serve as the primary sources of funding for R&D activities in Indonesia. Industries' lack

¹⁰ Presented by the Deputy for Strengthening Research and Development – MoRTHE, September 2017, in Jakarta

of willingness to innovate through R&D activities is still a major problem that has yet to be resolved. Industries in general consider research activities to be less attractive because of their nature of requiring a substantial investment value with high degree of uncertainty of success in the market.

On the other hand, a variety of innovative products that have been proven successful in the market, whether born from research activities or not, are still the first and foremost choice for industries to develop their businesses. This should not be surprising, given the main motive of the industry itself, which hopes to get maximum profit while minimizing the costs and business risks as much as possible (Aminullah, 2015). On the government side, the lack of various regulations to support and incentivize both R&D institutions and industries to collaborate, develop, and innovate through research activities is still the main obstacle that needs to be followed up.

Nevertheless, the enactment of Law No. 11 in year 2019 concerning the National Science and Technology System¹¹ brings new hope for accelerating socio-economic development based on the support of science and technology. Some fundamental improvements from the previous law¹² include the arrangements of science and technology planning and supervision, institutional governance, resources, networks, funding, and services.

Like the other countries adhering to the European Continental model's legal system, however, most of the regulations contained in the law require implementing regulations to be operational. The current condition is that no implementing regulations have been set, and it may be even longer due to the outbreak of COVID-19, which shift the focus of the government to handling the pandemic. Thus, the law has not yet been fully implemented.

In line with the improvement of various matters related to the national science and technology system as explained above, LIPI pays great attention to the governance of R&D funding. The lack of R&D funding in Indonesia can be seen from the GERD to GDP ratio, which stood at only 0.28% or Rp41.43 trillion in 2018, with more than 83% coming from

¹¹ Legalized and promulgated on August 13, 2019.

¹² Law No. 18 Year 2002 on the National System of Research, Development, and Application of Science and Technology.

government funds. Only a small portion comes from the private sector.

According to Prasodjo (2012), the importance of private sector involvement in national development is based on the premise that the breadth of problems and needs in the public sector are too big to be handled by the government alone. The limited resources owned by the government in responding to the breadth of problems certainly need to be supported by the participation of the private sector in order to realize optimal development.

In the context of R&D funding, by examining the data above, industry participation has not been well-developed. Increasing the private R&D funding is an inseparable part of strengthening the national science and technology system through regulatory support and incentives, as will be regulated in the implementing regulations derived from Law No. 11 Year 2019. Still, this is not the main focus of our attention in this study. Rather, our focus is on the mechanism and governance of R&D funding including how it is managed effectively through the research endowment fund. The concept of the endowment fund itself will be explained thoroughly in Section 3.

2. Status of R&D Funding Mechanism in Indonesia

Currently, the governance of R&D funding in Indonesia is managed through two mechanisms; first and foremost is through institutional funding; and second, through project funding. The institutional funding comes with no direct selection of projects or programs to be performed (Lepori, *et al*, 2007); it is given directly by the government to researchers through R&D institutions (Steen, 2012).

In Indonesia, the institutional funding mechanism follows the national planning rule set forth in Law No. 25 Year 2004 concerning the National Development Planning System. The budgeting process is based on the provisions set forth in Law No. 17 Year 2003 concerning State Financing. Each public R&D institution receives a funding share from the state budget (APBN) ¹³ attached to the institution for their R&D activities in accordance with the programs they have planned before; we then call it Entry List of Budget Implementation (DIPA). Thus, the treatment of public R&D activities has so far followed the usual scheme of the government's goods and services expenditure. This is deemed to hamper the development of R&D activities themselves; the characteristics are rigid, administrative, and technocratic, tending to be procedural.

The second R&D funding mechanism is project funding. Funding is attributed to a group or an individual to perform an R&D activity that is limited in scope, budget, and time, normally based on the submission of a project proposal describing the research activities to be done (Lepori, *et al*, 2007) and provided through intermediary institutions such as manager of R&D funds, and usually with some degree of operational independence (Steen, 2012).

The project funding in Indonesia currently comes from three main sources: 1) DIPA of MoRT/BRIN with several program schemes such as National System of Research Incentives

¹³ APBN, *Anggaran Pendapatan dan Belanja Negara* (in Indonesia); State Budget (in English): the state budget document for a period of one fiscal year

(Insinas)¹⁴, Industrial Technology Development (PTI)¹⁵, and many others; 2) R&D fund mechanism managed by LPDP under the Ministry of Finance, with the program called RISPRO¹⁶ (existing condition; the governance of research endowment fund thereafter will be our focus in discussion); and 3) Indonesian Science Fund (DIPI)¹⁷, a research funding institution under the management of an independent, non-structural organization¹⁸ called AIPI¹⁹.

The Ministry of Research and Technology/BRIN has a small portion of the budget allocated from its DIPA to fund research programs to be carried out by research institutions through several competition schemes or by forming a research consortia. Every scheme has terms, conditions, and policies set forth in ministerial regulations, and the distribution mechanism is generally competitive (through selection based on the feasibility of the proposal). Some schemes operated to date are as follows:

- Program of Insinas: intended for R&D between Technology readiness levels (TRL) 1 – 5
- Program of PTI: intended for R&D between TRL 4 – 6
- Program of Innovation: already has a prototype, with commercialization as the main goal (TRL 9)
- Programs intended for the university's R&D:
 - Program of National Competition
 - Program of Decentralization
 - Program of Assignment (top-down)

The other is DIPI, an independent research funding institution under the auspices of AIPI

¹⁴ Insinas, *Insentif Riset Sistem Nasional* (in Indonesia); National System of Research Incentives (in English).

¹⁵ PTI, *Pengembangan Teknologi Industri* (in Indonesia); Industrial Technology Development (in English).

¹⁶ RISPRO, *Riset Inovatif Produktif* (in Indonesia); Productive and Innovative Research (in English).

¹⁷ DIPI, *Dana Ilmu Pengetahuan Indonesia* (in Indonesia); Indonesian Science Fund (in English).

¹⁸ Institutions established through certain laws and regulations to support the implementation of state and government functions, which can involve elements of government, private sector, and civil society, and financed by APBN.

¹⁹ AIPI, *Akademi Ilmu Pengetahuan Indonesia* (in Indonesia); Indonesian Academy of Sciences (in English).

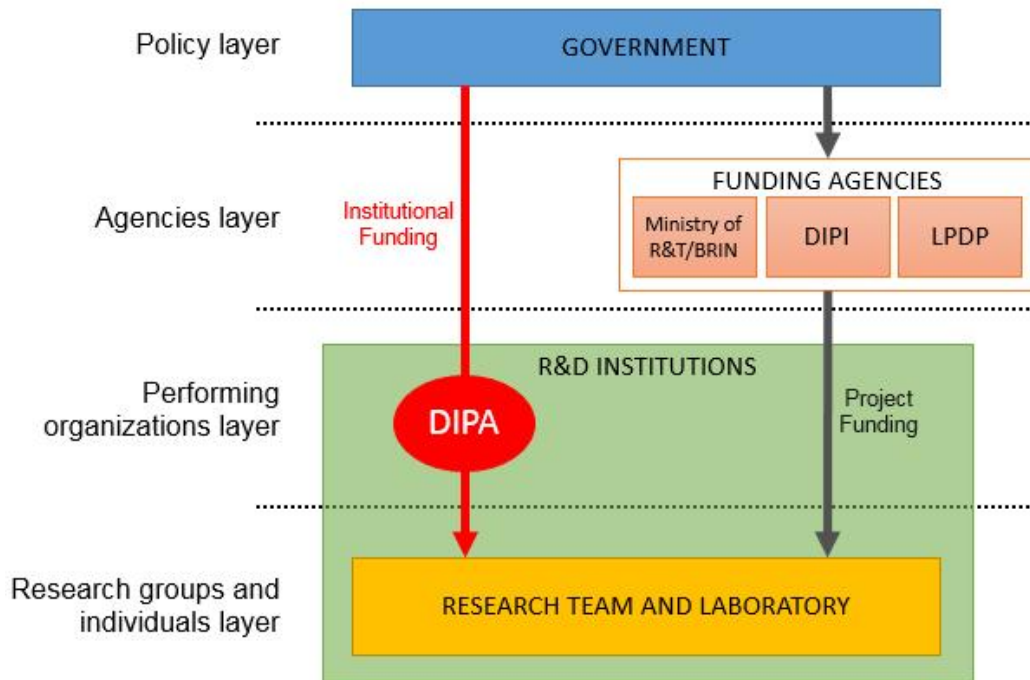
that aims to improve the national research ecosystem through competitive, flexible, and sustainable research funding. The funding scheme aims at improving the quality of cutting-edge fundamental research to build Indonesia's competitiveness on the world stage. DIPI was launched on March 30, 2016 by the Minister of Finance based on Presidential Regulation No. 9 Year 2016 concerning Amendments to the AIPI Statutes and Bylaws. The establishment of DIPI is supported by the Ministry of Research, Technology, and Higher Education (currently known as the Ministry of Research and Technology/BRIN), National Development Planning Agency, Ministry of Finance through LPDP, and several countries such as United Kingdom, United States of America, and Australia.

The establishment of DIPI is based on the results of a study conducted by Indonesian Science Academy and National Academy of Sciences of the US, i.e., funding and administration mechanisms that follow the annual cycle of APBN hamper the ability of researchers to demonstrate their best ability (DIPI, 2018). As an independent research funding institution, DIPI can raise funds from the national and international non-government sectors and manage it separately from the annual cycle of state budget. In carrying out its mission, DIPI provides research grants to Indonesian researchers through open competition based on the international standard propriety system involving national and international experts.

The other R&D project funding is run by Educational Fund Institute of Management (LPDP)²⁰, a unit under the Ministry of Finance in charge of managing the education endowment fund begun in 2010 and channeled through the mechanism of educational scholarship programs. Part of the return of investment (ROI) is allocated to fund research programs through a competitive mechanism with the program called RISPRO. The overall mechanism, regulation, and work design were managed by LPDP. To strengthen funding in the Productive and Innovative Research program, LPDP also cooperates with DIPI through collaboration schemes.

²⁰ LPDP, *Lembaga Pengelola Dana Penelitian* (in Indonesia); Educational Fund Institute of Management (in English).

[Figure 4-3] Indonesia's R&D Funding Mechanism



Source: Adapted from Lepori (2014), adjusted to the R&D funding mechanism in Indonesia

3. Governance of Research Endowment Fund in Indonesia

Since 2019, the government has allocated an endowment fund of Rp 990 billion to be used specifically for R&D activities; we call it research endowment fund. In 2020, it was agreed that another Rp 5 trillion would be allocated to this research endowment fund²¹. The nominal amount is not a value that can be directly used for R&D activities, only the investment value to be managed with the placement on deposits and state securities. Thus, the return on investment (ROI) will be a nominal value that can be used for R&D activities. The investment is managed for at least a year so that the utilization of ROI from endowment fund can be done by the following year.

The R&D budget, which is highly dependent on the annual cycle of state budget, is so far less than ideal to fund research programs. The limited ability of the government annual budget to fund research programs sufficiently on one hand, and the continuing increase in the need for R&D funding, prompted the government to launch a research endowment scheme. The existence of the research endowment fund, whose investment funds will continue to grow without having to rely on state budget, is expected to overcome the problem of lack of funding.

The next issue is related to the absence of mechanisms regarding the governance of research endowment funds. As a brand new scheme, the research endowment fund does not have a strong legal basis yet for operation at the moment. First, the mandate of the arrangement is actually stated in Article 62 of Law No. 11 Year 2019 concerning the National Science and Technology System and should be regulated further through a presidential regulation²², but the latter has not been established yet. Our study is expected to be an input of policy recommendations in the formulation of the presidential regulation.

²¹ In APBN Year 2020 (the second year), the government re-allocated Rp5 trillion, bringing the total investment to Rp 5.99 Trillion

²² More explicitly stated in Paragraph 6, Article 62

Second, arrangements related to the existing endowment fund are only set specific for education, as stated in Presidential Regulation No. 12 Year 2019 concerning the Education Endowment Fund. This regulation also becomes the main legal basis for LPDP to manage education endowment funds. As long as this presidential regulation is still valid, and without any other government policies, no single institution (including LPDP) can manage the research endowment fund due to the lack of regulations governing it. In other words, the research endowment fund does not have “a place to live” or the mechanisms, design, and arrangements to operate it. At the moment, the research endowment fund is still managed under the Minister of Finance as a State General Treasurer (BUN),²³ and the money itself is placed in the State General Treasury Account (RKUN)²⁴.

Presidential Regulation No. 12 Year 2019 is now confirmed to be in a renewal process by the Ministry of Finance. Still, the renewal focus of this regulation is to accommodate not just the research endowment fund itself specifically, but also the endowment fund for many other sectors such as cultural endowment fund and university endowment fund (both have begun to be budgeted this year), Islamic Boarding School (pesantren)²⁵ endowment fund, endowment fund for natural disasters, and many others that may emerge in the future. This new regulation is expected to be a legal basis for LPDP or any other institutions appointed/designated as “a house” to manage the endowment fund. Specific clarity regarding the governance of the research endowment fund is stipulated separately in implementing regulations derived from Law No. 11 Year 2019, and this must be in line with the renewal of Presidential Regulation No. 12 Year 2019.

3.1 Alternative Institutional Governance of the Endowment Fund

As we mentioned earlier, the research endowment fund does not have “a place to live” or the mechanisms, design, and arrangements to operate it at the moment. This is still managed under the Minister of Finance as a BUN, with the money itself placed in the RKUN.

²³ BUN, *Bendahara Umum Negara* (in Indonesia); State General Treasurer (in English).

²⁴ RKUN, *Rekening Kas Umum Negara* (in Indonesia); State General Treasury Account (in English).

²⁵ Islamic Boarding School (in English)

There have been a number of options related to the governance of research endowment fund. The latest discussion that developed in our government agreed that the endowment fund requires separating its management into three functions. Each function has its own competencies that are mutually supportive in managing the research endowment fund. The first function is responsible for managing investment portfolios. It is related to how this fund continues to develop its interest (ROI), which will then be used for R&D funding. It is good that this function is given to entities possessing competencies in investment management.

The second function is responsible for managing the R&D programs to be funded. This function is carried out by entities that have the authority with regard to policymaking, programs and research portfolios, which in this case should be the main task of the Ministry of Research and Technology/BRIN. The Ministry of Research and Technology/BRIN itself has so far had a National Research Master Plan²⁶, whose programs are spelled out in the National Research Priority²⁷ document. The last function is responsible for channeling the fund into the researchers based on the policy, research portfolios, and R&D programs set by the entity that is in the second function. This function relates to the management of research proposal selection and peer review process. The entity will determine proposals that are eligible for funding and may also decide whether the funding can be continued according to the review process.

There are three major options that are currently being raised in relation to the governance of research endowment fund.

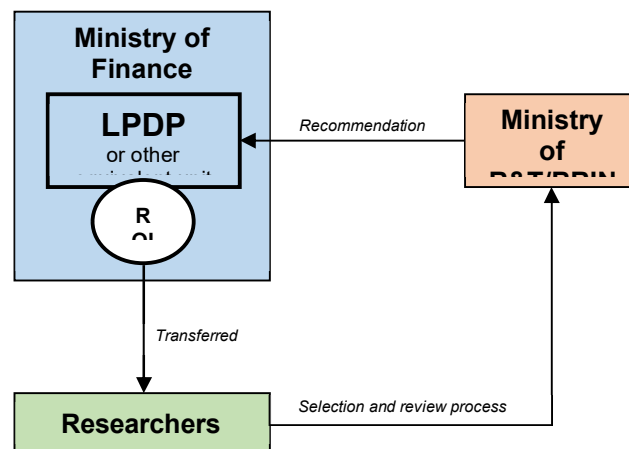
²⁶ RIRN, *Rencana Induk Riset Nasional* (in Indonesia); Master Plan of National Research (in English): document of long-term R&D programs; current document is a plan for the period 2017 – 2045.

²⁷ PRN, *Prioritas Riset Nasional* (in Indonesia); National Research Priority (in English): document of mid-term R&D programs and elaboration of RIRN; current document is a 2nd term of the existing RIRN and planned for the period 2020 – 2024.

3.1.1 Option 1: LPDP as Investment Manager and Ministry of Research and Technology/BRIN as R&D Program Manager and Fund Distributor

The investment of research endowment fund [function number one] is managed by LPDP or other equivalent unit under the Ministry of Finance. While the formulation of R&D programs and fund distribution [function numbers two and three] are fully managed by the Ministry of Research and Technology/BRIN, LPDP has the responsibility of earning a return in accordance with the predetermined targets, and the Ministry of Research and Technology/BRIN has the responsibility of utilizing the ROI to create a great R&D activity; creating great R&D programs are expected to lead to competitive selections and quality review processes. Fund management schemes can be done in two ways. First, the ROI is transferred directly from LPDP to researchers through the recommendations of the Ministry of Research and Technology/BRIN. Second, the ROI is transferred to the Ministry of Research and Technology/BRIN in order to manage it independently. This scheme can be carried out through the DIPA transfer mechanism by establishing a trust fund management unit²⁸ under the Ministry of Research and Technology/BRIN.

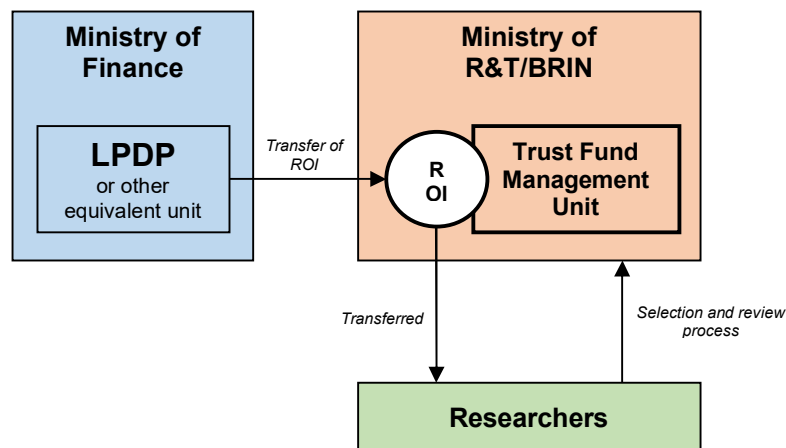
[Figure 4-4] First Way: Direct Transfer to Researchers (Option 1)



Source: developed by author

²⁸ Ruled in Presidential Regulation No. 80 Year 2011 concerning Trust Funds.

[Figure 4-5] Second Way: Establishing a Trust Fund Management Unit (Option 1)

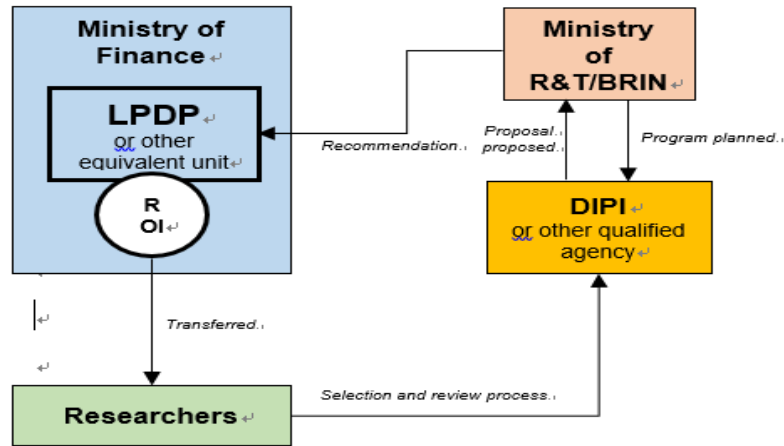


Source: developed by author

3.1.2 Option 2: LPDP as Investment Manager, Ministry of Research and Technology/BRIN as R&D Program Manager, and Independent Agency as Fund Distributor

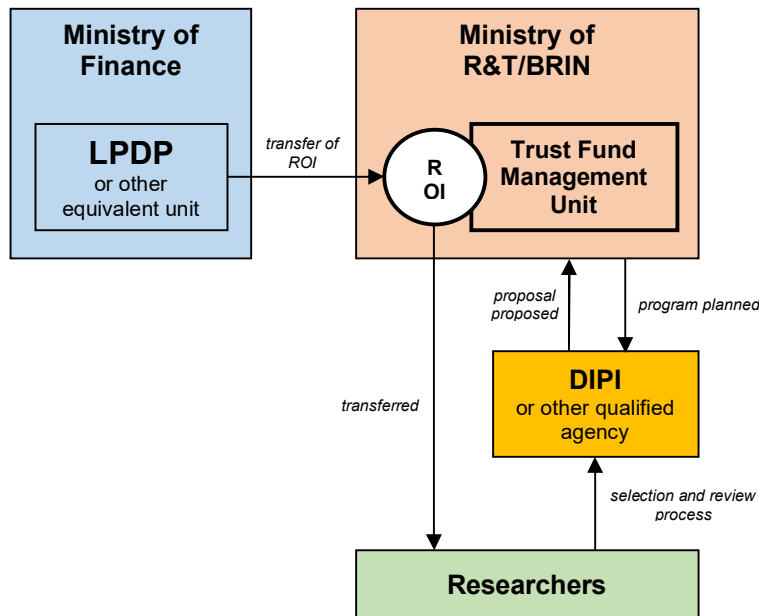
This option is similar to the first, but suggests that function numbers two and three be managed by different entities. In this case, LPDP or other equivalent unit under the Ministry of Finance remains the investment manager, with the Ministry of Research and Technology/BRIN remaining the R&D program manager. On the other hand, function number three related to the proposal selection and review process can be assigned to an independent agency outside the Ministry of Research and Technology/BRIN. This will guarantee the objectivity of funding because peer review is conducted directly by the research community. Combining function numbers two and three in one entity has the likelihood of hampering the independence of the process, because the proposal selection and review process is carried out by the policymaker itself. In this option, the government can establish/appoint an independent institution tasked with channeling the ROI to researchers, and DIPI might be the one recommended. This option can also be implemented through a direct transfer scheme or by establishing a Trust Fund Management Unit.

[Figure 4-6] First Way: Direct Transfer to Researchers (Option 2)



Source: developed by author

[Figure 4-7] Second Way: Establishing a Trust Fund Management Unit (Option 2)

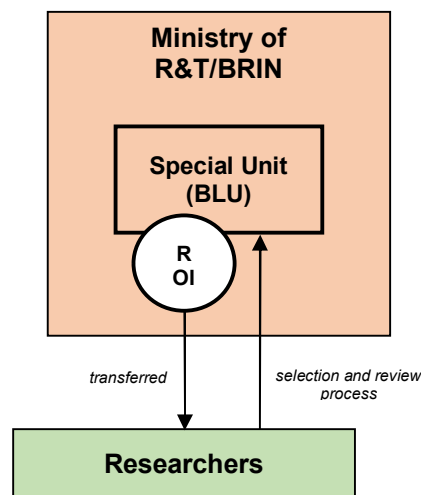


Source: developed by author

3.1.3 Option 3: Integrating all Functions under the Management of the Ministry of Research and Technology/BRIN

The last option is the integration of all functions under the management of the Ministry of Research and Technology/BRIN. In other words, investment [function number 1], policies and programs [function number 2], and proposal selection and review process [function number 3] are all run by the Ministry of Research and Technology/BRIN. The ministry will form a special unit called Public Service Agency (BLU)²⁹ under its organizational structure to manage all of these functions. In our opinion, this is the least recommended option for implementation. There is a separation of functions based on their respective competencies, and investment capability is truly not a competency possessed by the Ministry of Research and Technology/BRIN.

[Figure 4-8] Integrating All Functions under the Management of the Ministry of Research and Technology/BRIN (Option 3)



Source: developed by author

²⁹ BLU, *Badan Layanan Umum* (in Indonesia); Public Service Agency (in English): Government agencies established to provide services to the public in the form of goods and/or services that are sold without prioritizing for profit and in carrying out their activities based on the principles of efficiency and productivity. Ruled in Government Regulation No. 23 Year 2005 concerning the Public Service Agency.

[Table 4-1] Governance Options of the Research Endowment Fund

Governance Options	Option 1	Option 2	Option 3
Investment Manager	Ministry of Finance	Ministry of Finance	Ministry of R&T/BRIN
R&D Program Manager	Ministry of R&T/BRIN	Ministry of R&T/BRIN	
Selection and Review Process		Independent Institution	
Opinion by Author*	-	Most Recommended	Least Recommended

Source: developed by Author

*This neither reflects nor represents the opinion of other parties

3.2 Evaluating the Research Endowment Fund

The next question that needs to be addressed in the governance of research endowment fund is, “Who will be responsible for evaluating the program implementation and funded projects?” Currently, the evaluation of R&D programs and its projects in Indonesia’s public R&D institutions still follows the usual scheme of government goods and services expenditures, which focuses on achieving budget absorption, suitability of use of the budget between rules and the realized implementation, and achievement of R&D outputs (such as scientific technical reports, journals, proceedings, patents, etc.) according to the rigidly planned target. This is because the funding is attached to APBN, whose mechanisms and procedures are tied strictly to prevailing laws and regulations³⁰.

The independent management of ROI from the research endowment fund outside the annual cycle of APBN is now being sought. Therefore, the opportunity to use it as funding for R&D activities that are flexible, dynamic, and feasible for long-term research projects (not fixed on APBN) is widely open to be realized. Evaluation should be directed at achieving R&D programs and projects that are based on quality with cost-benefit analysis, leading to

³⁰ At least: Law No. 25 Year 2004 concerning the National Development Planning System and Law No. 17 Year 2003 concerning State Financing

results that have an impact on science, technology, and innovation and practically for sustainable socio-economic development.

By looking at the entities involved in the governance of research endowment fund as described in section 5, there are two alternative options related to the entity that can play a role in evaluating the research endowment fund.

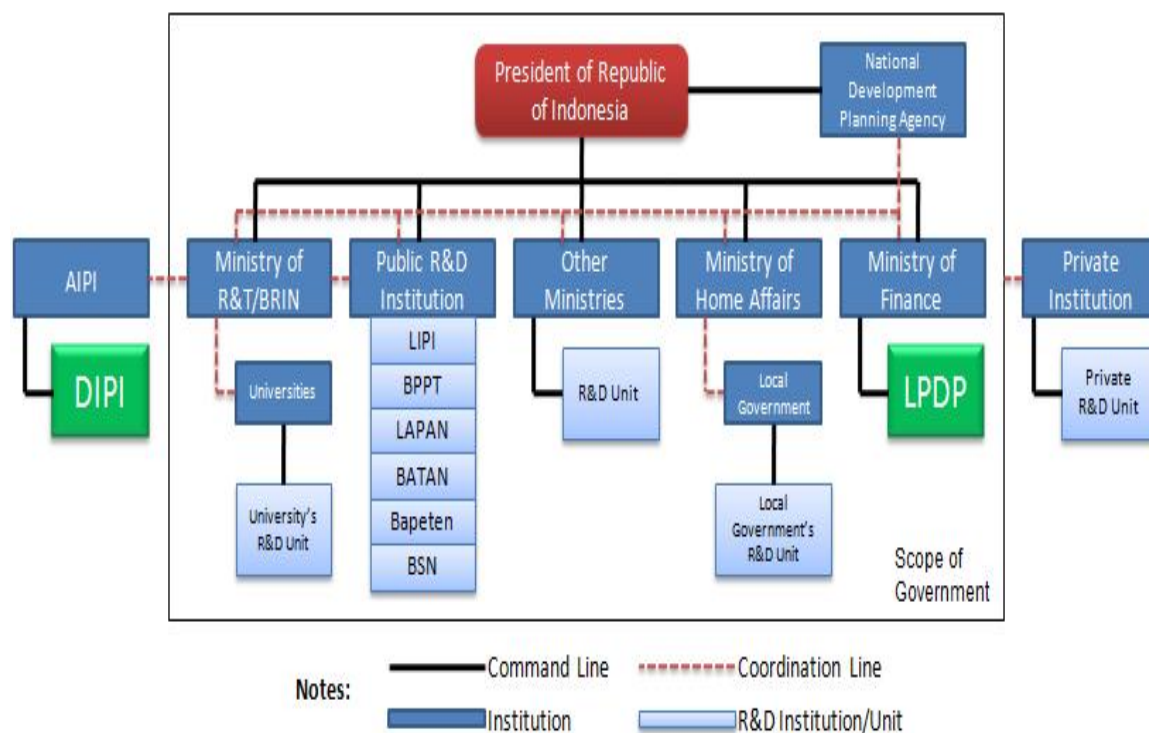
Option 1: The evaluation process can be carried out by the Ministry of Research and Technology/BRIN. As an actor in charge of managing R&D affairs in the government, the Ministry of Research and Technology/BRIN has the responsibility of ensuring that the various programs and policies implemented can support R&D activities, leading to the achievement of national development goals. The Ministry of Research and Technology/BRIN has adequate capacity and capability in conducting evaluations, supported by qualified resources.

Option 2: The Ministry of Research and Technology/BRIN can assign an independent institution to conduct evaluations. This option is based on the view that evaluations carried out by an independent institution other than the policymakers themselves can guarantee a more objective evaluation process. This evaluation becomes an input for the Ministry of Research and Technology/BRIN in making plans for the next program. In this case, AIPI might be the one relied on to evaluate programs and projects funded by the research endowment fund. This should be a better choice than the previous one.

4. Governance of Public R&D Institutions in Indonesia

There are two types of R&D organizations: 1) public R&D institutions directly under the president, coordinated by the Ministry of Research and Technology/BRIN (one of them is LIPI); and 2) R&D organizations under the ministries/institutions, universities and in local governments. There are at least six research institutions directly coordinated by the Ministry of Research and Technology/BRIN (four of them are in specific sectors), and almost every ministry/institution has an R&D unit within it.

[Figure 4-9] R&D Organization in the Government Structure of Indonesia



Source: developed by author

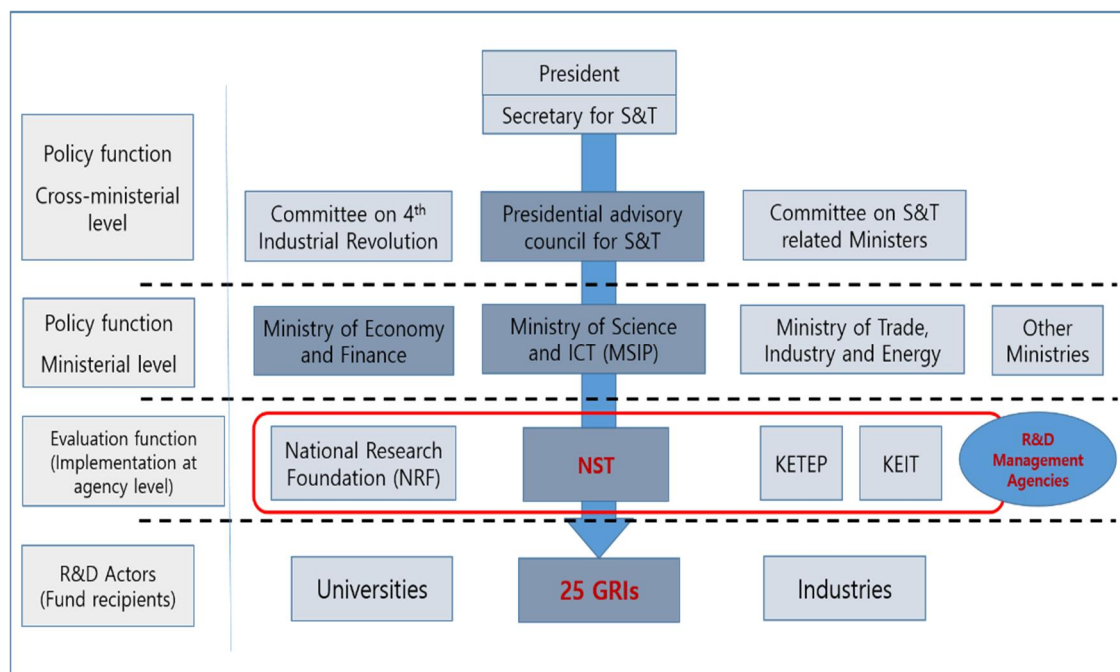
The coordination of R&D programs occurred only among research institutions coordinated by the Ministry of Research and Technology/BRIN. Nonetheless, it does not run optimally as the Ministry of Research and Technology/BRIN has no strong authority to make the research institutions it coordinates run R&D programs in accordance with the national R&D programs. This is because the authority with regard to the R&D program and its funding, DIPA entirely rests in each institution.

This results in programs overlapping among institutions, even competing with one another instead of collaborating. Such poor coordination has also led to the scattering of R&D funds in various research institutions without a clear direction toward the national R&D program objectives. The improvement of R&D institutional governance is part of the latest law of the national science and technology system, which is currently in the process of drafting its implementing regulations. Several governance options of public R&D institutions onward will be explained in the section about Policy Recommendations.

5. Korea's Experience in the Governance of Public R&D Institutions

The six public research institutions in Indonesia can be compared with government-funded research institutes (GRI) in Korea. GRI refers to a research institute wherein a certain portion of its operating resources come from government contributions. As of 2020, there are 25 GRIs in the field of science and technology. They are under the jurisdiction of the National Research Council on Science and Technology (NST) based on the "Act on the Establishment, Operation, and Promotion of Government-funded Research Institutions in the Field of Science and Technology."

[Figure 4-10] Governance Structure for GRIs in Korea



Source: processed by author

Historically, the 25 GRIs were classified into 3 groups in 1999, then into 2 groups in 2008, and finally into 1 group in 2014. Currently, the National Research Council of S&T, which is independent from government ministries, is managing all of the 25 government research institutes.

[Figure 4-11] 25 S&T-related GRIs in Korea

Fundamental S&T (11 GRIs)		Applied S&T (14 GRIs)	
	Korea Institute of Science and Technology		Korea Institute of Industrial Technology
	Green Technology Center-Korea		Electronics and Telecommunications Research Institute
	Korea Basic Science Institute		National Security Research Institute
	National Fusion Research Institute		Korea Institute of Civil Engineering and Building Technology
	Korea Astronomy and Space Science Institute		Korea Railroad Research Institute
	Korea Research Institute of Bioscience and Biotechnology		Korea Food Research Institute
	Korea Institute of Science and Technology Information		World Institute of Kimchi
	Korea Institute of Oriental Medicine		Korea Institute of Geoscience and Mineral Resources
	Korea Research Institute of Standards and Science		Korea Institute of Machinery and Materials
	Korea Aerospace Research Institute		Korea Institute of Materials Science
	Korea Atomic Energy Research Institute		Korea Institute of Energy Research
			Korea Electrotechnology Research Institute
			Korea Research Institute of Chemical Technology
			Korea Institute of Toxicology

Source: NST internal document; STEPI (2019), p. 119.

The vision of NST is to develop an open collaboration ecosystem for 25 GRIs, and its mission is to contribute to national R&D policymaking and S&T development including convergent technology by supporting and fostering GRIs. The main roles and strategy for realizing those vision and mission: i) establish a supporting and fostering system for GRIs; ii) facilitate connections between GRIs and industry; and iii) develop convergence technology between GRIs (STEPI, 2019, p.114).

To achieve this mission, NST conducts an annual institutional evaluation of GRIs in terms of research performance and institute management. The evaluation results are reflected on the coordination of research areas across the GRIs and improvement of each GRI's research

management system.

It should be noted that the influence of NST's institutional evaluation on budget allocation to GRIs is not significant. NST has a role as an administrative channel that collects each GRI's budget request and delivers it to MSIT, which in turn reviews the budget request with PACST from a national perspective and delivers it to MOEF. The final decision on the actual budget allocation is made by MOEF. To summarize, budget allocation to each GRI is determined through the order of GRI → NST → MSIT → PACST → MOEF as shown in Figure 4-11. The details of the budget process are as follows:

The annual budget of the 25 GRIs of NST is around 5 billion dollars. The institutional funding, as grant-in-aid, accounts for an average of 42% and comes from the Ministry of Economy and Finance, with a limited number of researchers and staff decided annually. The PBS projects from government departments constitute more than 50%. At least 5% of the budget comes from other sources including industry and self-earning finance. The institutional budget has to be submitted by GRIs to the NST Board of Directors and to MSIT. This goes to MOEF through the reviews of PACST at the end of June. MOEF submits the total annual government budget including R&D investment to the National Assembly, which then decides the budget until the end of December (STEPI, 2019, p.116).

6. Policy Recommendations

Three options were proposed, each of which relates to how to allocate the three functions. The three functions are: i) managing investment for ROI creation, ii) planning S&T policy and R&D programs, and iii) evaluating R&D performance and project proposals. Each of them is called investment, policy, and evaluation functions, respectively, for short. In addition, options 1 and 2 are classified according to the establishment of the evaluation function either within the Ministry of R&T/BRIN or outside the ministry as a separate evaluation agency. Option 3 is to bring all three functions and establish them within the Ministry of R&T/BRIN. Our team has mentioned in Table 4-1 that this is the most impractical option, so it will be excluded in the discussions below.

Option 1 is to delegate the investment function to specialized external investment institutions and to carry out policy and evaluation functions within the Ministry of Research and Technology/BRIN. The Ministry will assume the policy and evaluation functions altogether, whereas Option 2 is to separate both. Option 2 involves establishing an independent specialized agency outside the ministry and entrusting the evaluation work to it. External evaluation agencies are made up of civil experts from the R&D community, whereas public officials within the Ministry of Research and Technology/BRIN are responsible for policy function. By separating policy and evaluation works, the objectivity of evaluation can be improved because it can be compared to separating players and referees in a game.

For the comparison of Option 1 and Option 2, it is necessary to look first at the basic features of the three functions. Table 2 was prepared for this. In order to secure the connection between Table 4-1 and Table 4-2, the expression of the three functions was used as it is in Table 4-2.

[Table 4-2] Three Functions and Their Main Features

Function	Activities and considerations
Investment Manager (INVESTMENT)	- Financing activities for managing investment - Not directly concerned with R&D activities - No difference among policy (fund) sectors (education fund, R&D fund, etc.) -> Just creating ROI to be allocated to related fund sectors
R&D Program Manager (POLICY)	- Setting the annual sum of R&D funds to be allocated to national R&D programs and R&D institutions/units - Designing national R&D programs - Overseeing public R&D institutions (mission, performance, etc.) <- - Designing the S&T governing structure - Concerned with policymakers with formal authority (presidential office, government officials, etc.)
Selection and Review Process (EVALUATION)	- Evaluation of national R&D program performance - Evaluation of public R&D institutions/units' performance - Selection of proposals on a competitive base (organizing peer review system for review & selection) - Concerned with implementing operational tasks outlined by policymakers in the policy function area - R&D management agencies mainly composed of highly educated civil experts (PhDs/ not necessarily public officials)

Source: developed by author

Since the investment function is essentially concerned with the creation of financial return on investment, it is not directly related to R&D activities. Its mission is to create the necessary amount of R&D fund by ROI every year. Therefore, it is natural for Option 1 and Option 2 to separate the investment function.

The key question is whether to integrate or separate the policy and evaluation functions assumed by the Ministry of Research and Technology/BRIN. Here, we must take into account that the policy function is the work performed by formal policymakers, and the evaluation function is the executive work usually conducted by private experts, as shown in Table 4-2. This means that for Option 1 to work successfully, two conditions must be met:

first, to have competent human resources within the ministry who can perform evaluation tasks; second, being able to solve the inefficiency of integrating policymaking (referee) work and executive (player) work within the ministry.

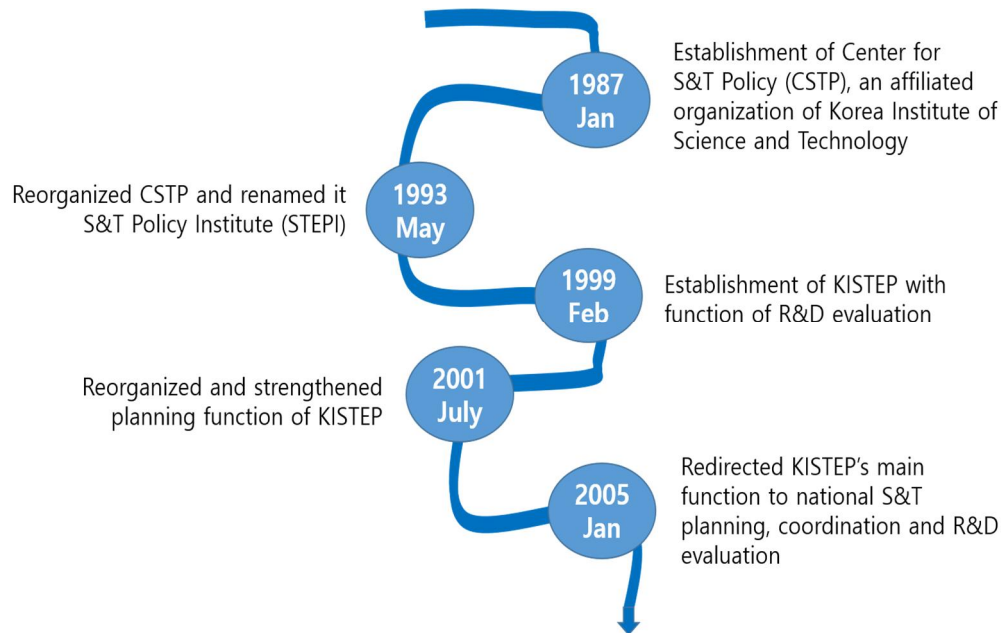
Considering these two conditions, option 2 may be a more rational approach than option 1 because the former covers these two conditions. Option 1 is an initial form before the governance structure became advanced in the form of Option 2. The barriers to the adoption of Option 2 are the selfish motive of the ministerial bureaucracy pursuing the expansion of its organization and power by internalizing the evaluation function.

Option 2 is very similar to the Korean experience. Looking back on the development of evaluation agencies in Korea, in 1987, the Ministry of Finance established the S&T Policy Center (CSTP) in KIST. As national R&D programs such as the HAN program began in earnest in the mid-1980s, the demand for evaluation work increased, so it was necessary to establish an evaluation agency.

At that time, planning and evaluation were the two main functions of CSTP. Inside CSTP were two divisions: a policy group (policy division) and an evaluation group (evaluation division). Looking at the two functions in terms of the relation between CSTP and Ministry of Science and Technology, CSTP has played a leading role in the evaluation and a supporting role in policy planning.

Subsequently, CSTP was upgraded to become an independent research institute as STEPI in 1993. Policy and evaluation functions were also the two main functions for STEPI. Then, the STEPI of 1993 was separated again into the remaining old STEPI and a new KISTEP in 1999. STEPI has kept focusing on policy work as it became independent from the Ministry of Science and Technology and got attached to the Research Council under the Office of the Prime Minister. KISTEP continued to remain under the Ministry of Science and Technology, taking charge of evaluation. KISTEP was reorganized in 2001 to strengthen the policy planning function; in 2005, the national science and technology planning and coordination function was added.

[Figure 4-12] Functional Expansion of an Evaluation Agency in Korea



Source: KISTEP (2017), p. 8.

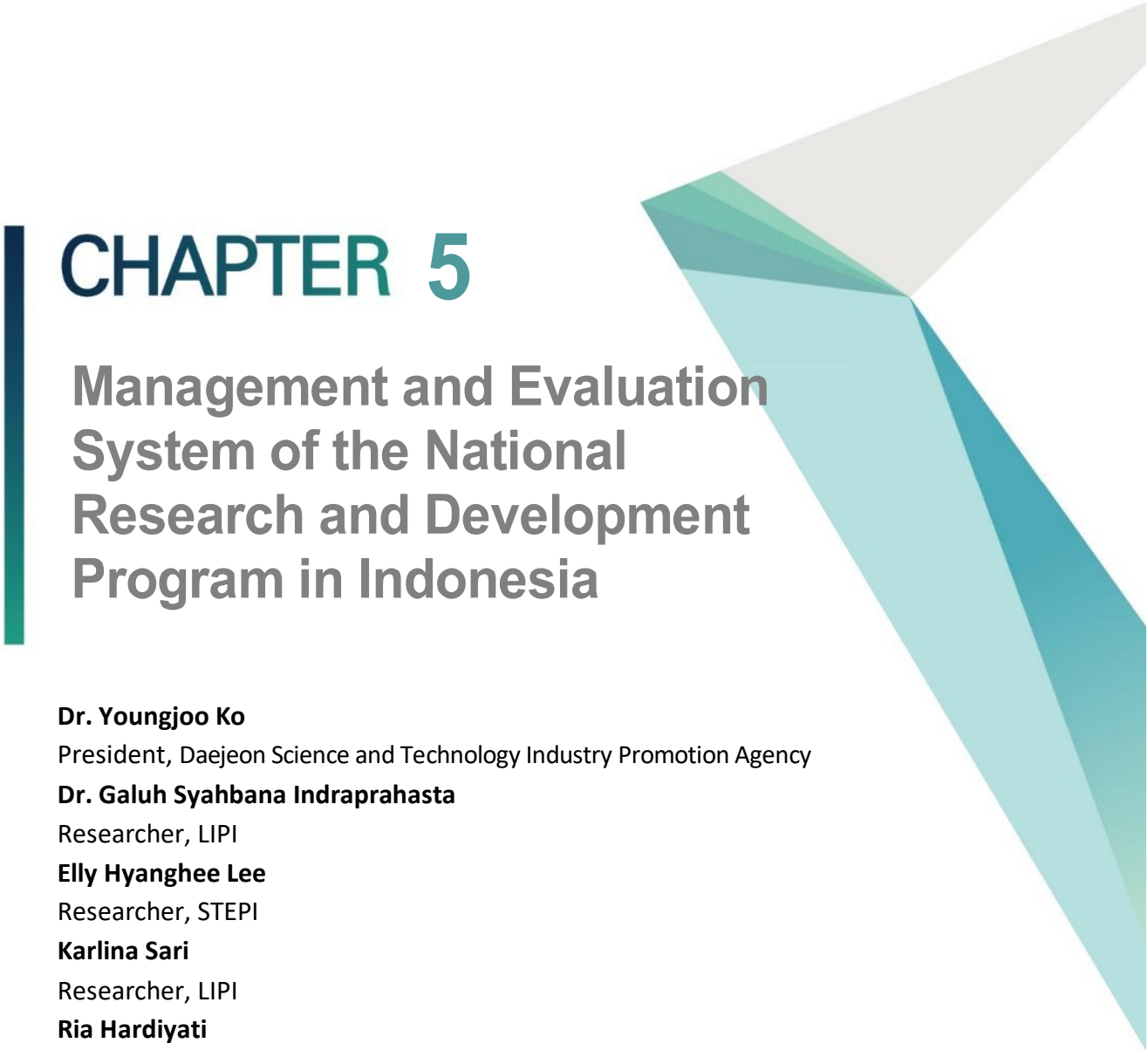
In this vein, there is some redundancy in the policy planning functions between STEPI and KISTEP. The difference is that KISTEP is closely connected with the Ministry of Science and Technology, whereas STEPI—as an independent organization from the Ministry of Science and Technology—keeps wider connections with various S&T-related ministries throughout the government.

In summary, Korea has established an independent evaluation agency outside the ministry from the initial stage, and external agencies such as CSTP, STEPI, and KISTEP have supported the Ministry of Science and Technology.

In the early stage of system building, who plays a leading role in the evaluation function may be of less importance. It could be either inside of the Ministry or a separate external agency. From a dynamic point of view, however, as the system becomes more complex and evolves with the increase of government R&D programs and increase in the size of the budget, evaluation agencies should be specialized with staff of professional training in economics, statistics, and technology since it is difficult for public officials to figure out the details of

evaluation work.

In addition, there is a very strong relationship between evaluation and policy functions. The policy planning should be based on empirical data and statistics generated from evaluation activities. This means that the evaluation agency collects and accumulates the data necessary for policymaking. For this reason, KISTEP has continuously expanded the policy function as well as the evaluation function as shown in Figure 4-12.

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CHAPTER 5

Management and Evaluation System of the National Research and Development Program in Indonesia

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Chapter 5. Management and Evaluation System of the National Research and Development Program in Indonesia

1. Introduction

This section aims to provide an overview of Indonesia's current R&D policy and a brief evaluation of the National Research Priority (*Prioritas Riset Nasional* [PRN]) 2020-2024. In doing so, we first perform this analysis within a brief historical narrative of the country's R&D policy and, to a certain extent, Indonesia's development. We believe that the STI direction of a country relates to its overall development progress and trajectory.

The R&D policy of Indonesia post-crisis can be seen as part of its envisioning of its national development within the changing environment, i.e., global economic integration and democratizing Indonesia. In terms of the first landscape, it is inevitable for any country to dodge from the global world. Economic globalization (and free market) can be viewed both as an opportunity and as a threat. Knowledge and technological transfer is one opportunity that can be gained by emerging economies; thus reaping benefits from the global world. Nonetheless, it is necessary for every country to have the right absorptive capacity to tap such global know-how. In addition to this, countries should be more competitive to be able to compete at the global level; thus entering the global market. In the free market mechanism, products from different parts of the world can penetrate the domestic market of a certain country more easily, thereby possibly hindering its domestic product (and technological) development.

In the context of democratizing Indonesia, several studies also show that inter-sectoral integration has become a challenging issue. R&D has so far been seen as a sectoral development and has not been utilized optimally as a result. As we will elaborate later in the discussion, efforts have been devoted to integrating this "sectoral" agenda into national development. The last two sections of this chapter will then specifically evaluate the country's recent R&D policy.

2. R&D Policy and National Research Program in Indonesia

As has been put forward by Aminullah (2019), the development of Indonesia's STI capacity and policy can be traced back to the 1970s, which is within the earlier part of the term of President Soeharto, Indonesia's second president. During his presidency, the Indonesian government mainly encouraged this capacity building through the establishment of several strategic state-owned industries and import substitution policy. Thus, R&D and later innovation were expected to be capitalized by these "national" industries. Once recognized as one of the emerging Asian tigers, Indonesia had been forced to shift its trajectory due to the economic crisis that badly hit most of the East Asian countries in 1997-1998. This unforeseeable economic (and later political) turmoil had direct implications on a wide variety of sectors. Many (if not most) of strategic state-owned industries were restructured, and some of them were even dissolved. Similarly, the direction of Indonesia's STI policy had changed substantially. It took almost one decade for Indonesia to recover from this crisis; after going through such period of recovery, the country finally had the opportunity to start building a new national system of development planning, notably marked by the promulgation of the National Long-term Development Plan (RPJPN)³¹ 2005-2025 and the first National Mid-term Development Plan (RPJMN)³² 2004-2010). This new development plan framework has hitherto been used as a development guide across different government sectors, both at the national and regional (provincial and local) levels.

In general, the recent (particularly over the past 10 years) R&D Policy in Indonesia can be divided into two policy settings. Before 2017, despite the existence of the so-called "White Paper" for Research, Development, and Application of Science and Technology, the country's R&D Policy was basically a medium-term policy and was more or less an

³¹ *Rencana Pembangunan Jangka Panjang Nasional in Indonesian.*

³² *Rencana Pembangunan Jangka Menengah in Indonesian.*

elaboration of the National Mid-term Development Plan (RPJMN)³³. With the issuance of the National Research Master Plan (RIRN)³⁴ 2017-2045, Indonesia has two layers of R&D policy. In this respect, while RIRN acts as the basic reference for Indonesia's R&D policy in the long term, the National Research Priority (PRN) is the medium-term R&D policy, elaborating RIRN. To understand the path dependency and path creation of Indonesia's R&D policy, the following discussion will briefly outline the landscape before 2017 first. We then zoom in on the new system afterward.

2.1 (Before 2017) National Research Agenda

2.1.1 Background

The government budget for supporting research, development, and application of science and technology is still very limited, and this limited budget should be used effectively to achieve the desired goals. In doing so, the research, development, and implementation of science and technology activities carried out by universities, ministerial, non-ministerial, or private R&D institutions as well as the community need to be directed toward certain topics that support the national agenda, i.e., economic competitiveness. For this reason, ARN was then formulated as a guide for conducting research, development, and application of science and technology in the country.

ARN is deemed to be very important because it becomes the basic guide and instrument for coordination and building of research consortia as well as a guide for conducting research activities consisting of different actors: researchers, engineers, academe, businesses, and government. Coherence in the context of direction and priorities of research is needed so that all efforts to support mastery and utilization of science and technology as well as to allocate the research budget can be carried out more effectively and efficiently.

The formulation of ARN is mandated by Law 18/2002 on the System of Research, Development, and Application of Science and Technology. According to this law, the

³³ *Rencana Pembangunan Jangka Menengah Nasional in Indonesian.*

³⁴ *Rencana Induk Riset Nasional in Indonesian.*

National Research Council (DRN)³⁵ has the main responsibility of formulating the direction and priorities of science and technology development. Since 2019, however, this law has been replaced by Law 11/2019, leading to several changes in the country's R&D landscape. For instance, a striking difference outlined by this law is that DRN has no explicit role and function in the new research agenda formulation, which we will elaborate further in some of the succeeding parts of this Chapter.

2.1.2 Scope

The scope of research area in every ARN period varies according to the focus mandated in RPJMN. The following table shows the scope of research area within each period:

[Table 5-1] Research Area of ARN

2006 – 2009	2010 – 2014	2015 - 2019
Food Security	Food Security	Food and Agriculture
New and Renewable Energy	Energy, New and Renewable Energy	Energy, New and Renewable Energy
Transportation and Management Technology	Information and Telecommunications Technology	Health and Medicine
Information and Telecommunications Technology	Transportation and Management Technology	Transportation
Defense Technology	Defense and Security Technology	Telecommunications, Information and Communication
Health and Medicine Technology	Health and Medicine	Defense and Security Technology
-	Advanced Material	Advanced Material
-	-	Social and Humanity

Source: DRN (2006), pp. 9-10; MORT (2010), p. 7; DRN (2016), p.4.

³⁵ *Dewan Riset Nasional in Indonesian.*

2.1.3 Parties Involved

As noted earlier, DRN is the institution with the main responsibility of formulating ARN as mandated by Law 18/2002. During the formulation process, DRN also formed a technical committee that consists of several relevant stakeholders, including industrial actors, government agencies, universities, and (public) R&D institutions. DRN also had a series of plenary sessions with the Parliament members (DPR)³⁶ during the formulation process.

2.1.4 Socialization, Operationalization, and Evaluation

ARN was socialized by DRN to all parties involved in research, development, and application of technology, including policymakers with the function of supporting science and technology development, both at the national and regional levels, most notably the National Development Planning Agency (NDPA/Bappenas), Regional Development Planning Agencies (BAPPEDA), National Representative Council (DPR), and Regional Representative Councils (DPRD).

In the latter process, ARN was operationalized through two approaches:

- As a reference of research agenda for R&D agencies under ministries, public R&D institutions, non-ministerial government institutions, regional governments, universities, and business actors so that these institutions can use their respective funding schemes according to the ARN direction;
- As a guide to implementing the research priorities carried out in the form of consortia. These consortia used funding originating from ministries / non-ministerial government institutions, grant schemes from the Ministry of Research, Technology, and Higher Education (MORTHE), and other sources such as the Endowment Fund for Education (LPDP), Palm Oil Plantation Fund Management Board (BPDP), and other legal funding.

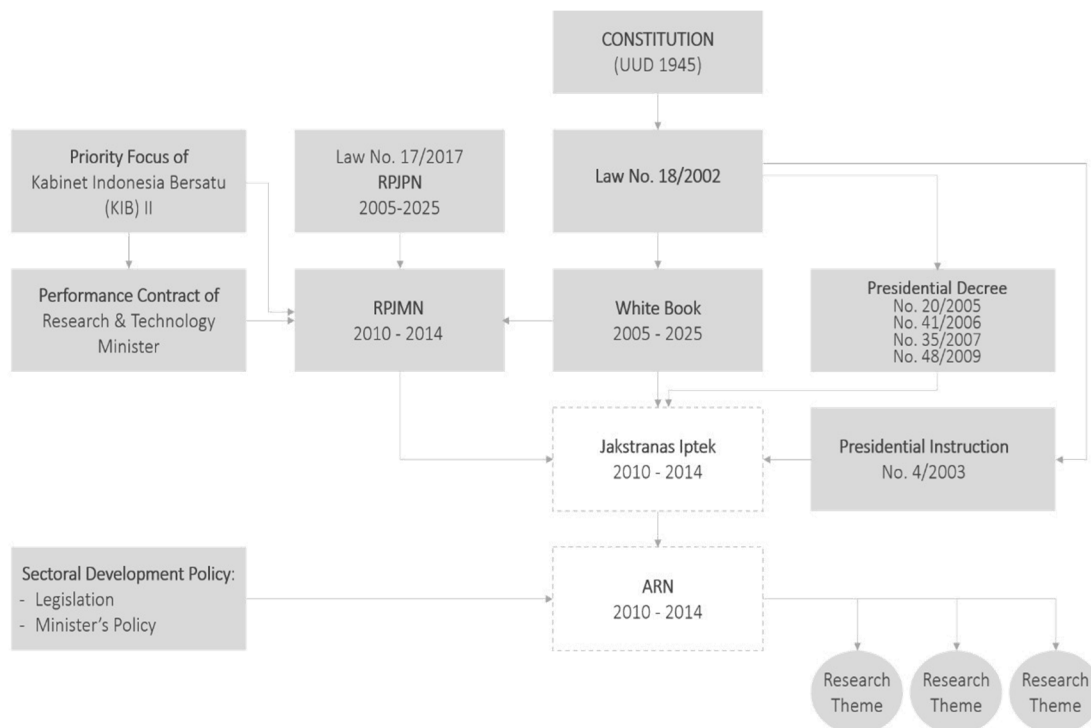
³⁶ *Dewan Perwakilan Rakyat in Indonesian.*

The last process, i.e., the ARN evaluation process, both in the context of ARN's implementation through research agenda and research priorities, was carried out by MORTHE with support from DRN.

2.1.5 Policy Structure

The formulation process of ARN refers to the direction of science and technology development as mandated in the RPJMN document (DRN, 2006, 2016; MORT, 2010). In addition to this, ARN was also prepared in line with several related policies such as the United Indonesia Cabinet (*Kabinet Indonesia Bersatu* – under the lead of President Susilo Bambang Yudhoyono) II Priority Focus, Minister of Research and Technology Performance Contract, and Strategic Policy for the National Development of Science and Technology (Jakstranas Iptek) (National Research Council, 2010). The science and technology policy structure where ARN is situated can be illustrated by the following figure:

[Figure 5-1] R&D Policy Structure Before 2017



Source: MORT (2010), p. 5.

2.1.6 Issues Related to ARN and DRN

Based on a study conducted by Oey-Gardiner (2011), there are a couple of important issues regarding ARN and DRN:

- ARN was theoretically aimed at functioning as a national reference for all Indonesian researchers in conducting their research projects. Nonetheless, it seems that the use of ARN as a reference remains minimal;
- The role of DRN was also weakened, as can be observed by the shift in its reporting hierarchy. Previously, DRN should report directly to the President of Indonesia; now, however, they only report to the Minister of Research and Technology (formerly known as the Minister of Research and Technology and Higher Education).

2.2 RPJMN 2015-2019: Science and Technology Section

2.2.1 Focus

The science and technology development plan only appeared in RPJMN 2010 - 2014 and 2015 - 2019, and it was discussed in Book 2 specifically in Chapter 4. In this chapter, the objectives, policy directions, and strategies are discussed. There are several differences in terms of development strategies between RPJMN 2010 - 2014 and 2015 - 2019:

- The 2010-2014 RPJMN strategy is focused on development priorities, namely the Strengthening of the National Innovation System that serves as a vehicle for science and technology development toward the vision of science and technology development in the long term, and the Improvement of Science, Research, and Application of Science and Technology (P3 Science and Technology) carried out in accordance with the directions outlined in RPJPN 2005-2025.
- Unlike the previous RPJMN, the science and technology development strategy developed in the 2015-2019 RPJMN tends to be more detailed and specific for each of its development policy targets and directions. The RPJMN of this period in the direction of science and technology development policy is seen as outlining the duties and

functions of the Public Research Institutes under the Ministry of Research and Technology (MORT). Thus, the development strategy also specifically mentions the assignment of each PRI.

The target of science and technology development is to increase the capacity of science and technology, which can be described as follows:

- a. Increased results of conducting research, development, and application of science and technology that support the following:
 - Competitiveness of the goods and services production sector;
 - Sustainability and utilization of biological and non-biological resources; and
 - Preparation of the Indonesian people for global life.
- b. Increased support for science and technology activities including the provision of human resources, infrastructure, institutions, and networks.
- c. Establishment of 100 Techno Parks in districts/cities and Science Park in each province.

2.2.2 Institutions

The general RPJMN preparation process was under the coordination of Bappenas, receiving inputs from the ministries / agencies and industries. Specific to the science and technology sector, the drafting process involved and collected data and information from PRIs under the coordination of MORT, namely LIPI, BPPT, LAPAN, BATAN, BAPETEN, and BSN, as well as the Eijkmann Research Institute. The document also mentions the relevant ministries (including the Ministry of Agriculture, Ministry of Health, Ministry of Industry, BPOM) because they act as regulators, not as participating research institutions. The science and technology development strategy merely compiled the assignments of these seven PRIs (including Eijkmann) without mentioning the coordination with the relevant ministries / technical institutions.

2.2.3 Obstacles for Implementation

As a national planning document, the objectives, policy direction, and science and

technology development strategies tend to be too detailed and apparently standalone. The documents do not specifically relate the science and technology field with other fields, especially industry. This has resulted in a lack of coordination between the seven implementing PRIs and the relevant technical ministries. In other words, the development plan in the field of science and technology is not well-integrated.

The Presidential Staff Office in 2016 evaluated RPJMN 2015 - 2019 and found a gap in the President's vision and mission (i.e., so-called “Nawacita” under President Joko Widodo). Basically, RPJMN is a translation of the President's vision, missions, and programs. This happened because RPJMN 2015-2019 was formulated at the time of the previous presidential period (President Susilo Bambang Yudhoyono), and the succession team of President Joko Widodo was not actively involved in the preparation process. Another obstacle is that the drafting process did not actively involve the industrial sector in order to find this sector's needs. Therefore, the development plan in the field of science and technology merely departed from the interests of the seven PRIs. In relation to the previous obstacle, another basic thing in the RPJMN document is that the determination of indicators of science and technology program achievements cannot be used to measure/assess the success of science and technology activities in supporting the competitiveness of the production sector.

2.2.4 Policy Structure

Before RIRN was enacted in 2017, RPJMN was established through a Presidential Regulation with reference to RPJPN (Presidential Regulation), outlining the President's vision, missions, and programs. Since the 2017 RPJMN should refer to the National Research Master Plan (RIRN)³⁷ 2017-2045, however, RIRN serves as a guide in the planning and implementation of the science and technology sector for the period 2017 - 2045.

³⁷ *Rencana Induk Riset Nasional in Indonesian.*

2.3 (After 2017) National Research Master Plan 2017-2045

2.3.1 Background

The evaluation of the White Paper/Book for Research, Development, and Application of Science and Technology 2005-2025, the National Strategic Policy in the Field of Science and Technology (Jakstranas Iptek) 2010-2014, and ARN 2010-2014 shows that these policies have not yet become a reference for research activities conducted by various ministries and public research institutions. Therefore, it has been argued that the country needs a sectoral master plan that is more structured and which has more legal strength in the form of RIRN with a time range of 2017-2045.

Thus, RIRN was prepared as the main reference for the planning research sector on the national scale. It complements the existing national output-oriented planning system in each ministry/non-ministry institution. This document is expected to be an operational guide for planning and evaluation for all stakeholders in the national scope. As an integrated document in research planning, RIRN considers and includes several previous documents related to research and development planning such as ARN, Jakstranas Iptek, 2015-2019 National Medium-Term Development Plan Books I and Books II, and Strategic Plan of the ministries/non-ministry institutions (Renstra K/L) concerned. This document was also prepared with regard to planning in the industrial sector (RIPIN: National Industrial Master Plan 2015-2035), National Energy Policy (KEN), and creative economy sector (National Creative Economy Master Plan [RIEKN]).

RIRN's Vision for 2017-2045 is "Competitive and Research-based Sovereign Indonesia in 2045." "Competitive Indonesia in 2045" implies that research becomes the main engine to produce inventions and innovations that ultimately have an impact on increasing the nation's competitiveness. On the other hand, "Research-based Sovereign" implies that RIRN is the starting point for establishing a socio-economically independent Indonesia through mastery of and competitive advantage in advanced science and high technology globally. To achieve the vision above, the mission of the 2017-2045 RIRN is:

- a. Creating an innovative Indonesian society through science and technology;

b. Creating a nation's competitive advantage in the global world through research.

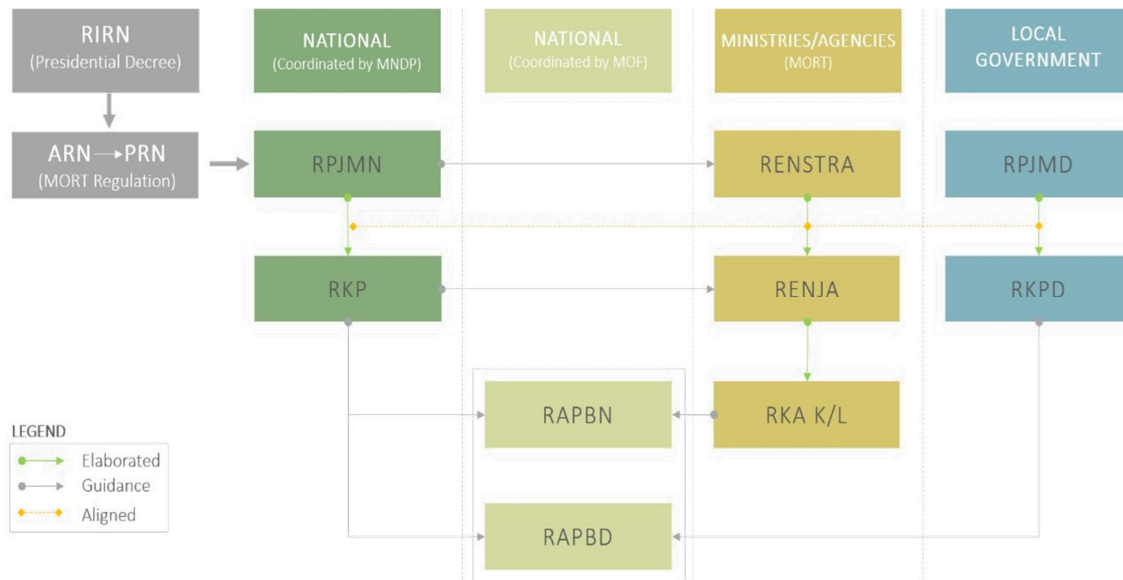
Based on the vision and mission, the objectives of the 2017-2045 RIRN are:

- Increasing the community's scientific and technological literacy;
- Increasing the capacity and competency of Indonesian research in the global sphere;
and
- Improving the economy through science and technology.
- To achieve this goal, the 2017-2045 RIRN targets are set as follows:
- Increase the quantity and quality of human resources related to research that is able to compete globally;
- Increasing the relevance and productivity of research and the role of stakeholders in research activities; and
- Increase the research contribution to national economic growth.

2.3.2 RIRN Position in the National Planning System

Unlike the positioning of ARN (which refers to RPJMN), RIRN is to be used as the umbrella referred to by RPJMN through its more operationalized plan: PRN [Figure 5-2].

[Figure 5-2] RIRN within the National Development Plan Framework



Source: MORTHE (2019), p. 4.

In addition to the current positioning of RIRN, this master plan is expected to have the following strategic role and function:

1. As a bridge to connect the long-term and annual developments
The focus of RIRN is on research priorities based on the current real power and capacity mapping. Therefore, RIRN is elaborated in the form of a National Research Priority with a 5-year period that includes focus areas expected to produce innovation products for a maximum period of 5 years. The preparation of RIRN is expected to serve as a bridge between the ideals of national development and operational steps that are evidence-based policy.
2. Synergize the development of national research
The development of RIRN is expected to build a synergy of national research, which not only improves efficiency but also increases its effectiveness.
3. As a means of reintegration of higher education and research
RIRN will encourage the harmonization of research activities in universities with the community and other research institutions.

2.3.3 Target and Focus

South Korea's condition is used as a benchmark to set targets in 2040 based on the fact that South Korea during this period was one of the countries reaching the peak of a science and technology-based economy. In addition, the demographic scale is not too far from Indonesia. A period of 32 years since 2017 is considered adequate for Indonesia to catch up with South Korea today. The 5-year target indicators are shown in [Table 5-2].

[Table 5-2] Five-year Targets

Indicators	2015	2020	2025	2030	2035	2040	2045
Researcher ³⁸	1,071	1,600	3,200	4,800	6,400	8,000	8,600
GBAORD/GDP (%)	0.15 ³⁹	0.21	0.42	0.63	0.84	1.05	1.26
GERD/GDP (%)	0.20	0.84	1.68	2.52	3.36	4.20	5.04
Ratio of researcher candidate ⁴⁰	5.6	20	40	60	80	90	100
Researcher productivity	0.02 ⁴¹	0.04	0.08	0.10	0.14	0.18	0.22
Multi-factor productivity (MFP) (%)	16.7	20	30	40	50	60	70

Source: MORTHE (2017), p. 34

As a further elaboration of RIRN, more technical planning was done in the form of National Research Priority (PRN) for a 5-year period (2015-2019). The following are the focus research priorities in this PRN 2015-2019 and some issues within each priority:

³⁸ Person(s)/million population

³⁹ Data in 2014, including R&D employee's salary, total allocation

⁴⁰ Ratio of master & doctoral students to bachelor students

⁴¹ Data in 2014, SCImago citation index

- Food and Agriculture: National food self-sufficiency has not yet been achieved
- Energy - New and Renewable Energy: High subsidies for certain types of fuel oil and electricity for certain consumers, which burden APBN
- Health and Medicine: Maternal and child health challenges; community nutrition / malnutrition; infectious diseases that are still prevalent (56%), such as dengue, malaria, diarrhea, and AIDS; non-communicable diseases that tend to continue to increase, such as cancer, heart disease, high blood pressure, and diabetes; environmental sanitation, especially those related to drinking water supply; and mental health that is quite common, such as depression that results in behavioral disorders.
- Transportation: Trade and transaction activities will be widely carried out through global networks; the role of transportation infrastructure will be very important, along with information and communication technology, in facilitating global and regional economic movements.
- Information and Communication Technology: The development of telecommunications in Indonesia has entered a new phase with the rapid development of the information technology industry.
- Defense and Security: The development of Indonesia's national defense force is an unavoidable necessity in addition to building a reliable TNI equipped with defense equipment; there is also a need to build a National Police that is equipped with special security equipment that is sophisticated and compliant with the standards.
- Advanced Materials: Advanced materials have now become a basic need in various advanced industries, and even such advanced materials have become one of the indicators of a country's industrial progress.
- Maritime: The maritime sector's extraordinary potential has not been optimally transformed as a source of progress and prosperity for the people of Indonesia.
- Disaster: In terms of geographical, geological, climatological, and demographic factors, Indonesia is a region prone to disaster risk.

- Social Humanities - Cultural Arts – Education: Encouraging science and technology and economic development as an inclusive innovation for national development, which also includes aspects of cultural development

The entire focus of this research is in accordance with the seven focuses in the ARN 2015-2019, plus three new focuses in accordance with the data obtained.

2.3.4 Actors

The RIRN arrangement was led by the Ministry of Research, Technology, and Higher Education, involving various stakeholders. Various actors are involved with regard to the strategic macro-policy to support national research activities, as shown in [Table 5-3].

[Table 5-3] Strategic Macro-policy to Support National Research Activities⁴²

Policy	Actor(s)
APBN allocation commitments to fulfill 25% of the country's contribution to research (GBAORD / GDP)	MoF/NDPA, MRTHE
Tax reduction of research budget allocations for private industry	MoF
Pure research budget as part of CSR funds	MoF
Integration and increase in the amount of budget for research on ISF	MoF, ISRFS, ISF
Venture capital incentives	MoF, FSA, CEA
Incentives for the establishment of venture company	Mol, MoCS, MLHR, CICB

⁴² MoF: Ministry of Finance; MRTHE: Ministry of Research, Technology, and Higher Education; NDPA (BAPPENAS): National Development Planning Agency; ISRFS (LPDP): Indonesian Scholarship and Research Funding Services; ISF (DIPI): Indonesian Sciences Fund; FSA (OJK): Financial Services Authority; CEA (BEKraf): Creative Economy Agency; Mol: Ministry of Industry; MoCS: Ministry of Cooperatives and SMEs; MLHR: Ministry of Law and Human Rights; CICB (BKPM): Capital Investment Coordinating Board; A3T (BPPT): Agency for Assessment and Application of Technology; IIS (LIPI): Indonesian Institute of Sciences; NNEA (BATAN): National Nuclear Energy Agency; NIAS (LAPAN): National Institute of Aeronautics and Space.

Policy	Actor(s)
Evaluation and revitalization of research grant schemes of the government (research, dissemination, infrastructure, institutional strengthening)	MRTHE
Incentives for using domestic innovation (domestic content level), both physical and intellectual components	MoI, MoT, MRTHE
Incentives for research collaboration with global partners	MRTHE, Non-ministry institutions /ISF, ministerial R&D agency
Establishment of and support for infrastructure centers of joint research at universities and R&D institutions based on their competencies	MRTHE
Realization of research infrastructure grant schemes for infrastructure center of joint research	MRTHE
Integration of research implementation in R&D institutions based on their domains	A3T, IIS, NNEA, NIAS, ministerial R&D agency
Integration and increase in the number of post-graduate scholarships through ISRFS	MoF, MRTHE, ISRFS
Incentives for domestic post-graduate study for fresh graduates	MRTHE, ISRFS
Research incentives for ex-diaspora and foreigners	MRTHE
Evaluation for increasing incentives for researchers and related professions (lecturers, researchers, engineers, various supporting institutions)	MRTHE, NNEA, A3T, IIS
Realization of post-doctoral schemes for PhD-qualified candidate researchers	MRTHE
Realization of research-based post-graduate programs in R&D institutions together with universities	MRTHE
Deregulation of IPR (Patent, Domestic Copyright, Plant Breeders' Right)	MLHR

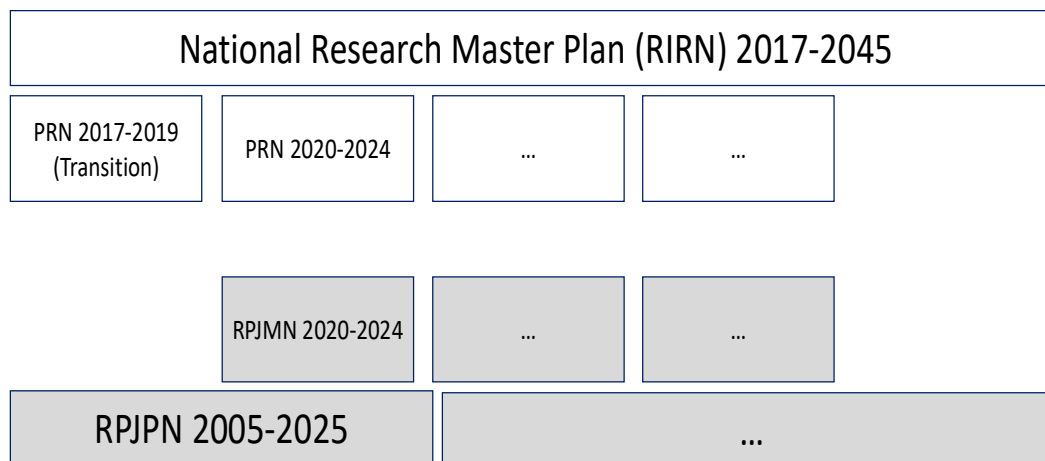
Policy	Actor(s)
Establishment of incubation centers (TP/STP) in various regions according to their potential	MoCS, MRTHE, MoI, CEA
Special funding scheme for dissemination (global-indexed publications) and refreshment (research visits, visiting ex-diaspora/ foreigners)	MRTHE
Determination of standard output indicators according to global standards for all research actors in all lines and domains	MRTHE, A3T, IIS
Globalization of domestic scientific activities (global-indexed conference)	MRTHE, A3T, IIS
Implementation of royalty system for government innovators	MoF

Source: MORTHE (2017), pp. 38-40.

2.4 National Research Priority 2020-2024

While RIRN 2017-2045 serves as the basic reference for Indonesia's R&D policy in the long term, the National Research Priority (PRN) 2020-2024 elaborates this master plan with a 5-year plan. Except for the first PRN (2017-2019), the periodization of PRN itself is adjusted to the national mid-term development plan (RPJMN) [Figure 5-3]. In this section, we do not focus on discussing PRN 2015-2019 (or some label this as PRN 2017-2019) as this PRN is a transition one (from the ARN framework to the RIRN framework). It should be noted here that the discussion of PRN 2020-2024 in this section is largely based on MORTHE Decree 38/2019 on PRN 2020-2024. Therefore, the discussion below does not capture the empirical execution of this mid-term plan. Rather, we aim to provide several ideal expectations for this mid-term policy as literally outlined by the decree.

[Figure 5-3] Periodization of PRN and RPJMN



Source: developed by author

As noted earlier, PRN is the operational plan of RIRN, so it consists of more concrete and detailed directions. In addition to several national-level documents, such as RIRN itself and White Paper of Research and Technology, PRN 2020-2024 also refers to research programs in different ministries, research institutions, and universities. Another important reference for PRN 2020-2024 is “Nawa Cita,” which is the vision of current Indonesian President Joko Widodo. PRN 2020-2024 is particularly directed to support the 6th Nawa Cita: “Increasing the Productivity of the Indonesian People and the Country’s Competitiveness in the International Markets.” So basically, the development of R&D and innovation is expected to contribute to the national economic growth and improvement of people’s welfare.

In terms of research focus, PRN 2020-2024 has 9 focuses. Nonetheless, these focuses are slightly different from those mentioned in the previous PRN’s direction. Each research focus has its own target(s) to be achieved within the next five years. The following points briefly outline this particular matter:

- Food: It is expected to be able to produce superior types of food commodities (agriculture, horticulture, plantations, and livestock) that are adaptive to certain conditions.

- Energy: It is expected to be able to produce and utilize renewable energy sources, such as clean fuels based on new and renewable energy and electricity technology based on new and renewable energy.
- Health: It is expected to develop and apply the following: (a) production technology for medicinal preparations (based on natural raw materials) and drug raw materials for import substitution; (b) medical equipment and instrumentation technology to reduce dependency on imports; and (c) technology of biological raw materials based on natural resources, stem cells, and conditioned stem cell medium.
- Transportation: This research focus deals with infrastructure and means of land, sea, and air transportation to increase capability, safety, reliability, and competitiveness.
- Engineering: This research focus includes the following: (a) technology to increase the added value of salt products; (b) handling, processing, and packaging technology for food products; (c) building construction technology for disaster mitigation, prevention & preparedness; (d) ICT technology development; (e) development of advanced materials for health; (f) textile technology development; and (g) development of machine technology for industry.
- Defense and Security: This research focus is primarily aimed at supporting the implementation of policies of the National Defense Industry Policy Committee (KKIP), including the development of defense support technology and mobility support technology.
- Maritime: Research in this area focuses on (a) strengthening of infrastructure and maritime connectivity and (b) protection and utilization of maritime resources.
- Social Humanities – Education – Art and Culture: This research focus is aimed at the synergy of science and technology development where the resolution of problems related to technology development research (hard technology) is in line with solving social and humanities problems, including evaluating the national development policy system. In turn, the research carried out is useful for strengthening community institutions and overcoming problems of access to resources and addressing inequality

and poverty, including strengthening national social capital through innovative curricula as well as cultural arts research that supports tourism as a driver of the national economy. Research in education and culture is directed at expanding access to and improving the quality of education, developing innovative curriculum and learning as well as computer-based education assessment models, and promoting culture and developing language.

- **Multi-discipline and Cross-sector:** This research focus is aimed at conducting research activities that are considered important to address various national strategic issues and issues that require approaches from various disciplines and sectors, such as disaster, stunting, and climate change.

Each of these research focuses in PRN 2020-2024 is further detailed into several research themes, each of which is then broken down into several research topics. Collaboration is expected to occur within each topic as the decree explicitly denotes the participating institutions (i.e., the coordinating institution and its member institutions) for each research topic. Still, the institutions stated in this PRN document seem to be a preliminary picture. For instance, in many research topics, universities and business entities are often listed, but not their specific names.

Another important thing that can be extracted from this decree is that PRN 2020-2024 also provides opportunities for research projects that are not directly related to the research focuses above as long as these projects can support strategic development in certain ministries. Nonetheless, it is not fully clear whether ministerial-based research should also be carried out in a collaborative fashion or not.

3. Status of Monitoring and Evaluation System of the National Research Program in Indonesia

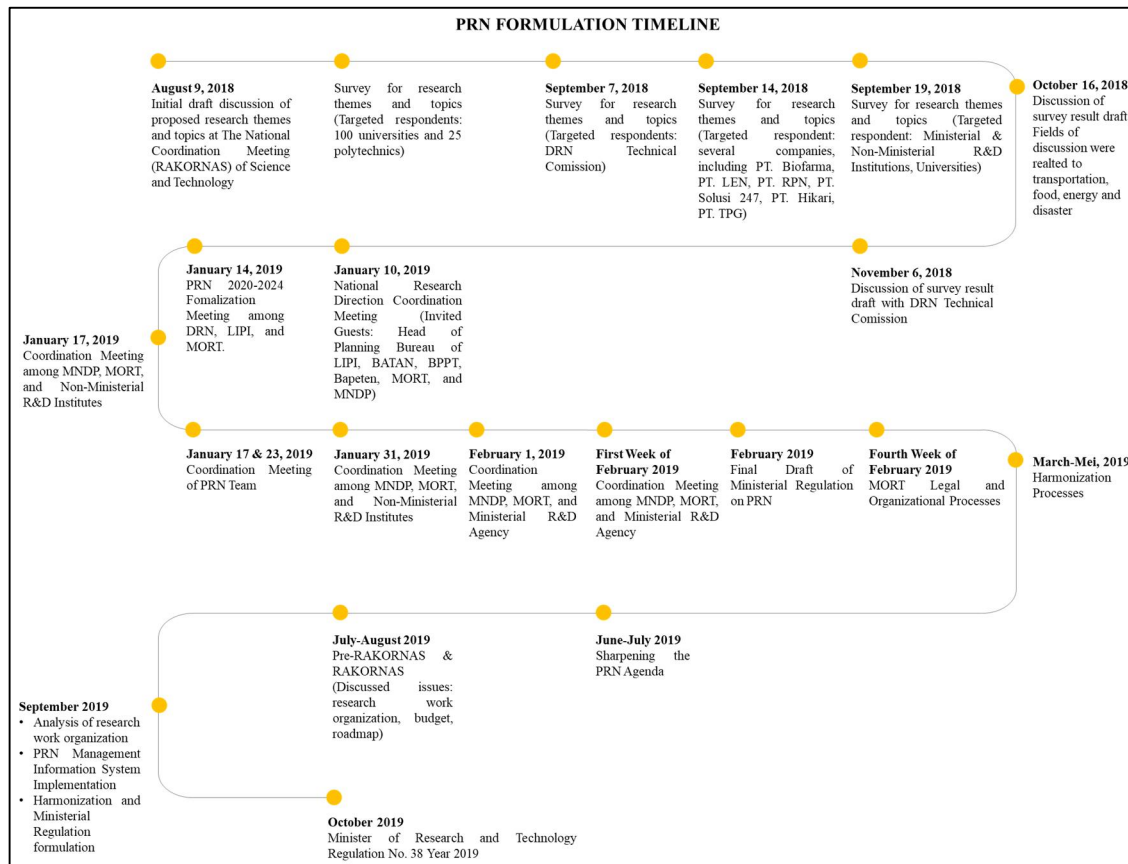
As noted earlier, a new R&D research framework was proposed less than 5 years ago, and its full implementation has just started in 2020. In order to grasp its potential values fully and anticipate its possible pitfalls, it is necessary to investigate the policymaking process of Indonesia's National Research Program or what, in the Indonesian context, is formally called the PRN. Here, we focus on the first mid-term period of PRN, i.e., PRN 2020-2024, as we could call here the first policy experimentation within the new R&D research framework. In doing so, we investigate this process by outlining each stage of this process, i.e., formulation, implementation, and evaluation. In our perspective, a good national research program should be supported by a good policymaking process at every stage of this "cycle": the interconnectivity of each stage and the extent to which the overall process may (or may not) be able to address the recurring issues (e.g., lack of inter-actors' networks, disjointed economic-industrial and R&D policies, etc.) and support the desired outcomes as mainly outlined by RIRN. The vignette of the policymaking process of PRN 2020-2014 below provides a starting point, which will be critically discussed in the last section. It is worth emphasizing at the outset that, given the fact that this NRP is still in its infancy stage, a comprehensive evaluation of the process cannot be carried out.

3.1 Formulation

Traditionally playing a central role in R&D governance and development in Indonesia, the Ministry of Research and Technology (MORT)/BRIN was also responsible for coordinating the entire PRN 2020-2024 formulation process. The initial process of PRN formulation, starting at the end of 2018 as shown in the PRN Formulation Timeline in Figure 1, discussed the details of research themes and topics of PRN 2020-2024 through several meetings with related stakeholders. A set of surveys were also conducted to gather different perspectives and views regarding themes and topics from these stakeholders, such as universities,

National Research Council, public R&D institutions, and industries. These series of discussions continued until 2019.

[Figure 5-4] PRN Formulation Timeline



Source: Adapted from MORTHE (2019), p. 8; Author's Analysis

The PRN agenda contains nine main research focuses: food; energy; health; transportation; engineering; national defense & security; maritime; social, law, cultural art, & education; and multidisciplinary & cross-sectoral research. Most of these focuses are in accordance with RIRN 2017-2045, although the latter is more inclined toward the response to the needs and recent issues. Furthermore, each research focus is organized and broken down into several themes, programs, and products, referring to the results of the former survey that initially generated 80 themes and 416 programs. In order to narrow down and make it more specific, an analysis was performed based on the level of research importance and resources

availability along with the various national strategic documents. Thus, the final PRN Agenda contains 9 research focuses emphasizing 30 themes, 47 programs, and 49 products as shown in the following table:

[Table 5-4] Research Focus in PRN 2020-2024

No.	Research Focus	Themes	R&D Programs	Products
1	Food	1	8	8
2	Energy	3	4	4
3	Health	3	9	9
4	Transportation	1	3	3
5	Engineering	7	7	7
6	National Defense and Security	4	4	4
7	Maritime	2	2	4
8	Social, Law, Cultural Arts, Education	5	5	5
9	Multidisciplinary & Cross-Sectoral (Disaster, Biodiversity, Stunting, Environment, Water Resources, Climate) Research	4	5	5
TOTAL		30	47	49

Source: MORTHE (2019), p. 13; Author's analysis

In general, the PRN research agenda has been formulated via both top-down and bottom-up approach. The top-down process refers to the context through which MORT was formulated based on inter-ministerial meetings and on RPJMN as the national inter-sectoral development guidelines. In this case, the Ministry of National Development Planning (MNDP/ Bappenas) was also involved in the PRN formulation process as it is the sole authority having the responsibility of formulating and coordinating the implementation of RPJMN. In this sense, MNDP is responsible for ensuring the conformity of the national research agenda to the RPJMN framework. Meanwhile, the bottom-up process represents the research proposals and needs provided by several stakeholders.

Although MORT has tried to administer a bottom-up process during the formulation process, it seems that this process was not running optimally as indicated by the minimum

involvement of the industrial actors, i.e., limited room to gather their ideas or assess their needs. A similar condition also applied to universities as these actors apparently have no sufficient and independent budgets to be allocated for PRN 2020-2024. This makes the identification and determination of needs and research agenda less optimal; thus relying more on the perspectives and capabilities of researchers, particularly those coming from PRIs.

Furthermore, because PRN has an integrated research agenda, the research teams consist of various institutions and agencies, and the same is true for budgets. Although research under the PRN agenda is to be funded mostly by state budget, its source does not incessantly originate from MORT. In fact, it depends more on other institutions or agencies involved in the research team. Therefore, the availability of funds from the members is also one of the main criteria in determining the themes, programs, products, and team members selected in the PRN Agenda.

3.2 Implementation

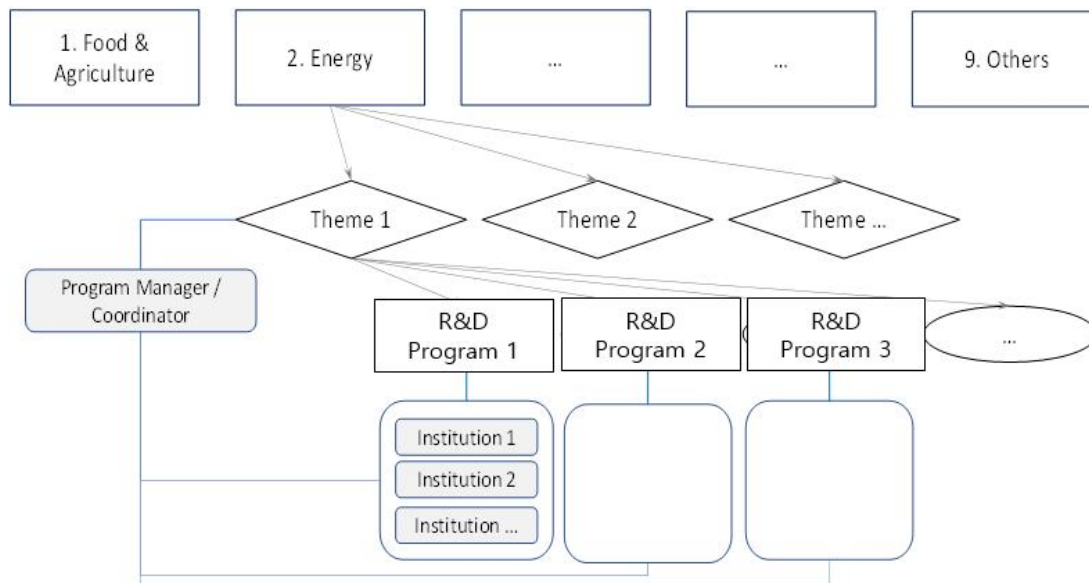
In the implementation stage, PRN 2020-2024 has a set of diverse themes and programs. The nine focus areas of PRN 2020-2024 were determined based on RIRN 2017 – 2045 and mainly used as the underlying umbrella of these themes and programs. These nine areas are Food, Energy, Health-Medicine, Transportation, Product Engineering, Defense and Security, Maritime, Social Humanities, and Multi-disciplinary Research. Each area is divided into themes, each of which is further divided into national research and development programs. Under each program, a research project is designed to be integrated (conducted by different institutions). As mentioned earlier, these 9 focus areas in PRN 2020-2024 can be divided into 80 research themes and 416 R&D programs. Of these 80 themes, 30 themes (which can be divided into 47 R&D programs) are selected as the integrated national flagship that is expected to produce 49 products. Meanwhile, the other 50 themes (which can be divided into 369 R&D programs) are institutional/ministerial flagship. Research themes and R&D programs within this national flagship are executed across different ministries and institutions (Figure 5.5). For instance, research theme 1 has a number of R&D programs, and

two or more institutions can execute each program. Each institution then carries out a specific research project, which is different from the one that other institutions implement under the same R&D program. Thus, each research project carried out by each institution can complement each other. According to our findings, although each research project (under the same R&D program) is (strongly) related, there is lack of evidence of direct collaboration (in terms of researchers' mobility, etc.) among research projects/ institutions. In other words, a researcher tends to do his/her research project under his/her own institution even as cross-institution collaboration is also encouraged.

Basically, PRN 2020-2024 consists of two schemes: the integrated national flagship and institutional/ministerial flagship. For the national flagship, the activities will be done through consortium, producing targeted products/outputs. Each theme has one coordinator (program manager), mostly from public research institutes or ministries. The main responsibility of the coordinator is to ensure that all members of his/her team can work together to achieve the desired targets. Thus, while the integrated national flagship aims to involve different actors—public research institutes, ministries, universities, and state-owned companies—the institutional/ministerial flagship has research carried out internally by each institution/ministry.

Although PRN 2020-2024 has somewhat diverse themes and R&D programs, the opposite picture applies in the context of the performers (the ones executing research projects). Research projects are mainly executed by public R&D institutions (PRIs), such as LIPI, BPPT, BATAN, and those belonging to ministries. Meanwhile, there is lack of participation from industries and universities, mainly due to the fact that industries and universities were not much involved in the formulation stage. Although some industries are indeed involved in the research collaboration, most (if not all) of these industries are state-owned companies.

[Figure 5-5] Implementation of Research Projects under the National Flagship

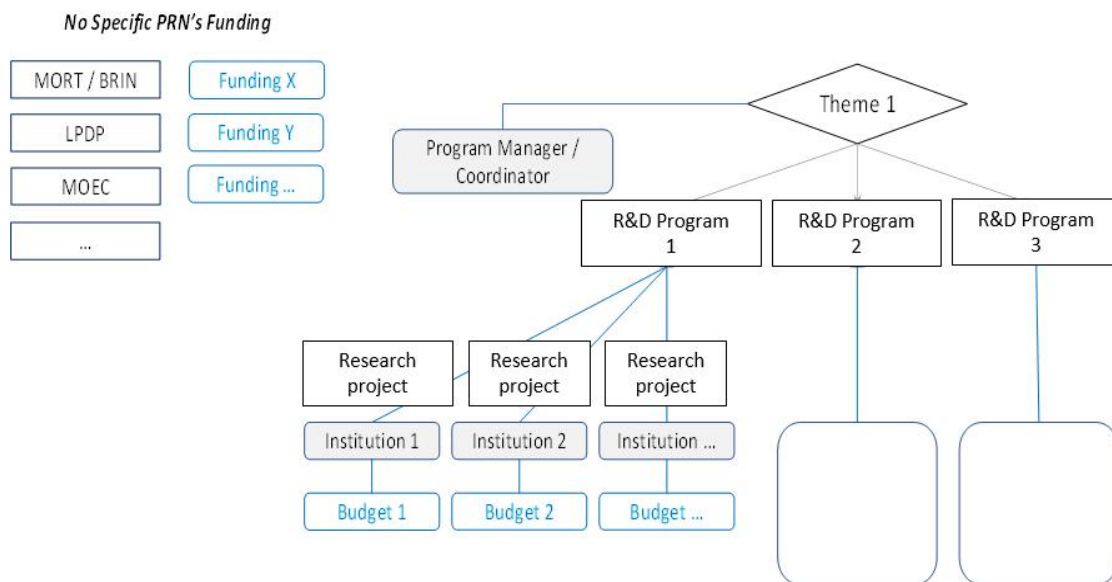


Source: Author's analysis

The budget mechanism for implementing these research projects under the integrated national flagship may add to the cacophony of the collaboration process above. One thing is for sure though—there is no specific open call for proposals particularly dedicated to PRN's projects, despite the recent funding mechanism provided by the Indonesian Endowment Fund for Education (LPDP), a semi-autonomous body under the Ministry of Finance. Every institution involved in each R&D program has to adjust its decentralization research fund/block grant to fit a particular R&D program under the PRN 2020-2024 research agenda. Researchers interested to join this research agenda have to propose their research proposals that will be selected/discussed based on each institution's own mechanism. In the context of PRIs, researchers or research groups wishing to propose their projects have to be part of institutions that are already involved in this research agenda. Based on our observations, however, new institutions (including non-government research institutions) can also join the existing networks under the approval and coordination of the program manager.

Although these institutions, mostly PRIs, initially have to utilize their own budgets to execute research projects under the PRN 2020-2024 agenda, they can also apply for funding from external sources (competitive grants) both nationally and internationally (Figure 5.6). The selection process is executed by the funder itself; therefore, the review process differs depending on the mechanism of each funding institution. Although collaboration research is intended to be carried out within a multi-year vision, the existing funding disbursement is allocated only for a one-year project, both in the context of block grant and competitive grant. Basically, the budget system—except for the case of LPDP—for research projects provided internally (by own institution) and externally (by external institutions, particularly national ones) is arranged based on the national fiscal year mechanism.

[Figure 5-6] Budget Mechanism of Research Projects under the National Flagship



Source: Author's analysis

3.3 Evaluation

According to the Regulation of the Ministry of Research and Technology No. 36 Year 2018, MoRTHE/BRIN has the authority to evaluate RIRN by assessing the target indicators of RIRN and the implementation of PRN (Article 8). The evaluation of PRN is conducted by, at the lowest level, echelon 1 officials of ministries, non-ministerial institutions, local governments, and other stakeholders at least once a year. They report the results of PRN monitoring and evaluation to the Director General of Strengthening for R&D (DGRS) under MoRTHE, and DGRS will then report it to the minister. At least once a year, the Minister makes a report of PRN implementation to the President. The result of monitoring and evaluation is used as a basis for the PRN follow-up and drafting as well as to decide whether the PRN executors should be given reward or punishment. Institutions obtaining the “reward” will be given priority to acquire research fund. On the other hand, the “punished” institutions will be subject to fund reduction, or worse, termination.

On the other hand, for RIRN performance indicators, the Minister can ask the DGRS to carry out the evaluation. He also can appoint a data and information unit under MoRTHE to collect data on the achievement indicators of the RIRN targets. The data will be utilized to evaluate RIRN performance. DGRSD should report the evaluation to the Minister at least once every 5 years, with the Minister reporting to the President. This RIRN evaluation result will be used for the next PRN drafting recommendation.

Besides the PRN and RIRN evaluation, evaluation is also conducted for each research project. MoRTHE/BRIN evaluates the research projects conducted by ministries or non-ministerial institutions, whereas the Ministry of Education and Culture (MoEC) evaluates research projects led by universities. In addition, funding agencies (both government and non-government) also have the authority to evaluate how researchers spend the research fund.

Since it has only been three years since RIRN was formulated, there is no evaluation yet on RIRN performance. Based on our interview with several PRN executors, the PRN evaluation still emphasizes the management area rather than the substantial one. The management evaluation focuses on how the budget is spent and what output can be achieved after the

research is finished. The output from the PRN program can be in the forms of product, intellectual property, scientific publication, or policy recommendation. The evaluators have to ensure that the quantity and quality of output are worth the budget spending. A standard operation procedure for the PRN evaluation has not been established, according to our interviewees. Therefore, every program coordinator has his/her own guide for evaluation.

4. Korea's Experience in the National Research Program Focusing on the Monitoring and Evaluation System in Indonesia

4.1 Formulation

While Indonesia has a 32-year long-term RIRN master plan for R&D policy and a 5-year period PRN plan to prioritize research targets, Korea produces a five-year Basic Plan for Science and Technology Development by each five-year term government. Even though the current Moon Jae-in administration set up the 4th Basic Plan for Science and Technology in Feb. 2018, which initially included a long-term vision of science and technology to 2040, there are no close relations between the 2040 vision and those 5-year target technologies in the Basic Plan. Instead, the Plan selected 120 focus technologies by sectoral expert committees for the period 2018 - 2023 based on five criteria: contribution to economy and society; contribution to science and technology; technology competitiveness; innovativeness of technology; and engagement with the Basic Plan.

The 120 technologies categorized into 11 areas: health, energy and resources, ICT and SW, construction and transportation, environment, machines and manufacturing, agriculture and food, space, materials, defense, and disaster. As the highest decision-making body of the Basic Plan, the Presidential Advisory Council on Science and Technology (NSTC) is chaired by the President, Moon Jae-in. The Ministry of Science and ICT is responsible for the entire management of process and contents supported by KISTEP, a think tank of government R&D policy, and participated by nearly all government departments.

The national R&D programs of government departments should be connected to those technology areas and agendas institutionally. Nonetheless, the actual situation is different in many departments, which usually pursue their own path-dependent trajectories. That is why the Ministries' committee to coordinate the duplication and connectedness of departmental R&D programs was established in 2018, supporting the NSTC's role.

The selection process of technology areas and relevant R&D programs and governance to decide the national plan in Indonesia are similar to those of Korea. The differences between the two countries lie not only in the function of long-term vision and mid-level coordination process but also in the selection criteria of focusing technologies and existence of a think tank. Korea needs to think about introducing the role of a long-term vision in Indonesia to overcome the short-term perspective. Indonesia may do well to establish not only more coordinated process and governance but also more rational criteria to select specific technologies and R&D programs to develop them.

Another point for more consideration is the participation of the private sector in the formulation process. Korea usually takes into account the current and mid-term needs of the industry and has it participate in all levels of the formulation process. Indonesia should actively foster the industry to create a sufficient pool of private-sector experts from the industry in the formulation process.

4.2 Implementation

The implementation process of R&D in Korea usually starts with the establishment of program plan, announcement of programs, call for proposals, establishment of selection plan, completion of reviews and selection projects, notification of selection, and finalization of agreements. The establishment of program plan is a starting point of implementation for the current year, identifying changes of strategies and improvements. The next stage is to provide a description of the types of projects to be funded on the Ministry of Science and ICT (MSIT), NRF, and NST websites. After accepting proposal submissions, agencies establish a plan for each review phase, including information about the review methods, budget allocation, and selection quota. The multi-level review of preliminary, expert, and comprehensive type results in proposal selections. The selected proposals are notified, and agreements are finalized, and research grants for them are distributed. Key review processes consist of preliminary process focusing on qualifications, expert review such as online, panel review, and presentation based on review supported by expert reviews. The first stage reviews the qualifications of the principal investigator including application

eligibility. The expert review is conducted by expert reviewers with industry-academe research backgrounds or by a separate review panel. In this stage, they usually focus on creativity, spirit of challenge, researcher capability, and achievements according to the review indicators and criteria. The next stage is review by specialist organizations for compilation, coordination, and review of selection (proposal) regarding the outcomes of expert review by the PM consultative group and preparation of the allocation (proposal) of the number of selected projects while adjusting the research grants. The program implementation committee conducts a committee review by including the comprehensive review group to finalize confirmation and to check the validity of review outcomes, selected projects, policy considerations, etc. The last stage is to notify the selection results and adjust the details of research grants and to conclude the agreement with access to the project management system and guidelines for the required documents.

The selected research project teams should submit an annual performance plan until one or two months before the termination of research period in the pertinent year, and a PM must conduct interim review. Projects that produce inadequate research performance can be discontinued, or their research grants may be reduced for the following year.

4.3 Evaluation on Final Performance

The final (phase) review is to determine continuation of support, decide the progress management results and scale of research grant, and notify the research institutions of the progress management results. The final (phase) report for review purposes and the self-review report are submitted one month before the termination of the research period. Evaluators with industry-academe-research background conduct an expert review utilizing the PM of the National Research Foundation and pool of evaluators including evaluators who participated in the selection screening. Specialist organizations then conduct a review to determine the project ranking and research grants based on scores and opinions from the review, establishing the grade (proposal) according to expert review results and maintaining statistics on review results, incentives, and penalties.

Each department has to conduct autonomous meta-review as to whether the review and process are rational and appropriate and report the review results to MSIT. KISTEP conducts annual investigation and analysis on the efficiency and effectiveness of all national R&D programs based on the multi-level reviews.

Korea's review process is highly systematic with multi-layer stages. Those kinds of review system in different stages have contributed to expanding R&D programs and investment in the fast-changing strategy toward advanced countries. Nonetheless, too many reviews are conducted, and commercialization ratio is low compared to R&D investment and number of patents. Korea has to transfer the review system with much more qualified reviews related to a more qualified index to be on a par with advanced countries. On the other hand, Indonesia needs to establish more systematic review systems with more rational evaluation index to arrive at a long-term vision approximating Korea's status.

5. Policy Recommendations for the Management and Evaluation System of the National R&D Program in Indonesia

Based on our “quick” ongoing (*ex-ante*) evaluation as well as lessons from the case of South Korea, there are a number of important issues worth emphasizing. One recurring issue that we try to discuss here is whether PRN 2020-2024 has the potential to be used as the national reference for conducting research across different institutions. This issue also relates to the main rationale of the dissolution of ARN. In this respect, PRN 2020-2024 may face the same pitfall as its predecessor did, although there is a sign of stronger commitment to use PRN as the “ultimate” reference. Based on our observation, to a certain extent, PRN 2020-2024 has been used as a reference for doing research. This particularly applies when the participating institutions commit to allocating their research budgets to address the particular research topics. This somewhat strong commitment has been formed through the formulation process that involved longer discussions and assessments, including more detailed mapping on the strength of each participating institution. This particular process should be further strengthened by the collaborative mode of doing research (within each topic) particularly in the context of the “National Integrated Flagship” (*Flagship Nasional Terintegrasi*). As mentioned earlier, however, the PRN 2020-2024 has another mode of doing research, called “Ministerial/Non-ministerial Institution Flagship” (*Flagship Kementerian/Lembaga*), and this seems to have the likelihood of repeating the “failure” as ARN was judged to be.

To the best of our knowledge, there is weaker commitment from ministries, particularly those who do not participate in the National Integrated Flagship. Although research in these ministries may fit into one or more research focus (es) as outlined by PRN 2020-2024, their outcomes and targets potentially deviate from the “national” agenda. It is true that, in essence, (non-research) programs in each ministry should support the overall national agenda. Still, the recurring issue is that these programs are often executed in a scattered manner, creating some overlaps between different ministries. A similar condition of research projects may also apply, thereby leading to inefficiency in research budget

allocation.

Another important issue relates to the applicability and benefit of the research projects within the broader national development. As previously discussed, the development of R&D and innovation in PRN 2020-2024 is expected to contribute to national economic growth and improvement of people's welfare. Nonetheless, this kind of achievement can be hampered if the needs assessment in the formulation process is not carried out comprehensively. In this respect, most of the research projects (and needs assessment) currently under implementation were mainly based on the researchers' perspective. This happens because there was lack of participation from the business entities. Therefore, it is possible that the current research products are actually not the ones that are really needed by users (industries) or other beneficiaries (people in general). Although collaboration among different institutions (particularly between research institutions) has been improved, collaboration between different types of actors (in this case, business entities and other non-governmental entities) remains limited (at least within the first year of PRN's implementation). This condition is, of course, strikingly different from the one in South Korea where needs assessment was carried out based on the actual needs of users. Still, one important thing to note here is that we also have to acknowledge the difference between Indonesia and Korea in terms of economic-industrial setting. What we mean here is that Korean industries have been developed so well so that they do need R&D to make their products more globally competitive.

The discussions above imply that R&D policy should go hand in hand with economic policy. The involvement of business entities starting from the formulation process in PRN 2020-2024 is one way of bridging this policy gap. Another means of utilizing R&D products is to formulate an economic policy that can nurture local enterprises and later using R&D in their production system to enhance their overall competitiveness. As this particular subject needs its own elaboration, further studies are recommended to investigate Indonesia's economic development and policy and the extent to which this can be related to STI development and policy. One important aspect that should be considered is ensuring the inclusiveness of economic development as, based on different studies, trickle-down effect has not fully worked in the context of manufacturing industries (particularly those that are integrated

into the global production network).

As noted earlier, collaboration (in the implementation phase) seems to be far from ideal. This is due to the fact that limited (inter-actor) collaboration was found in the formulation process. Rather, collaborative works are particularly more pronounced in the context of the researchers' dimension. This limitation should be addressed to prevent a potential mismatch between R&D's outputs and actual needs. Of course, these outputs are not specifically aimed at achieving economic goals alone but also addressing environmental and social problems. In the context of South Korea, environment-friendly and lower carbon development has been mainstreamed within the STI agenda. In Indonesia, however, the proportion may be somewhat different as its economic development has not reached a mature level, particularly in terms of industrialization.

Ideally, the evaluation process can serve as a feedback to improve the quality of R&D being conducted. Still, this process cannot be fully utilized in that certain respect given its administrative nature. There should be an evaluation system to ensure that the overall process of R&D activities can achieve both the more "technical" and "direct" targets (i.e., R&D's products and outputs) and the "indirect" desired outcomes (i.e., national economic growth and improvement of people's welfare). The establishment of index or indicators in the monitoring and evaluation system is something that can be considered accordingly. So far, substantial evaluation is conducted by the program manager /coordinator (and his/her team) without any proper guideline on how R&D activities under PRN 2020-2024 should be evaluate substantially.



CHAPTER 6

Conclusion

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Chapter 6. Conclusion

1. Policy Recommendations

1.1 Policy Recommendations for STI Think Tank

For the feasible governance of STI policy, the creation of a think tank in Indonesia should consider the existing regulations, norms, and systems. Nonetheless, creating a new think tank unit on STI policy in the broader context will face big challenges or even clash with other regulations or agency interests. Indeed, designing an STI policy think tank unit is not separated from the intent of top management, or even political will. This study provides four alternative STI policy think tank units that can be planned and designed in Indonesia.

[Table 6-1] Suggested STI Think Tank Governance in Indonesia

Option	Funding source	Governance	Status of the members	Function
1	Government budget (Ministry of Research and Technology/BRIN)	Government-affiliated	Government officers	Planning, Implementing, Evaluating
2	Budget of non-government organizations	Autonomous & independent body	Private sector	Planning, Implementing, Evaluating
3	Budget of government agencies and donors	Quasi-independent	Private sector	Planning, Implementing, Evaluating
4	Governmental budget under the Ministry of Finance (LPDP scheme)	Quasi-governmental	Contracted public officers	Planning, Implementing, Evaluating

Source: Developed by author

The ideal condition cannot be reached due to the current obstacles in Indonesia. Before one of the options can be implemented, there are important issues that must be addressed. First, there is a need to reform the rigidity to enable more flexible bureaucracy. A less flexible system may affect finance, law, and administrative procedures, for instance. Managing funds from both government and other sources requires creating a new form of think tank such as quasi-independent and independent think tanks in Indonesia. Second, a favorable coordination system should be structured among public agencies. In addition, the STI Policy think tank needs coordination with wide range of stakeholders. Third, Indonesia needs to increase the number and capacity of human resources in STI policy. The limited number of researchers performing functions in STI policy studies and lack of data on STI mean lack of capacity of human resources in STI policy. Fourth, evidence-based policymaking on STI should be strengthened. In most cases, policymaking without evidence tends to lead to trial judgment and errors. Fifth, it is necessary to develop an enabling environment for promoting innovation. In Indonesia's case, government bodies should have the same frequency for synergy to create an innovation system. Kemenristek/BRIN can be the main actor for constructing NIS among the institutions.

1.2 Policy Recommendations for R&D Funding and Management of Endowment Fund

Public R&D institutions are currently in the form of 1) research institutions directly under the president coordinated by the Ministry of Research and Technology/BRIN and 2) R&D units within the ministries/institutions, universities and in local governments. Practically, the Ministry of Research and Technology/BRIN only coordinates research institutions but has weaknesses, since the authority of the R&D program and the DIPA still entirely rests with each institution.

"The new" Law No. 11 Year 2019 has mandated the integration of all public R&D institutions in Indonesia under the auspices of the Ministry of Research and Technology/BRIN to ensure the sustainability of one-way R&D activities in order to increase R&D capacity in national development. Nonetheless, the meaning of "integration" in this mandate gives rise to some

perceptions as to how it should be done, which in turn produced two major options of governance.

The mandate of integration is defined as institutional integration. In this way, all (or maybe several) public R&D institutions (including LIPI) will be merged into the organizational structure of the Ministry of Research and Technology/BRIN. The centralization of R&D institutions under the auspices of the Ministry of Research and Technology/BRIN will facilitate control of directed R&D activities in accordance with the national R&D program. This option is the most likely one to be implemented, because it comes from the political will of the elected president. Nevertheless, we realize that structural integration needs a long and comprehensive process to be done well. The integration of public R&D institutions from various sectors with very broad field and scope is the main challenge; if this is not properly planned and designed, it can become a threat to the continuity of R&D activities in Indonesia.

[Table 6-2] Governance Options of Public R&D Institution

Governance Option	Option 1 Structural Integration	Option 2 Program Integration
Strength	Facilitate control of the directed R&D activities in accordance with the national R&D program	Integration carried out without changing the existing institutional structure
Weakness	The integration of public R&D institutions from various sectors with very broad field and scope needs a long and comprehensive process to be done well If it is not properly planned and designed, the cost will be very high	This has been proven unsuccessful so far. It cannot integrate the ministry's R&D units due to the absence of a command and coordination line, since R&D units will be working based on programs given from their own ministries.
Opinion by Author*	Preferable	Less Preferable

Source: Processed by Author

*This neither reflects nor represents the opinion of other parties.

The mandate of integration is not defined as institutional integration, but only the integration of programs and its funding. This means that public R&D institutions remain in their current position, but the authority in the formulation of the R&D program and its budget is entirely taken over by the Ministry of Research and Technology/BRIN. In other words, public R&D institutions (including LIPI) will only work to carry out what has been planned by the Ministry of Research and Technology/BRIN. Public R&D institutions do not have R&D programs and funding; the DIPA attached to the institution will only be used for the development of R&D resources, such as infrastructures, researchers and other human resources competencies, and networks. Some parties also call this option “strengthening the coordination.” This option could be the best choice to implement for now because integration is carried out without changing the existing institutional structure. Still, this option has a major weakness, i.e., it cannot integrate R&D units under ministries (the proportion is larger than the amount of research institutions like LIPI) due to the absence of a command and coordination line between the Ministry of Research and Technology/BRIN and ministry’s R&D units. Nonetheless, R&D units will be working based on programs and duties given from their own ministries. The experience in Indonesia shows that, without any clear lines of coordination, each institution will tend to run on their own, and collaborative processes are very difficult to carry out. This will hamper the sustainability of one-way R&D activities in order to increase R&D capacity in national development.

Option 1 is to integrate all research institutes and place them under the jurisdiction of a single ministry (structural integration, making a super-ministry). Such approach of structural integration needs to consider the following points:

First, it is necessary to choose the scope of the institution to be integrated. Since various types of public research institutes, such as 6 independent research institutions, research centers under the Ministries, and university units, are involved, it is necessary to consider how they are classified and grouped. According to the type of research institute, it is possible to group them into three: public research institutions, ministerial R&D units, and university R&D units. What is very important in this approach is that R&D management agencies should be established as an intermediary organization. Otherwise, the ministry will be overloaded, and it cannot manage the integrated R&D institutions efficiently. These

agencies mediate between ministries and R&D institutions/units, and they are commissioned by the ministries to conduct evaluations of institutional management or R&D performance. Second, the discussion of intermediary organization (R&D management agencies) implicitly assumes that the R&D management agencies should be established outside the single super-ministry. This is the same logic as building an independent institution (option 2) in terms of governance of the research endowment fund. Third, structural integration requires strong political support because there may be opposition and conflict from target institutions/units. If the conflict worsens, we can access sequentially among the three types in a step-by-step manner. In this case, one alternative is to choose first the 6 public R&D institutions and make a benchmarking case that can improve Indonesia's R&D performance.

Option 2 is to leave the current structure untouched and strengthen the coordination of R&D institutions/units through R&D program formulation and budget allocation. Option 2 (System operation: Coordination by program and budget) may be an easier approach compared to option 1 because there will be little resistance to reform. Four things need to be mentioned.

First, formulating R&D programs for the institutions/units is a huge challenge. The establishment of national R&D programs is a key task in policy planning, requiring good policy competency from the Ministry of Technology and Research/BRIN in charge. The program should be designed according to the national development strategy and future industrial development direction; it should respond quickly to the changing technology environment and continuously adjust the program portfolio from a long-term perspective. In the case of Korea, based on periodic technology forecast projects, the government, R&D management agency, and industry-university-GRI experts jointly participate in the program planning committee. Logistics work on the composition and operation of the planning committee is carried out in such a way that the R&D management agency supports the S&T Ministry. Second, the key to coordinating the activities of the institution/unit through the program is to control the overlapping research among research institutions/units. Redundancy of R&D occurs in programs of S&T-related ministries as well as projects of R&D institutions/units. In Korea, ministerial-level redundancy is controlled by inter-ministerial

PACST, and GRI-level redundancy is controlled by NST. Third, for successful budget allocation, an evaluation system for R&D institutions should be established. This means that budget allocation and institutional evaluation are closely related. It will not be successful if performance evaluation is not integrated into the budget and only budget allocation is emphasized. Fourth, in addition to the importance of the evaluation system, a balanced proportion of institutional and project support should also be emphasized. There is a similarity between the fact that NST oversees the GRIs in Korea and the fact that AIPI oversees the 6 R&D public institutions in Indonesia. Nonetheless, it seems that AIPI, as an Umbrella Organization, is not strong, and the affiliated institutions are enjoying a great deal of independence and autonomy. This is mainly because the proportion of funding for institutional support is excessively high, about 95%, so the affiliated research institutions can decide on their own programs without external influence. In Korea, the proportion of institutional and project support has changed from 70:30 in the 1990s to 40:60 after the 2000s. Since project support is based on competition, it has more performance-stimulating properties. There should be a productive balance between institutional autonomy and system performance.

1.3 Policy Recommendations for the Management and Evaluation System of the National R&D Program

The Indonesian government has long- and mid-term R&D policy such as RIRN (2017 ~ 2045), which is the Master Plan on National R&D Program and PRN (National Research Priority) to be revised every five years. This indicates that Indonesia has a structure for formation, implementation, and evaluation of an R&D program and a strong will to implement an R&D policy. Still, it seems that Indonesia remains at the early stage of implementation of national R&D programs, and ex-ante and ex-post evaluation processes are not clear. In addition to this analysis, the investment target for R&D announced by the Indonesian government is not enough to operate R&D programs. Besides, funding sources to implement R&D programs are not determined. The cooperation mechanism between industry and national research institutes remains weak. Industries are not innovation-oriented due to their lack of R&D capacity. R&D activities in Indonesia are mainly driven by some state-owned

companies.

Although national flagship programs are being implemented on a joint research basis, tangible outcomes cannot be ensured due to the lack of industry participation. They could only produce academic papers and patents. The same is also true for research projects implemented by national institutes and universities. Other issues include lack of research, which can lead to innovation, and apparently weak cooperation mechanism among national research institutes, which facilitate voluntary cooperation. Therefore, Indonesia is advised to take further actions such as increasing R&D investment as long as the R&D funding sources are clearly indicated. Programs should be developed to induce the private sector to make R&D investment. Above all, there is a need to restructure the governance for R&D policy and programs so that the institutional framework for ex-ante and ex-post evaluation and implementation of R&D program and projects is effectively applied, and cooperation and coordination among research actors (national research institutes and universities) can be effectively facilitated.

Based on the Basic Plan for S&T (five-year plan), Korea was able to expand its R&D budget, enabling competitions among Ministries for national R&D programs and having a well-functioning system for ex ante and ex-post evaluation. In addition, the Korean government has been supporting research programs for public research institutions and universities. Various public programs also helped establish the system for industry-driven R&D investment, which led to the success of industry-driven technology innovation in Korea. Still, Korea has to address various issues such as lack of original technology and regional innovation capacity and ineffective cooperation mechanism of industry-university for convergence research innovation. These issues are mainly attributed to Korea's tendency of fast-follower research and government-driven research. Therefore, learning from Korea's lessons, Indonesia should undertake the implementation of a national R&D strategy.

2. Conclusion

Indonesia put its national priority on building STI capacity, aiming to achieve a knowledge-based economy in the mid- and long terms. The 2020 K-Innovation program contributes to Indonesia's STI development by providing policy consulting on the national innovation system of Indonesia, which is one of the key partner countries in Korea's southern policy. Policy directions are expected to be drafted from K-Innovation's consulting led by Korean experts through consultations with Indonesian researchers to improve the STI system in Indonesia. More specifically, Koreans shared insights and policy directions from the Korean and Indonesian practices regarding STI governance, management and evaluation system of the national R&D program, and management of R&D funding, which Indonesia considers in restructuring the framework on such focus areas. STEPI supported the capacity building of LIPI, Indonesia's largest research institute, over the last several years through the K-Innovation program. STI cooperation at the organizational level is believed to contribute to strengthening cooperation at the national level.

ABBREVIATIONS

AIPI	Akademi Ilmu Pengetahuan Indonesia (Indonesian Academy of Sciences)
APBN	Anggaran Pendapatan dan Belanja Negara (State Budget)
Bapeten	Nuclear Energy Regulatory Agency
Bappenas	Badan Perencanaan Pembangunan Nasional (National Development Planning Agency)
BATAN	Badan Tenaga Atom Nasi
BLU	Badan Layanan Umum (Public Service Agency)
BPPI	Agency for Research and Development of Industry
BPPT	Badan Pengkajian dan Penerapan Teknologi (Agency for the Assessment and Implementation of Technology)
BRIN	Badan Riset dan Inovasi Nasional (National Research and Innovation Agency)
BSN	Badan Standarisasi Nasional (National Standardization Agency)
BUN	Bendahara Umum Negara (State General Treasurer)
CIPG	Centre for Innovation Policy and Governance
CSTP	Center of S&T Policy
DIPA	Entry List of Budget Implementation
DIPI	Dana Ilmu Pengetahuan Indonesia (Indonesian Science Fund)
Iptekdikbud	Directorate of Science-Technology, Education, and Culture
KEIN	National Committee of Economics and Industry, Indonesia
KISTEP	Korea Institute of S&T Evaluation and Planning
KOSIME	Korean Society of Innovation Management and Economics
KOTIS	Korean Technology Innovation Society
KSI	Knowledge Sector Initiative

LAPAN	Lembaga Antariksa dan Penerbangan Nasional (National Institute of Aeronautics and Space)
LIPI	Lembaga Ilmu Pengetahuan Indonesia (Indonesian Institute of Sciences)
LPDP	Insentif Riset Sistem Nasional (Educational Fund Institute of Management Insinas)
MOEF	Ministry of Economy and Finance, Korea
MOI	Kementerian Perindustrian (Ministry of Industry)
MORT	Ministry of Research, Technology
MORTHE	Kementerian Riset, Teknologi dan Pendidikan Tinggi (Ministry of Research, Technology, and Higher Education)
MOTIE	Ministry of Trade, Industry, and Energy
MSIT	Ministry of Science and ICT, Korea
NST	National Research Council of Science & Technology
PACST	Presidential Advisory Council on Science and Technology
PPKMI	Centre for Science-Technology-Innovation Policy and Management
PRN	Prioritas Riset Nasional (in Indonesia); National Research Priority
PTI	Pengembangan Teknologi Industri (in Indonesia); Industrial Technology Development
PTKSSI	Centre for Specific Technology Area and Innovation System
RIRN	Rencana Induk Riset Nasional (National Research Master Plan)
RISPRO	Riset Inovatif Produktif (in Indonesia); Productive and Innovative Research
RKUN	Rekening Kas Umum Negara (in Indonesia); State General Treasury Account
RPJMN	Rencana Pembangunan Jangka Menengah Nasional (National Medium-Term Development Plan)
RPJPN	Rencana Pembangunan Jangka Panjang Nasional (Long-Term National Development Plan)
R&D	Research and Development
S&T	Science and Technology

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APPENDIX

1. Policy Think Tank Units in Indonesia

Indonesia has roughly 50-100 policy think tank units dispersed in many institutions affiliated with government units, private/business units, political parties' units, and universities.

No	Lembaga Think Tank	Kategori	Fokus Masalah
1	Centre for Strategic and International Studies (CSIS)	Autonomous and Independent*	Politics, Economy, & Security
2	Economic Research Institute for ASEAN and East Asia (ERIA)	Quasi Independent	Economy
3	Center for International Forestry Research (CIFOR)	Quasi Independent	Welfare & Environment
4	Indonesia Corruption Watch (ICW)	Autonomous and Independent	Corruption Issue
5	Center for Regional Information and Studies (PATTIRO)	Autonomous and Independent	Public Governance
6	Center for Strategic and International Studies ASEAN Institutes of Strategic and International Studies (ASEAN-ISIS)	Quasi Independent	Security
7	Indonesian Institute of Science (LIPI)	Government Affiliated	Economy
8	Habibie Center	Autonomous and Independent	Modernization and Democratization
9	Center for Indonesian Policy Studies (CIPS)	Autonomous Independent	Welfare, Economy Food Security, Education
10	Institute of National Capacity Studies (INCS)	Quasi Independent	Security, Health, Education, Democratization
11	Knowledge Sector Initiative (KSI)	Quasi Governmental	Education, Equality, Food Security
12	Center of Reform on Economics (CORE)	Quasi Independent	Macro Economy, Thematic
13	Reformed Center for Religion & Society	Quasi Independent	Religion, Minority
14	Center for Indonesia Taxation Analysis (CITA)	Autonomous Independent	Tax
15	The Indonesian Institute, Center for Public Policy Research	Autonomous Independent	Economy, Social, Politics, Law
16	The Prakasa	Quasi Independent	Fiscal, Social, Sustainable Development
17	Akatiga	Autonomous Independent	SocialPolicy, Economy Growth, Labour, Migration and Job , Government & Transparency, Agriculture and Food Security
18	Independent Study Center for Acehese Democracy (SCAD Independent)	Autonomous Independent	Democratization and Human Rights
19	Wahid Institute	Quasi Independent	Pluralisme, multikulturalisme, democracy, human rights
20	Yudhoyono Institute	Political Party Affiliated	Prosperity and Security
21	Megawati Institute	Political Party Affiliated	Pancasila (Indonesian Ideology)
22	Paramadina Public Policy Institute (PPPI)	University Affiliated	Public Governance, Corruption Issue
23	Center for Political Studies UI	University Affiliated	Politics
24	The Center for Public Policy Transformation (Transformasi)	Quasi Independent	Climate change, Sea Affairs, Economy
25	Lembaga Penyelidikan Ekonomi dan Masyarakat – FEB UI	University Affiliated	Economy

Source: McGann (2020), and authors' findings.

2. Advocacy Activities through Media

2.1. Kick-off Seminar: 19 February 2020

■ ET News on 20 February 2020

STEPI, 인도네시아와 과기정책 협력의 장 펼쳐

발행일 : 2020.02.20

기사번역의뢰



가

가



데이터 분석의 무한 가능성 - 다양한 산업/부서별 혁신 사례



<2020년도 한-인도네시아 국제기술혁신사업 착수세미나 참석 단체사진>

과학기술정책연구원(STEPI·원장 조항희)은 지난 18~19일 인도네시아 자카르타에서 열린 '한국 전문가 초청 과학기술정책 강연'에 참여해 '2020년도 한-인도네시아 국제기술혁신사업' 착수세미나를 개최했다.

한국 전문가 초청 과학기술정책 강연은 주인도네시아 한국대사관과 인도네시아 연구기술부가 공동 주최했다.

■ Herald Corporation on 20 February 2020



19일 인도네시아에서 열린 2020년도 한-인도네시아 국제기술혁신사업 착수세미나 참석자들이 기념촬영을 하고 있다.(STEPI 제공)

- "허리통증"완화입증! "이것"으로 간...
- [단독] "이것" 지금 당장 시작하면, "급..."
- 로또용지 찢지마 "뒷면"에 1등당첨...
- 로또당첨, "이것"만 기억하면 당첨가...

[헤럴드경제=구본혁 기자] 과학기술정책연구원(STEPI)은 18~19일 양일간 인도네시아 자카르타에서 '한국 전문가 초청 과학기술정책 강연'과 '2020년도 한-인도네시아 국제기술혁신사업' 착수세미나를 개최했다.

먼저 18일 열린 '한국 전문가 초청 과학기술정책 강연'에서는 임덕순 STEPI 선임연구위원이 '한국의 과학기술정책의 과거와 미래'란 주제로 강연을 진행했다.

임 선임연구위원은 이번 강연에서 한국의 과학기술정책 경험을 공유하고, 인도네시아의 과학기술정책 현황 분석 및 진단결과를 바탕으로 한-인도네시아 과학기술협력방향을 제시했다.

■ Facebook on 26 February 2020

 Stepi iicc님이 Deok Soon Yim님, Jong-in Choi님과 함께 있습니다. ...
 어제 오후 1:28 · 🌐 ▼

STEPI continues to work with Indonesian government on STI policy through 2020 K-Innovation ODA Program. At the policy briefing on STI hosted by the Embassy of Republic of Korea in Indonesia and the Ministry of Research and Technology, Dr. Deok Soon Yim, Senior Research Fellow of STEPI shared his expertise on Korea's STI Policy Development with Dr. Bambang Brodjonegoro, Minister of Research and Technology of Republic of Indonesia and heads of Indonesian national research institutes. On 19 February, 2020, STEPI and LIPI co-organized a kick-off seminar to launch 2020 K-Innovation ODA Program with Indonesia. 2020 Program will support Indonesia to improve STI governance particularly on coordination mechanism, institutional framework, financial and investment system.



  Deok Soon Yim, Ashraf Gardizy, 외 13명 댓글 2개 공유 1회

2.2 STEPI Special Session during the 2020 ASIP Conference

■ Herald Corporation on 25 September 2020

헤럴드경제

STEPI, 인도네시아 국가혁신시스템 연계 강화 지원

입력 2020.09.25. 오후 12:30

구본혁 기자 >

공감 댓글

기사



STEPI와 인도네시아 국립과학원(LIPI) 공동주관으로 진행된 '인도네시아 특별세션'에서 임덕순(가운데) STEPI 선임연구위원이 토론을 진행하고 있다.[STEPI 제공]

[헤럴드경제=구본혁 기자] 과학기술정책연구원(STEPI)은 24일 제주도 한화리조트에서 열린 '2020 아시아기술혁신학회 국제컨퍼런스'에서 '한국 파트너십을 통한 인도네시아의 혁신시스템 연계'를 주제로 인도네시아 특별세션을 개최했다.

STEPI와 인도네시아 국립과학원(LIPI) 공동주관으로 진행된 '인도네시아 특별세션'에는 메고 피네티토 LIPI 연구기술혁신부 과학기술정책보좌관·부원장을 포함해 인도네시아 및 국내 과학기술정책 전문가 50여명이 온라인으로 참석했다.

■ STEPI Homepage

STEPI 동정

HOME > 연구원 소식 > STEPI 동정

Print + - 원대로

제목	STEPI, 아시아기술혁신학회 "인도네시아 특별세션" 진행(9/25)		
작성일	2020-09-29	조회수	420



과학기술정책연구원(STEPI)은 24일 제주도 한화리조트에서 열린 '2020 아시아기술혁신학회 국제컨퍼런스'에서 '한국 파트너십을 통한 인도네시아의 혁신시스템 연계'를 주제로 인도네시아 특별세션을 개최했다.

STEPI와 인도네시아 국립과학원(LIPI) 공동주관으로 진행된 '인도네시아 특별세션'에는 메고 피네티도 LIPI 연구기술혁신부 과학기술정책보좌관 부원장을 포함해 인도네시아 및 국내 과학기술정책 전문가 50여명이 온라인으로 참석했다.

인도네시아국립기술혁신협력사업(K-Innovation)의 정책자문 결과를 공유하기 위해 기획된 이번 특별세션은 하나의 주제를 통해 국내 전문가와 인도네시아 전문가가 힘을 미뤄 발제하는 방식으로 진행됐다.

권기석 한밭대 교수와 유카 마스마라 LIPI 연구원은 '인도네시아의 과학기술정책 쟁점크 설정방안, 김성수 한국외대 교수와 두 디 히다얏 LIPI 과학기술정책센터장은 '인도네시아의 연구진흥기금 관리방안 설계' 등을 발제했다.

갈루 인드라프라하스타 박사는 한국과 인도네시아의 국가연구개발프로그램의 평가시스템 비교분석을 통한 '인도네시아 국가연구개발프로그램 평가시스템에 대한 제언'이란 주제로 발표했다.